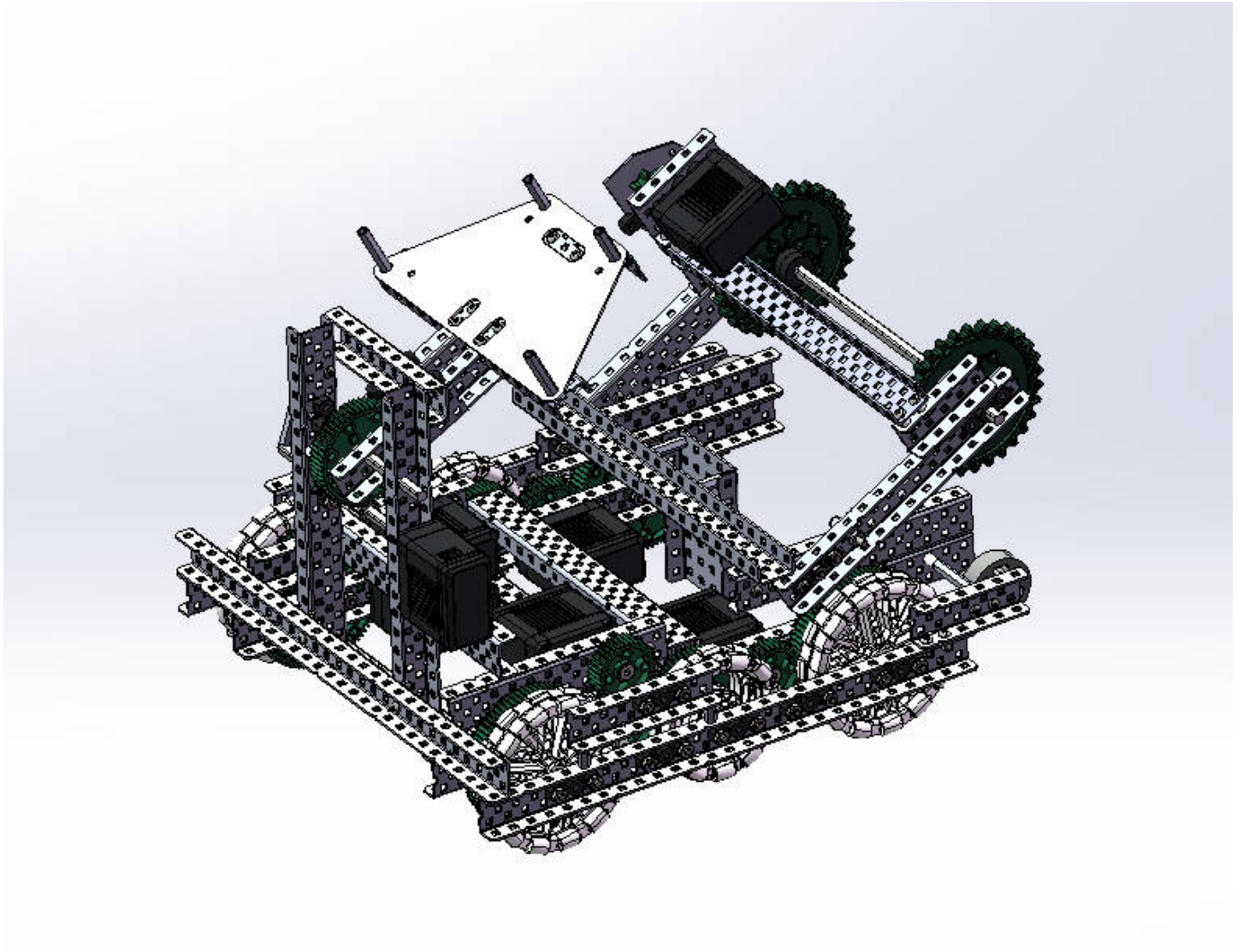


V2.5 257RPM cata



抛投机

分为3个部件

- 底盘

6电机，36-84 齿轮传动，7：3传动，257RPM，6轮

- 抛投结构

1电机，12-60齿轮传动，5：1传动，减速传动

- 吸取结构

1电机，600RPM直传

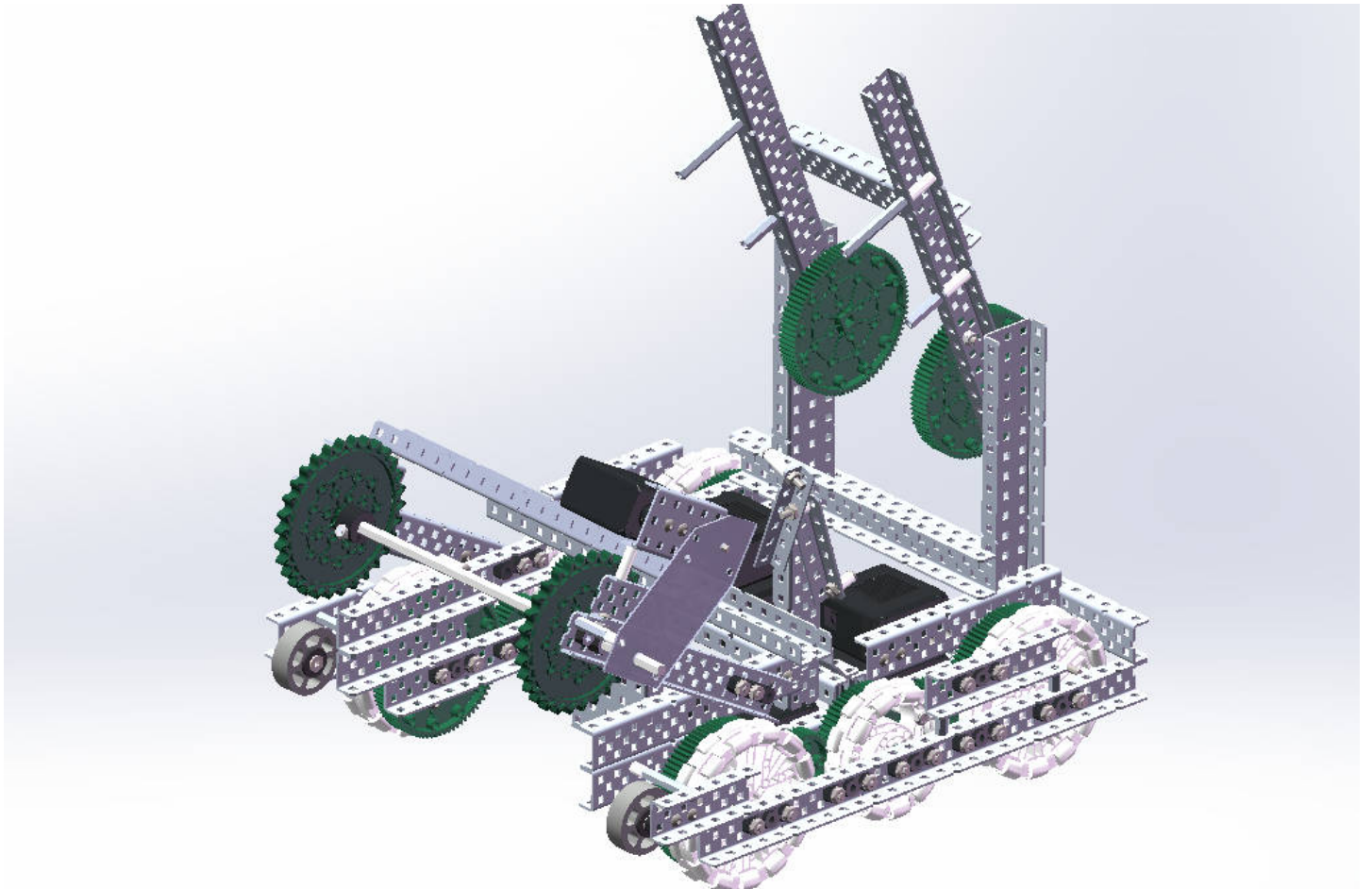
- 爬升结构

侧面爬升，可挂A

- 翅膀

双气动活塞

V2.5 257RPM climb



爬升机

分为3个部件

- 底盘

6电机，36-84 齿轮传动，7: 3传动，257RPM，6轮

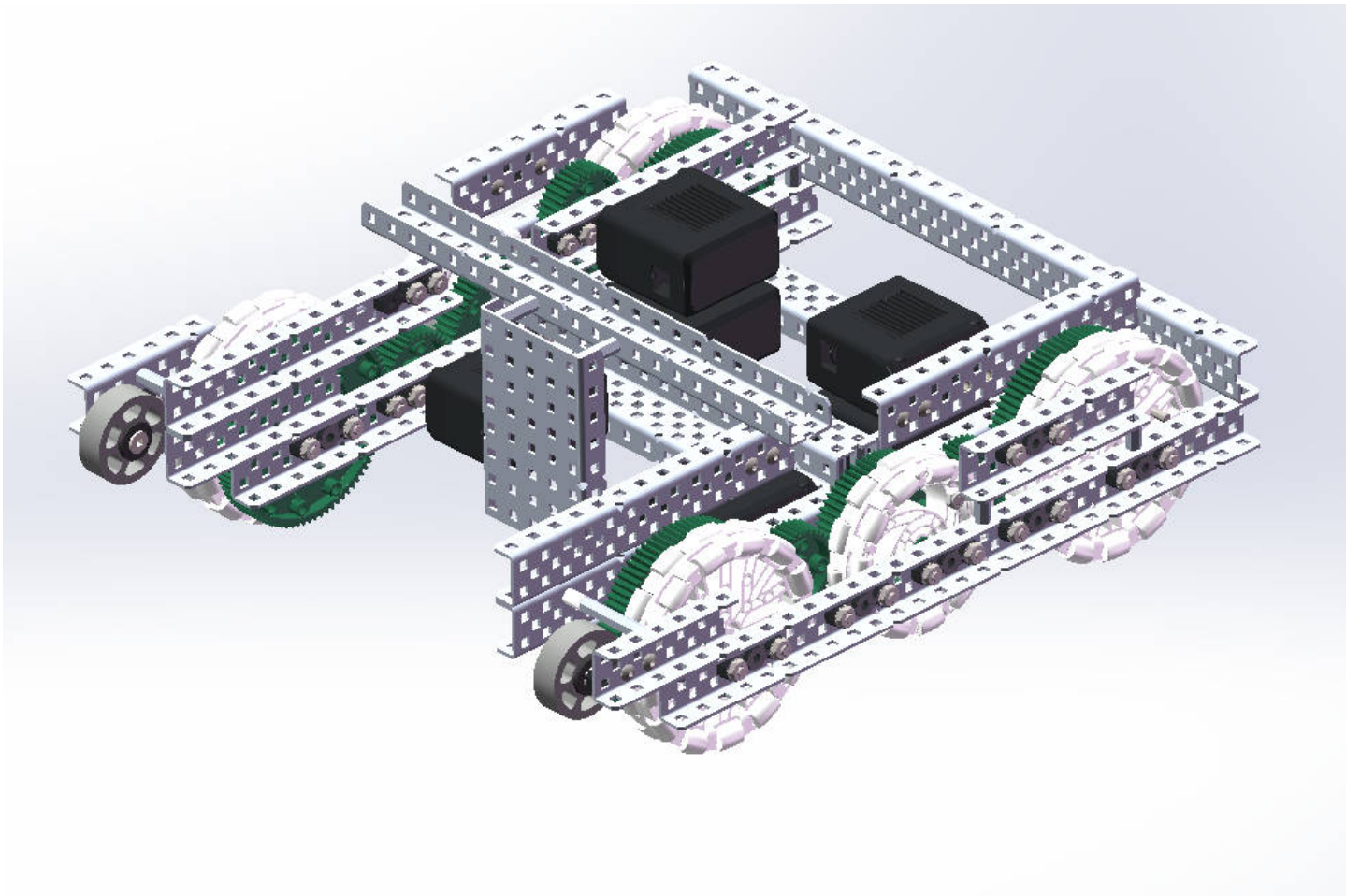
- 爬升结构

1电机，12-84齿轮传动，7: 1传动，减速传动，可挂等级B

- 吸取结构

1电机，600RPM直传

01 底盘搭建

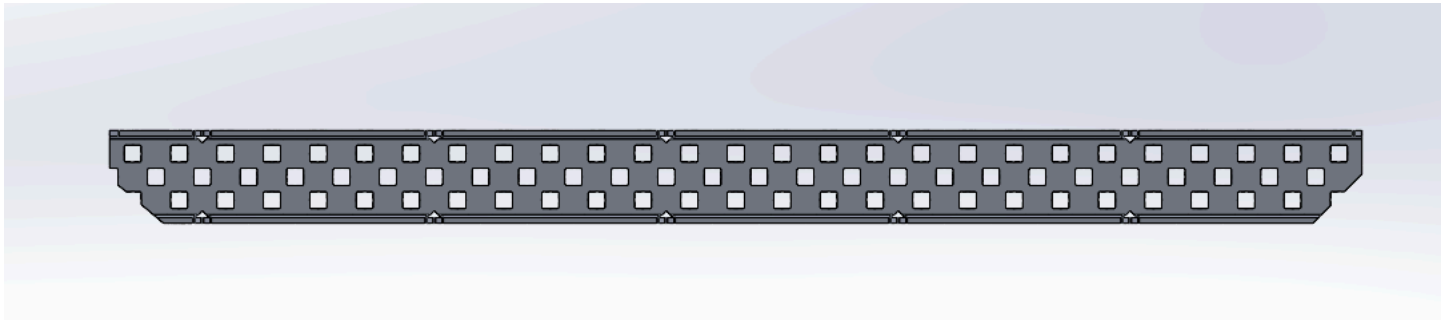


铝件长度清单

铝件名称	数量
窄C 27	5
窄C 17	4
窄C 21	1
窄C 13	2
窄C 8	4
窄C 12	2
窄C 6	2

1.铝件加工

前后都需要切割一个角，方便越野

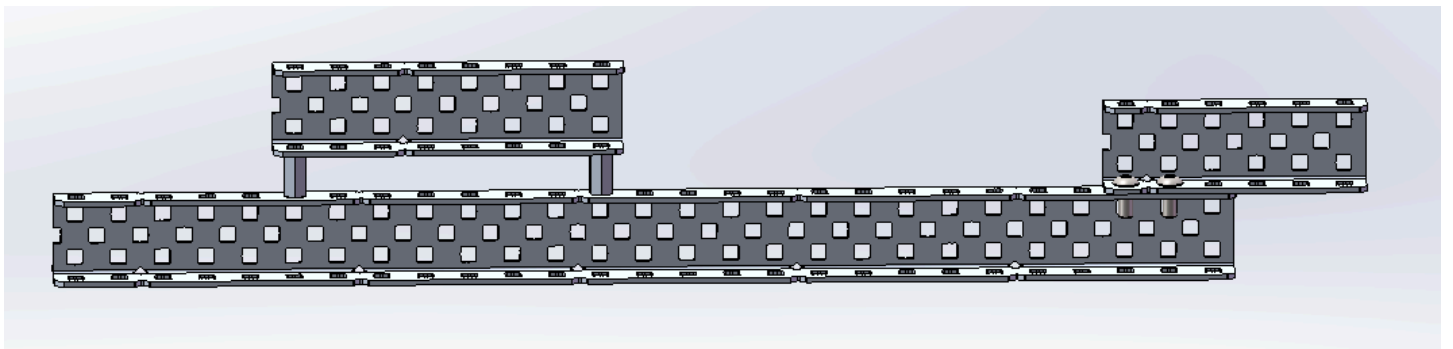
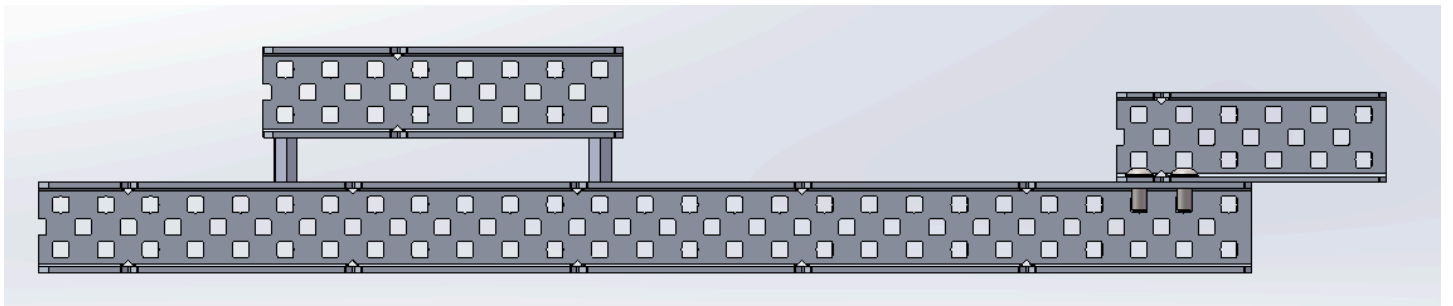


注：下面的搭建步骤图把切割的角隐藏起来了，读图时需要注意

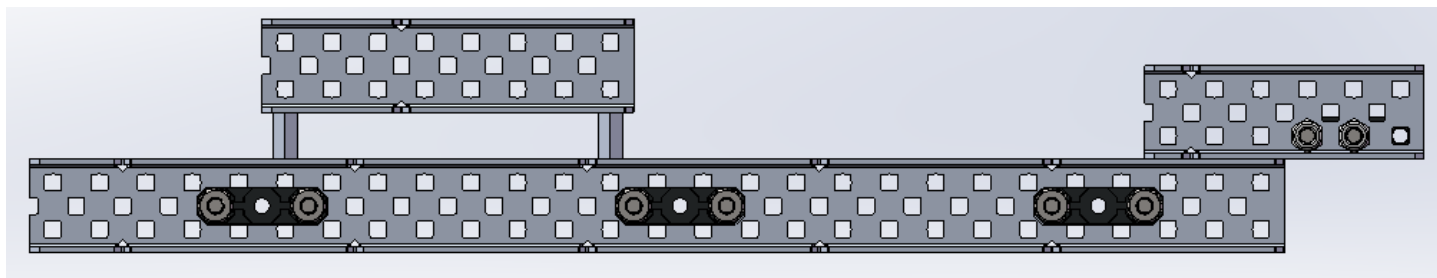
2.安装底盘外侧

注：下图安装的铝件已经隐藏了切割部分

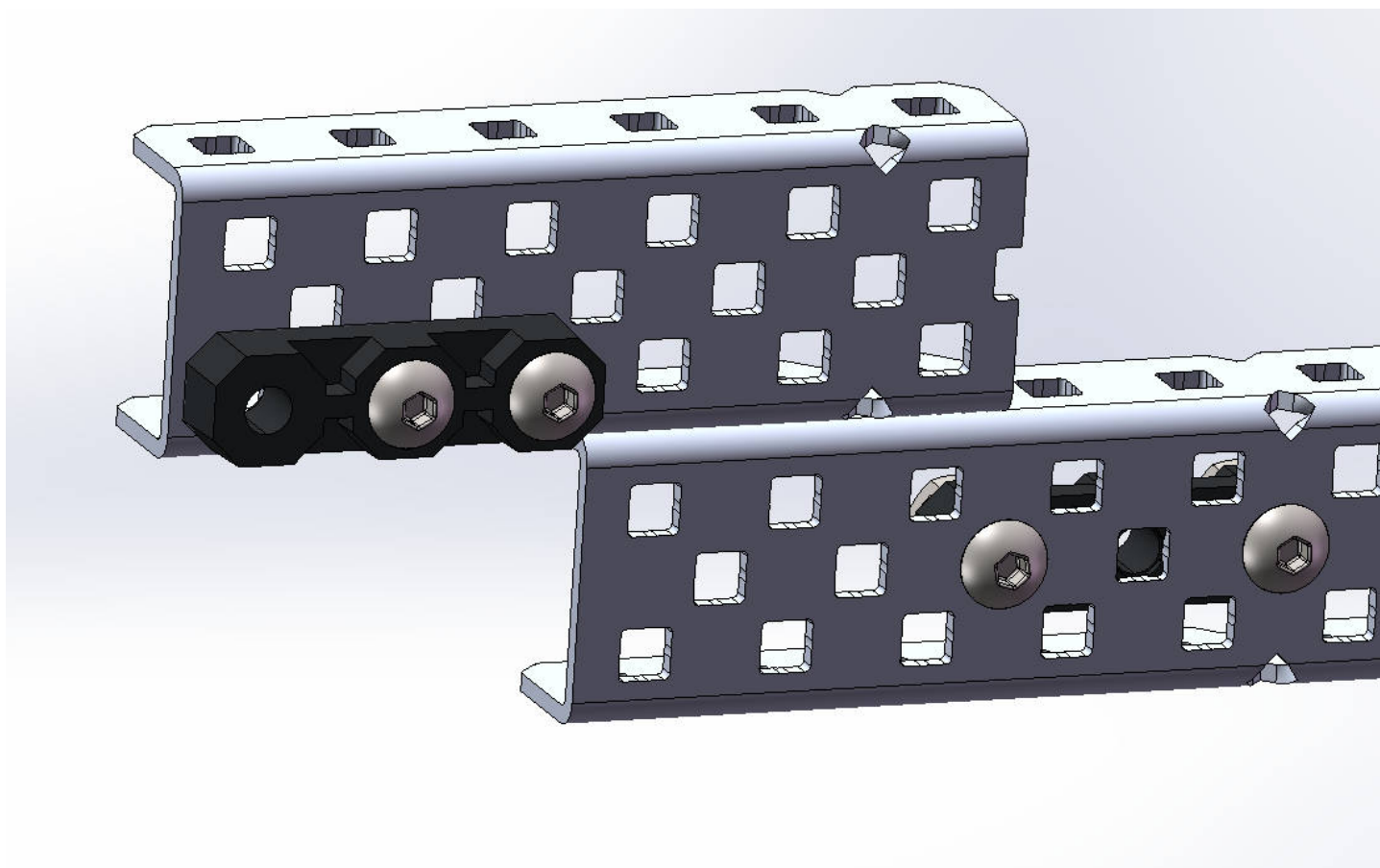
需要安装2个如下图所示的装配体（注意2个一定要对称）



安装定位轴承座

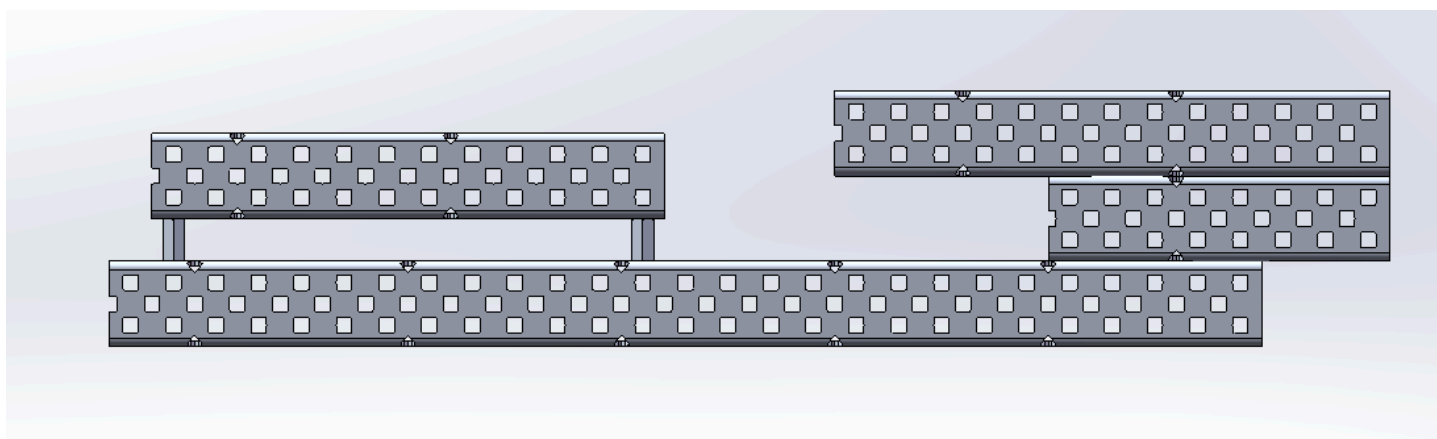


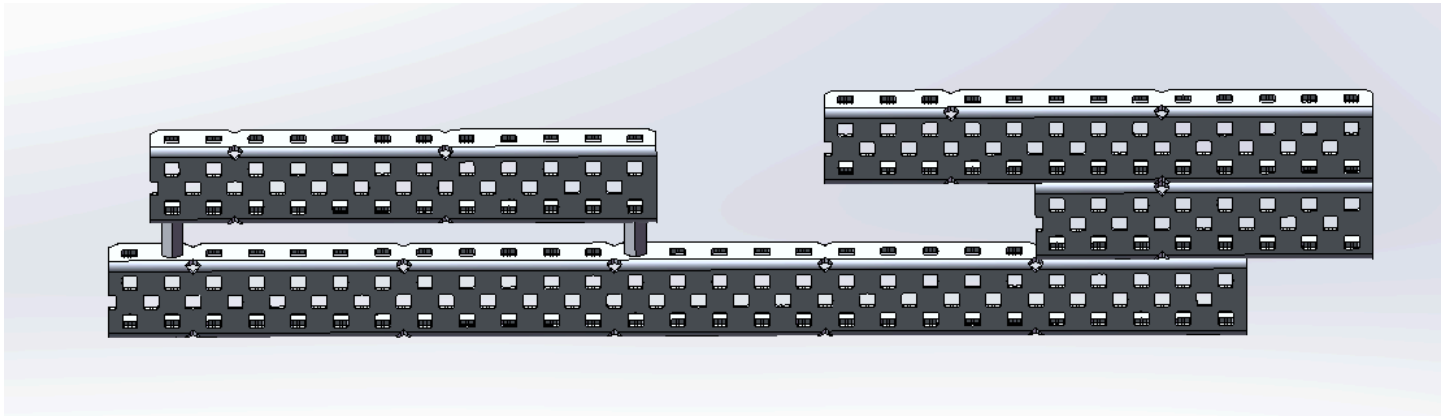
安装滑轮轴承座



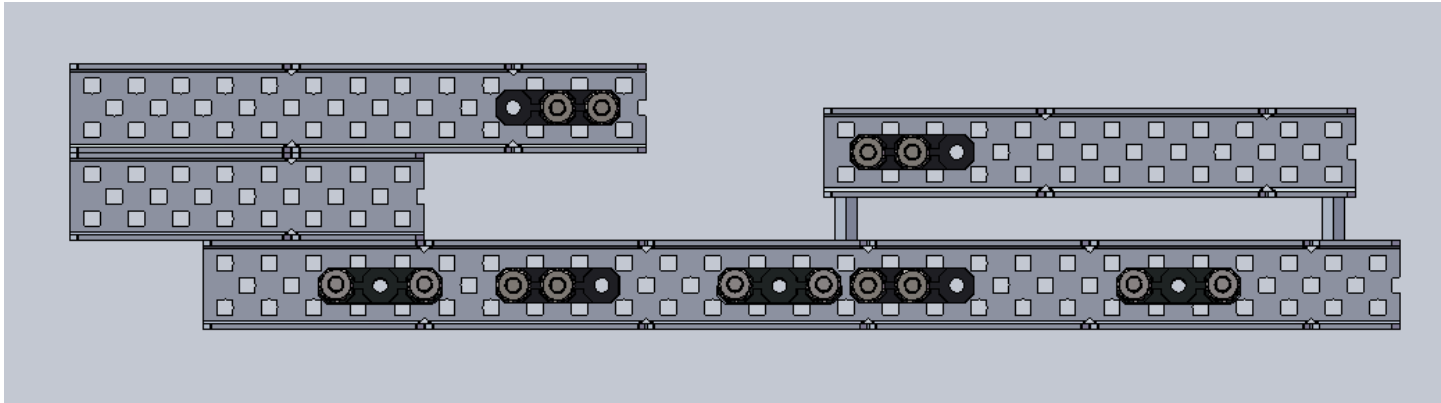
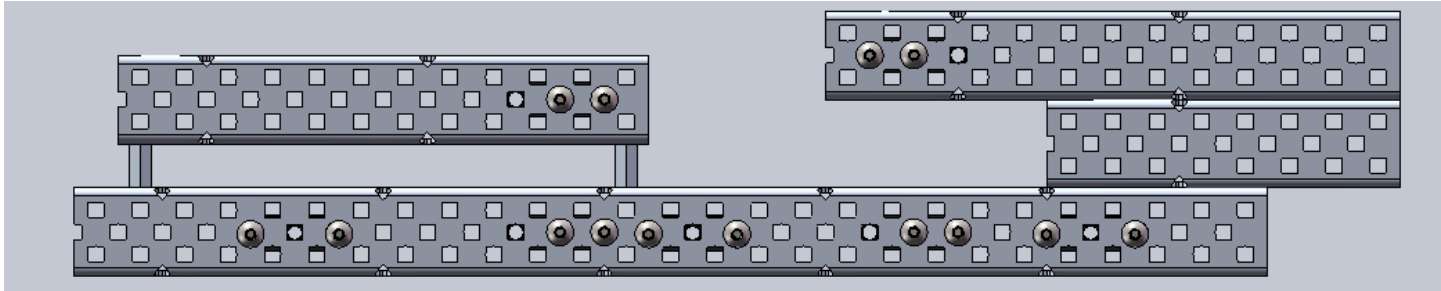
3.搭建底盘内侧

需要安装2个如下图所示的装配体（注意2个一定要对称）





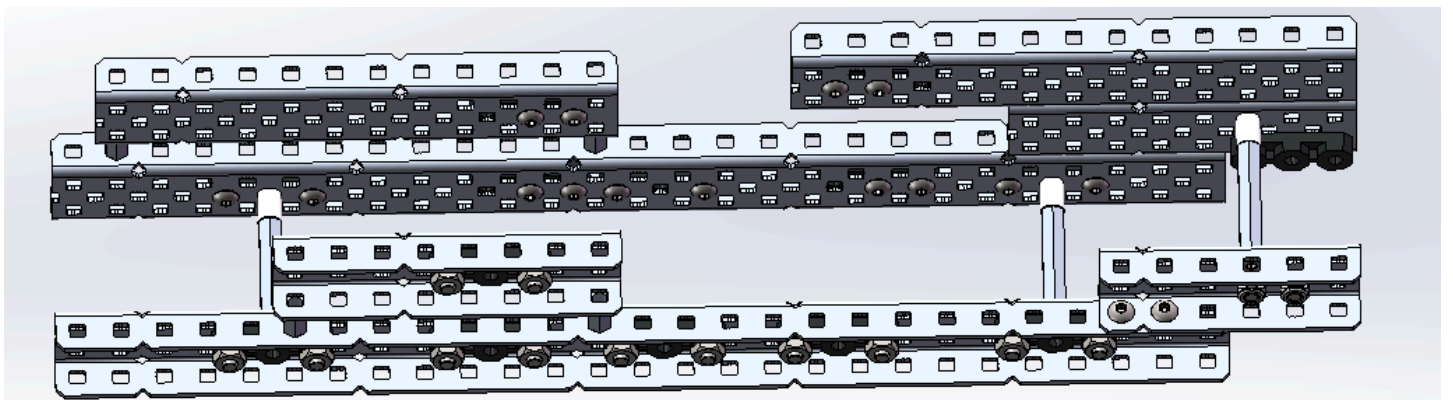
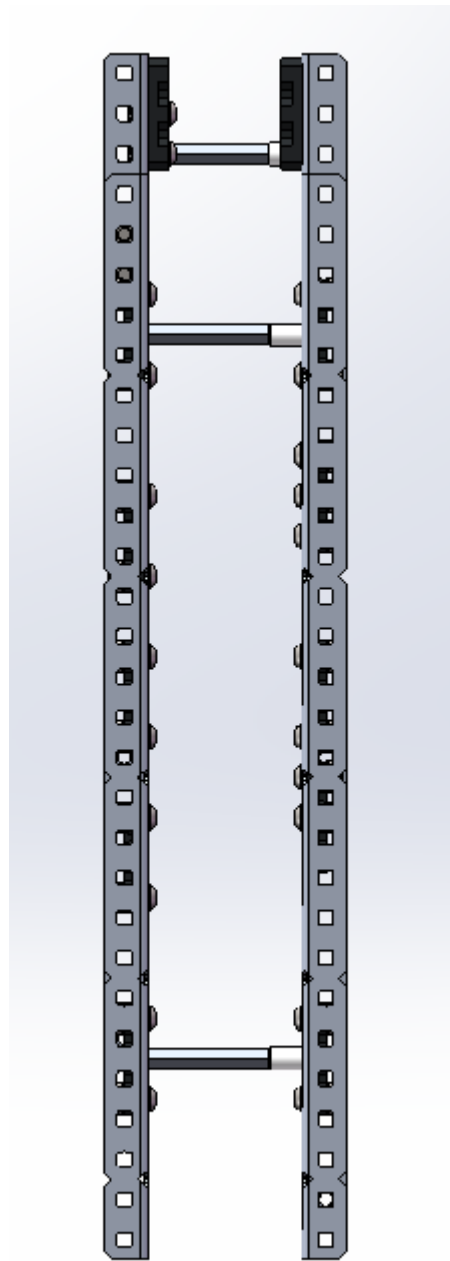
安装定位轴承座

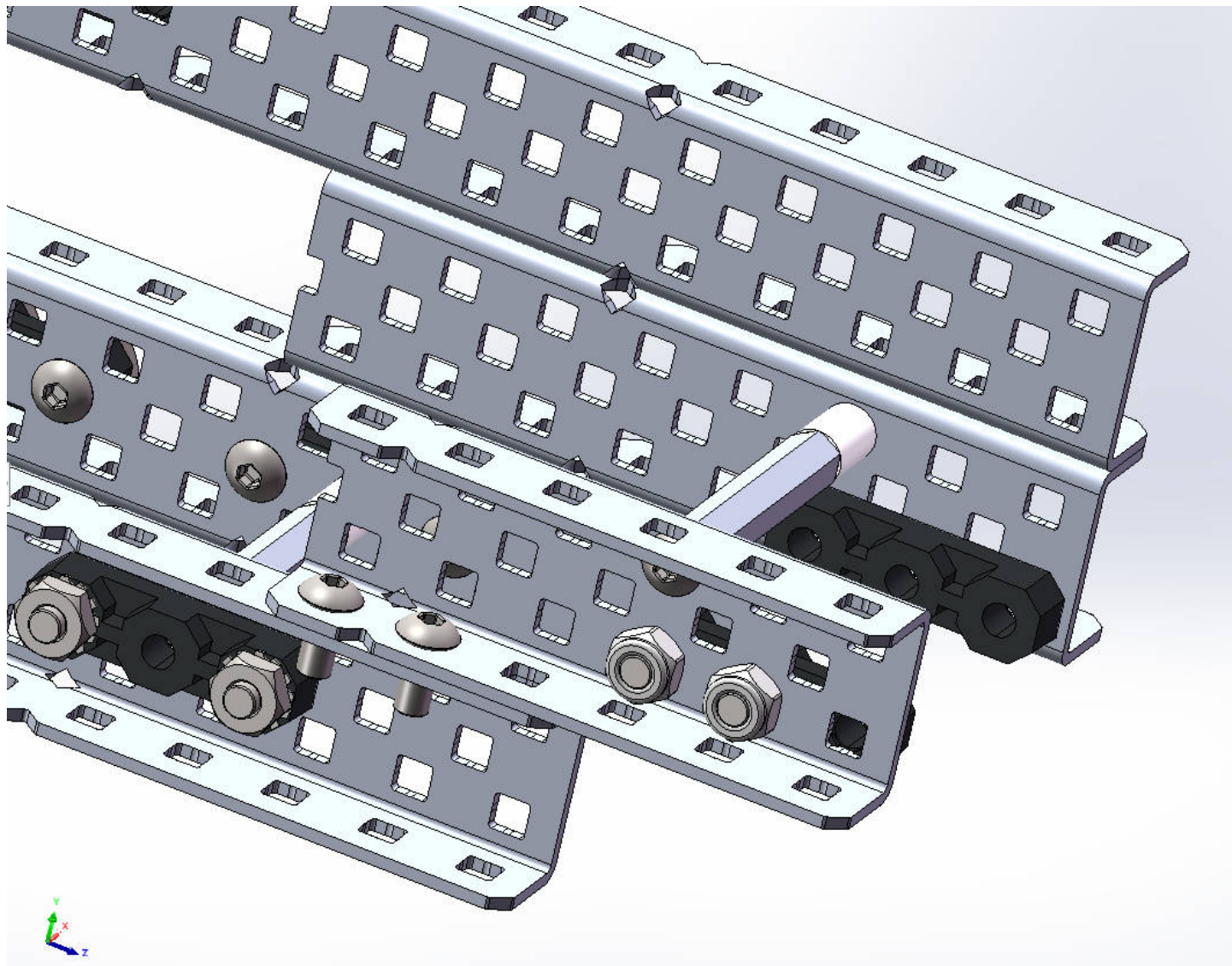


4.安装框架

- 用螺母柱安装（1.5in螺母柱 + 10mm垫圈）

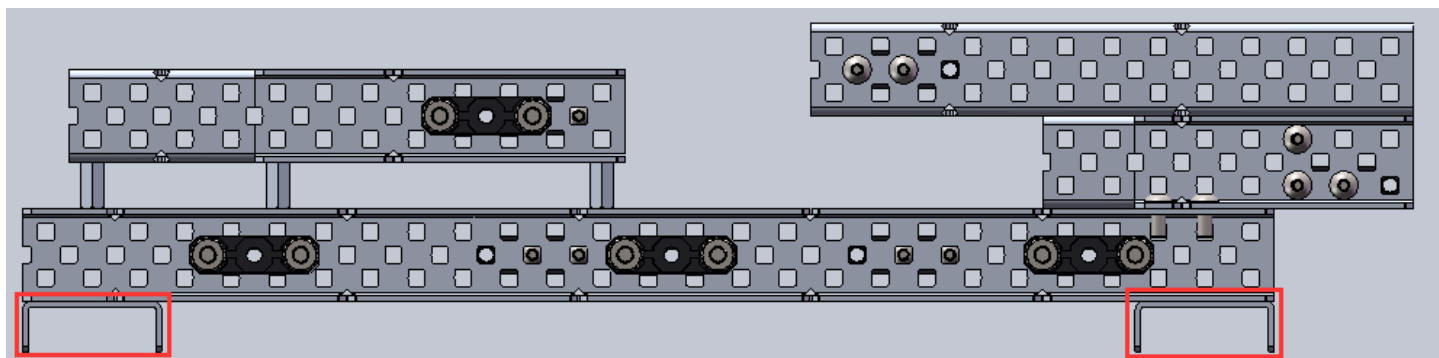
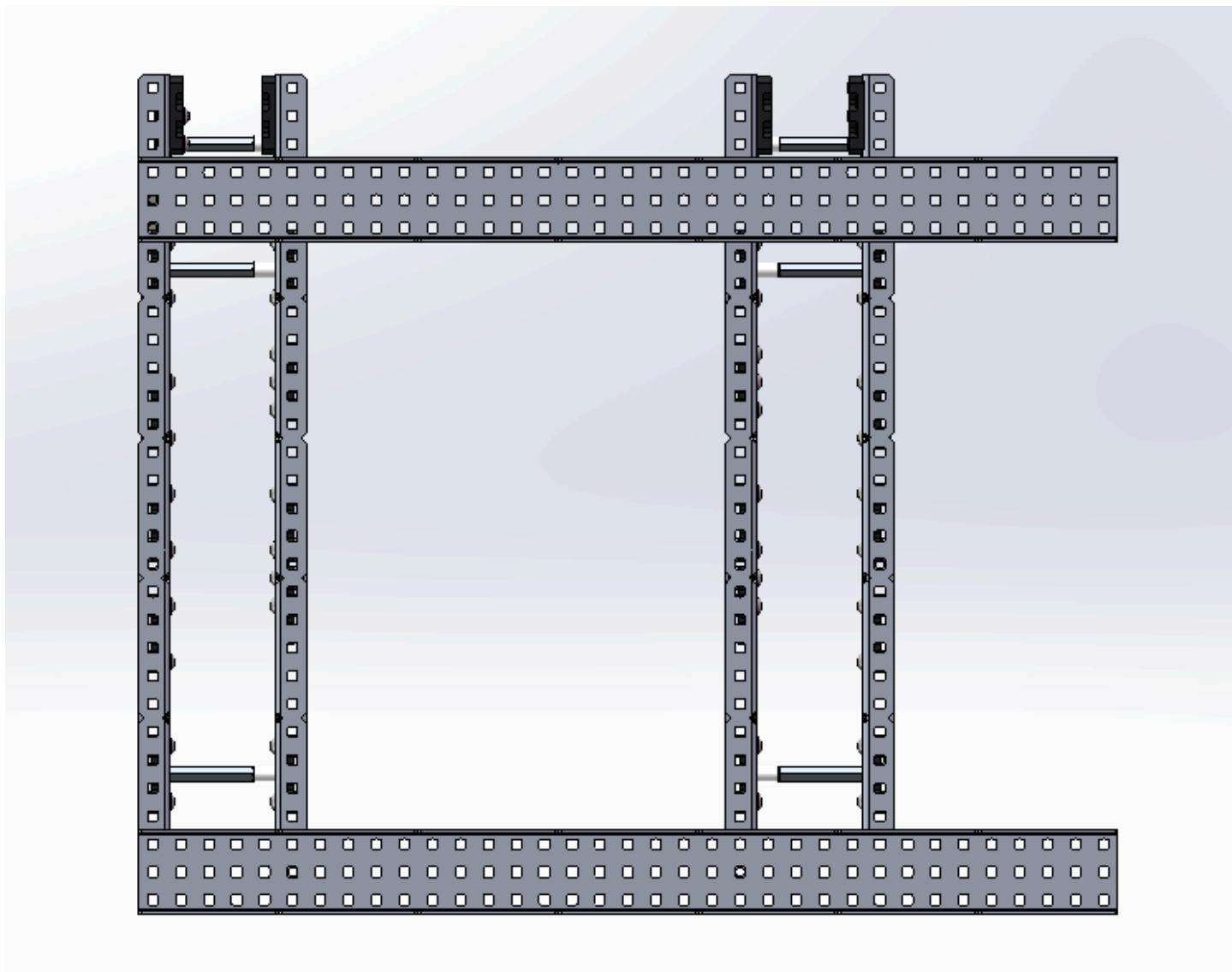
安装两个，需要对称





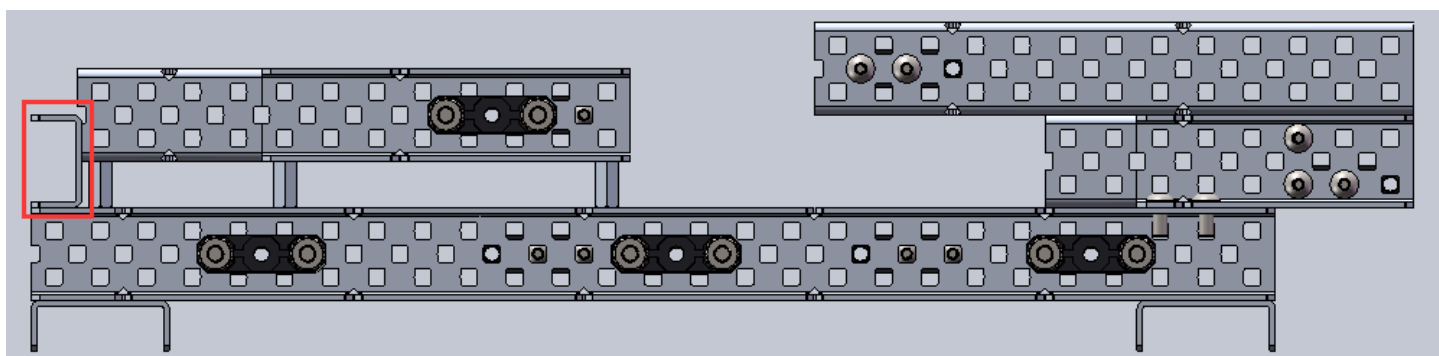
- 安装定位横梁

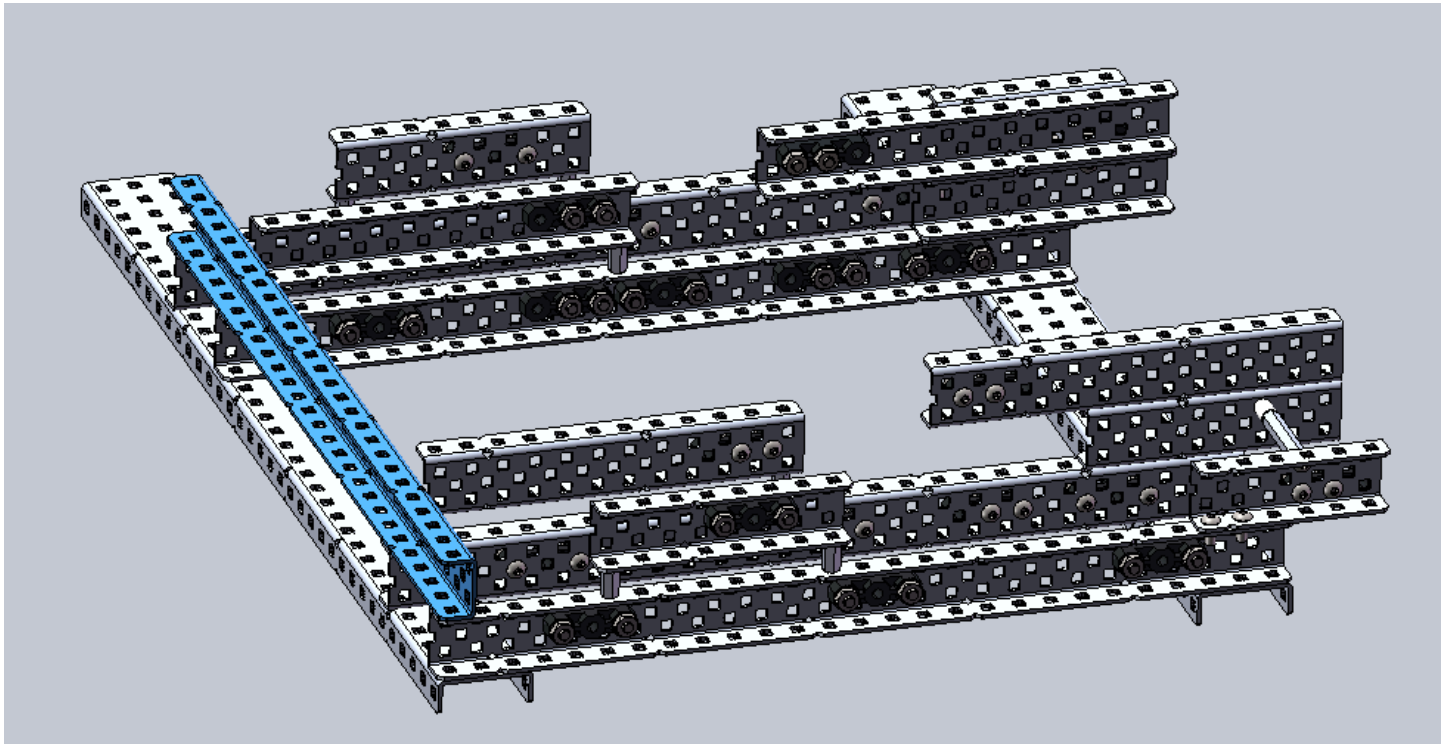
底盘总宽度是27格



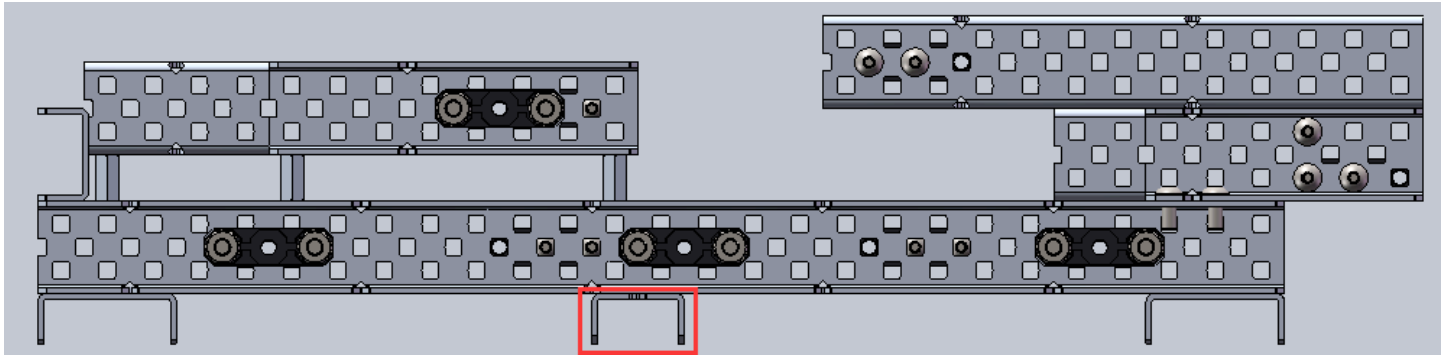
- 安装固定横梁

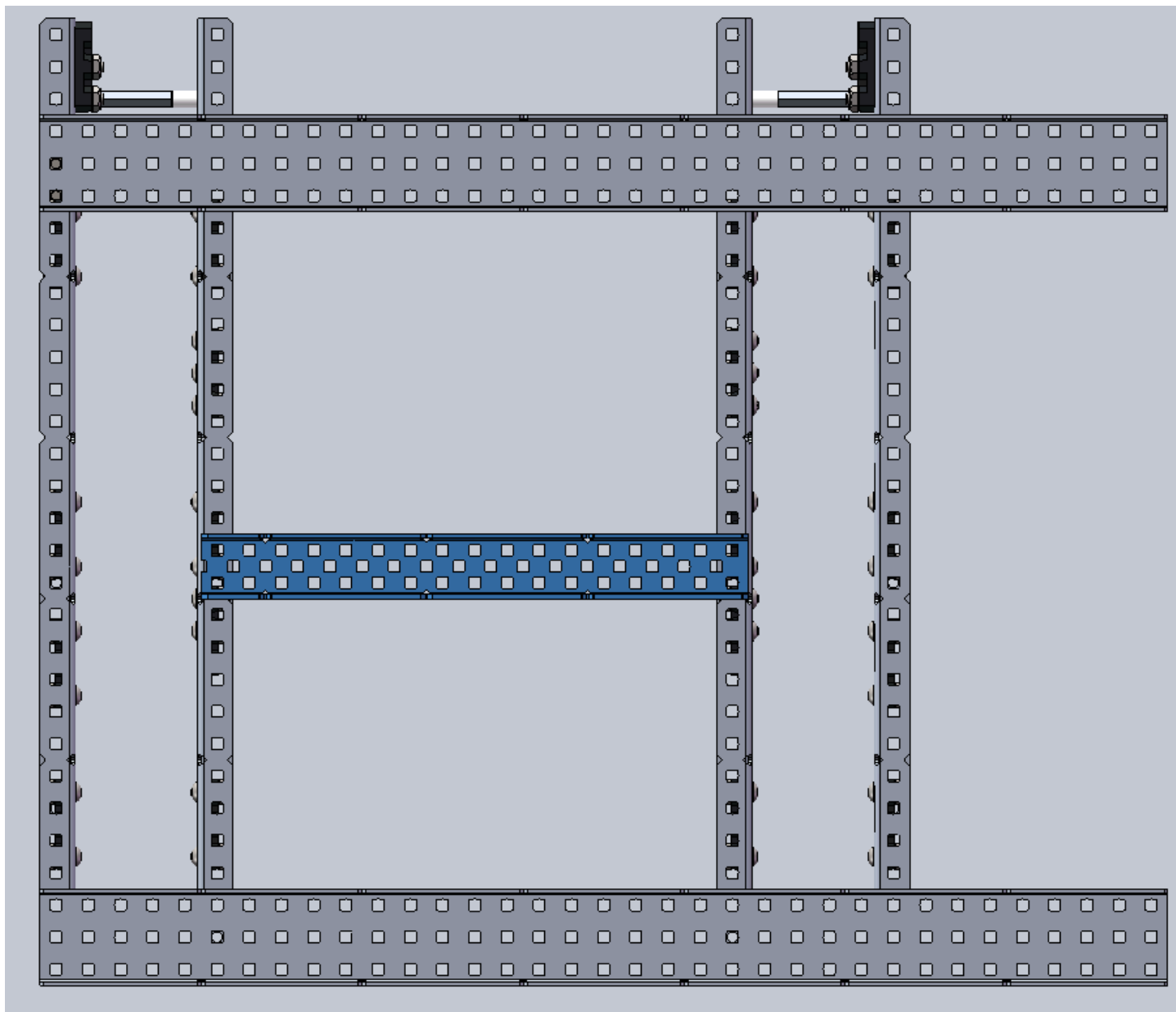
固定横梁1





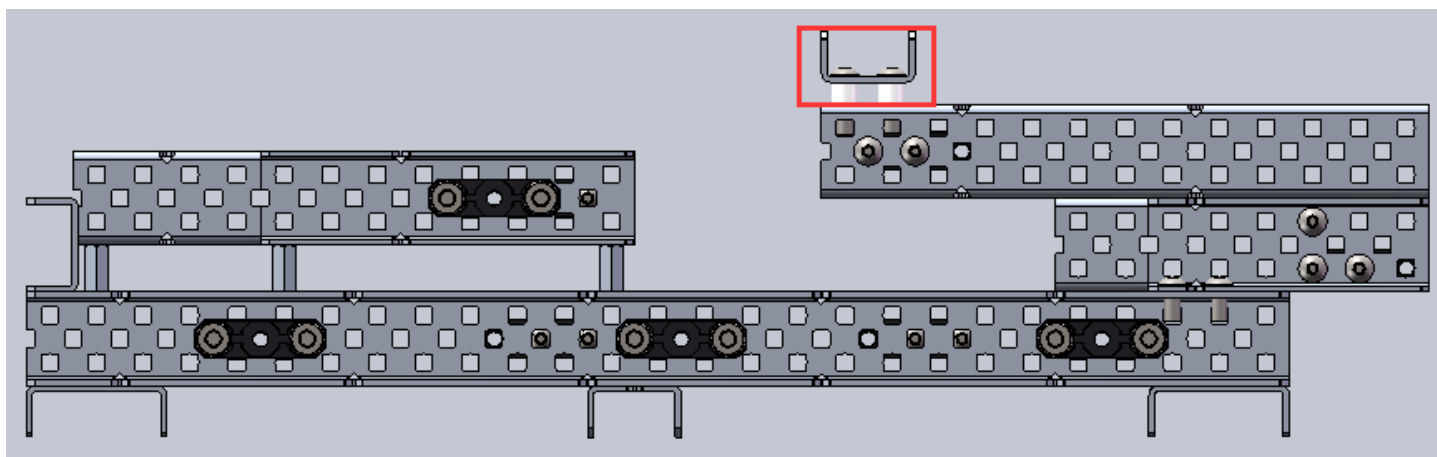
固定横梁2

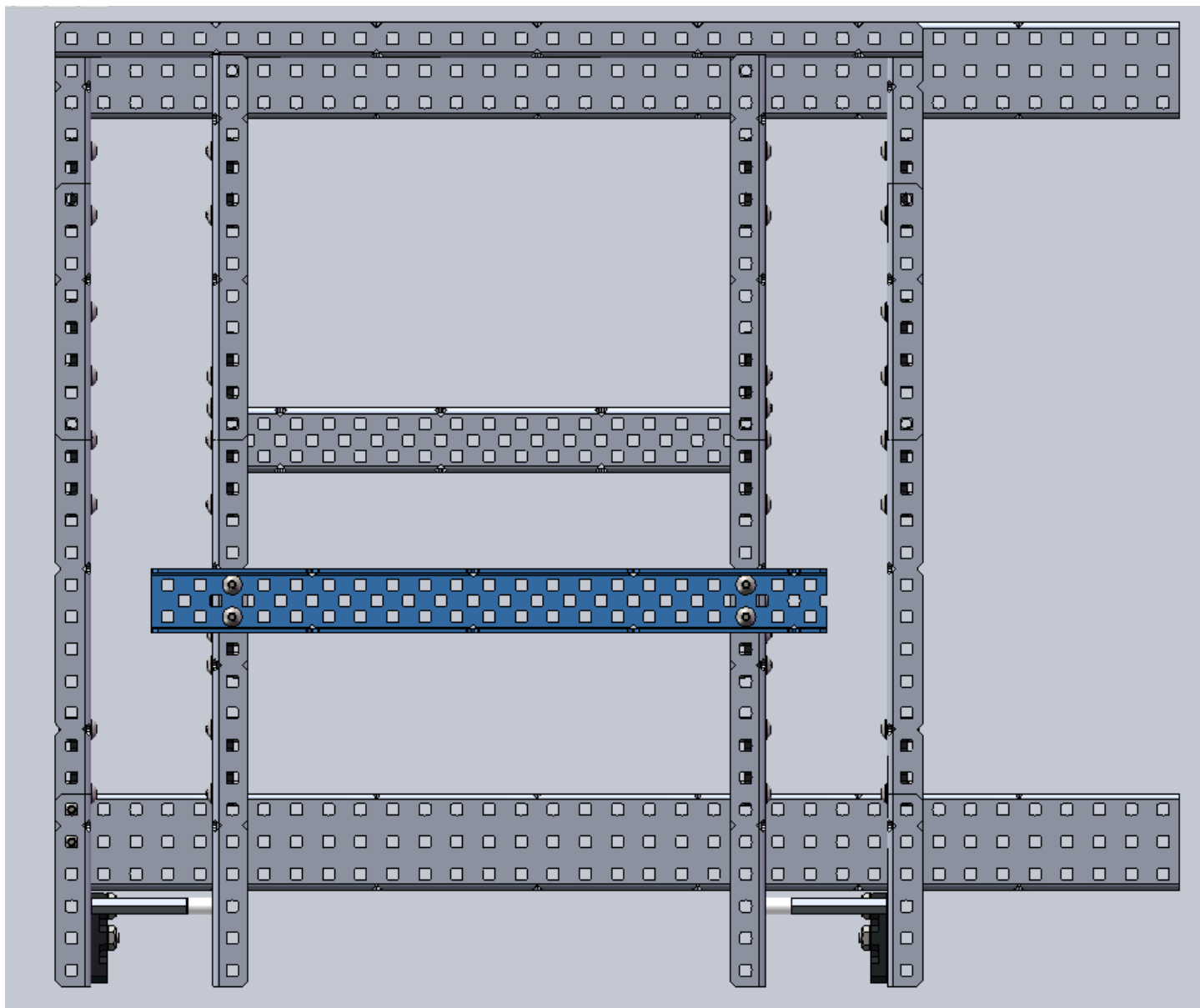




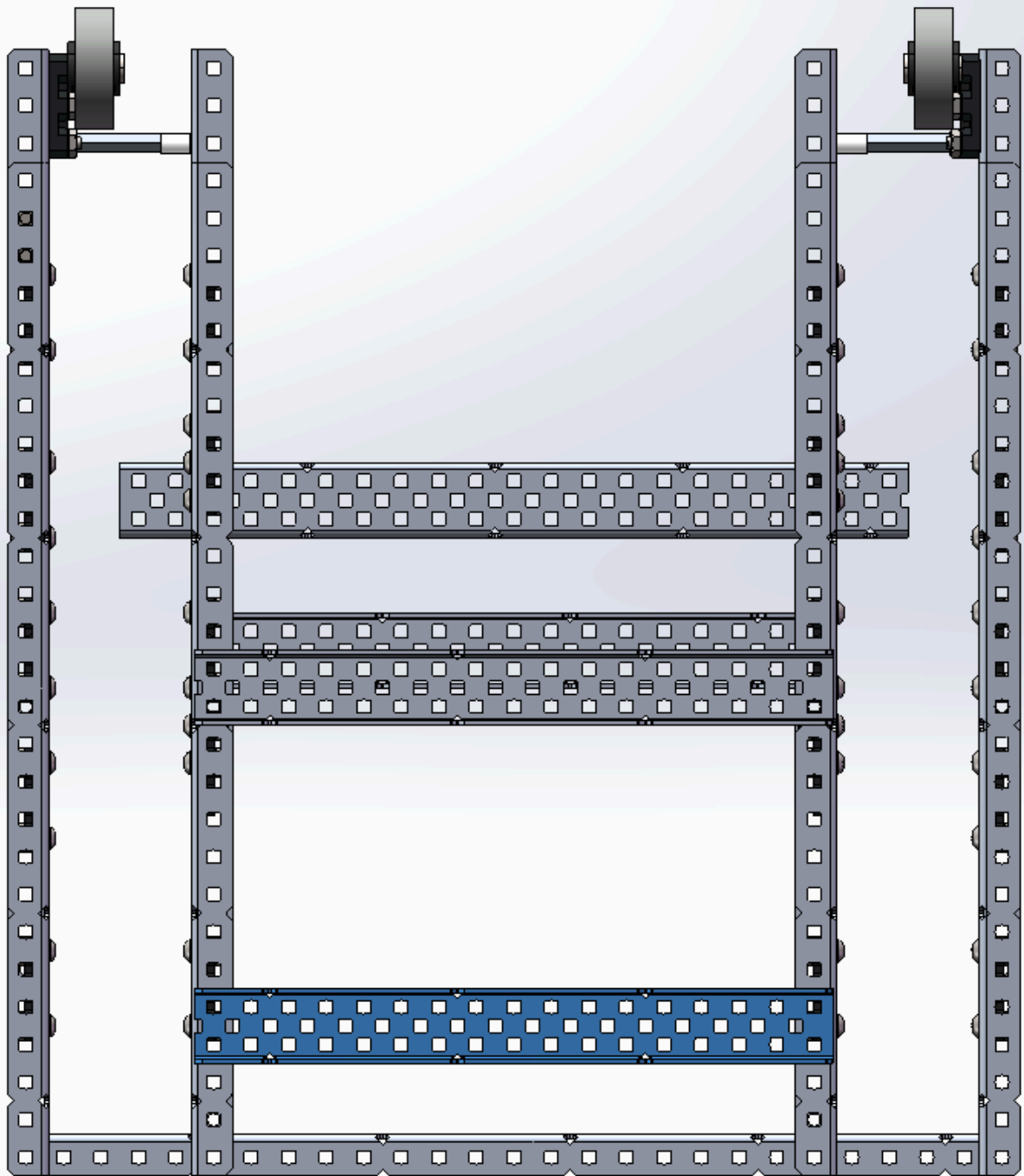
固定横梁3

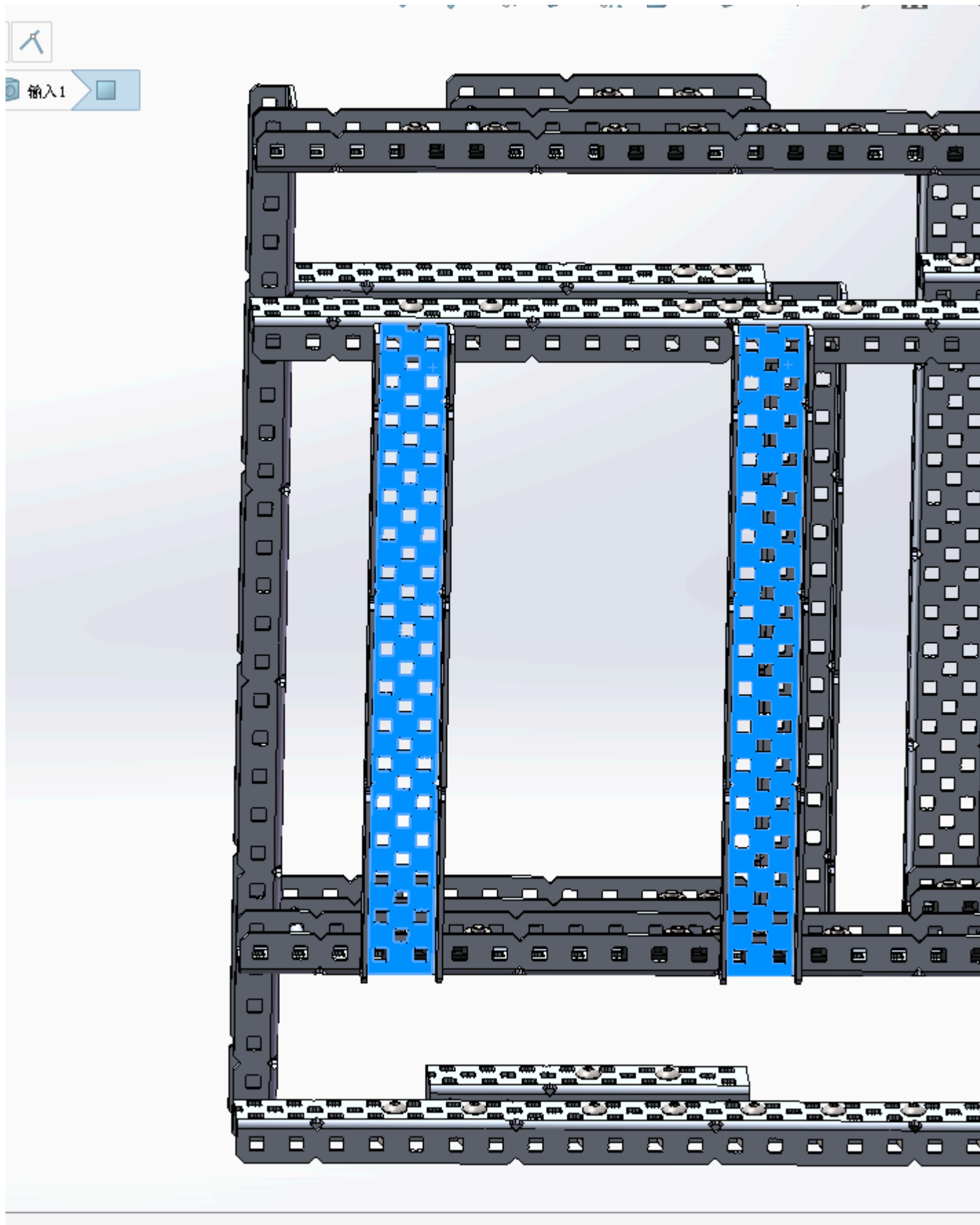
需要垫入8mm垫圈





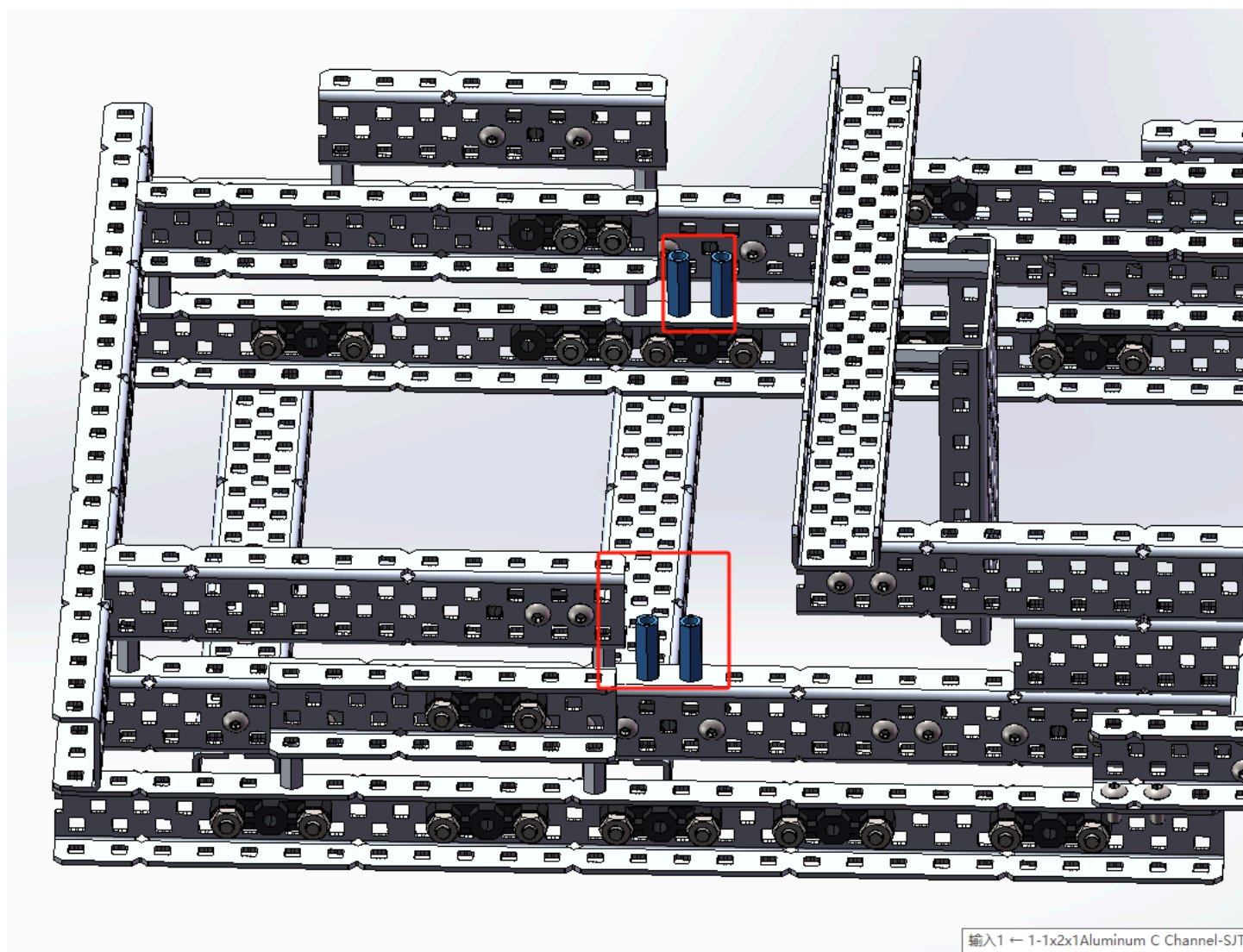
- 固定横梁4



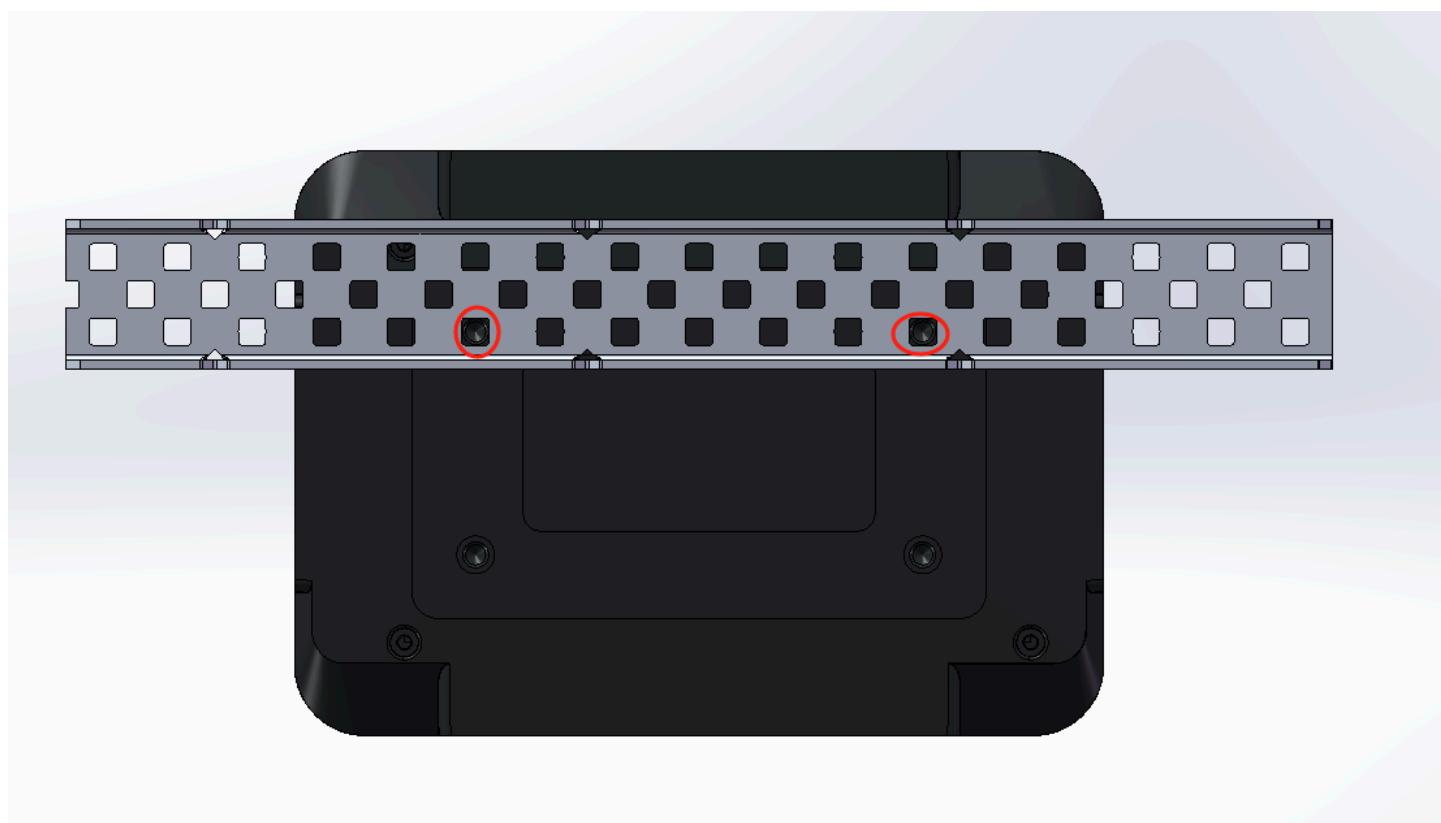


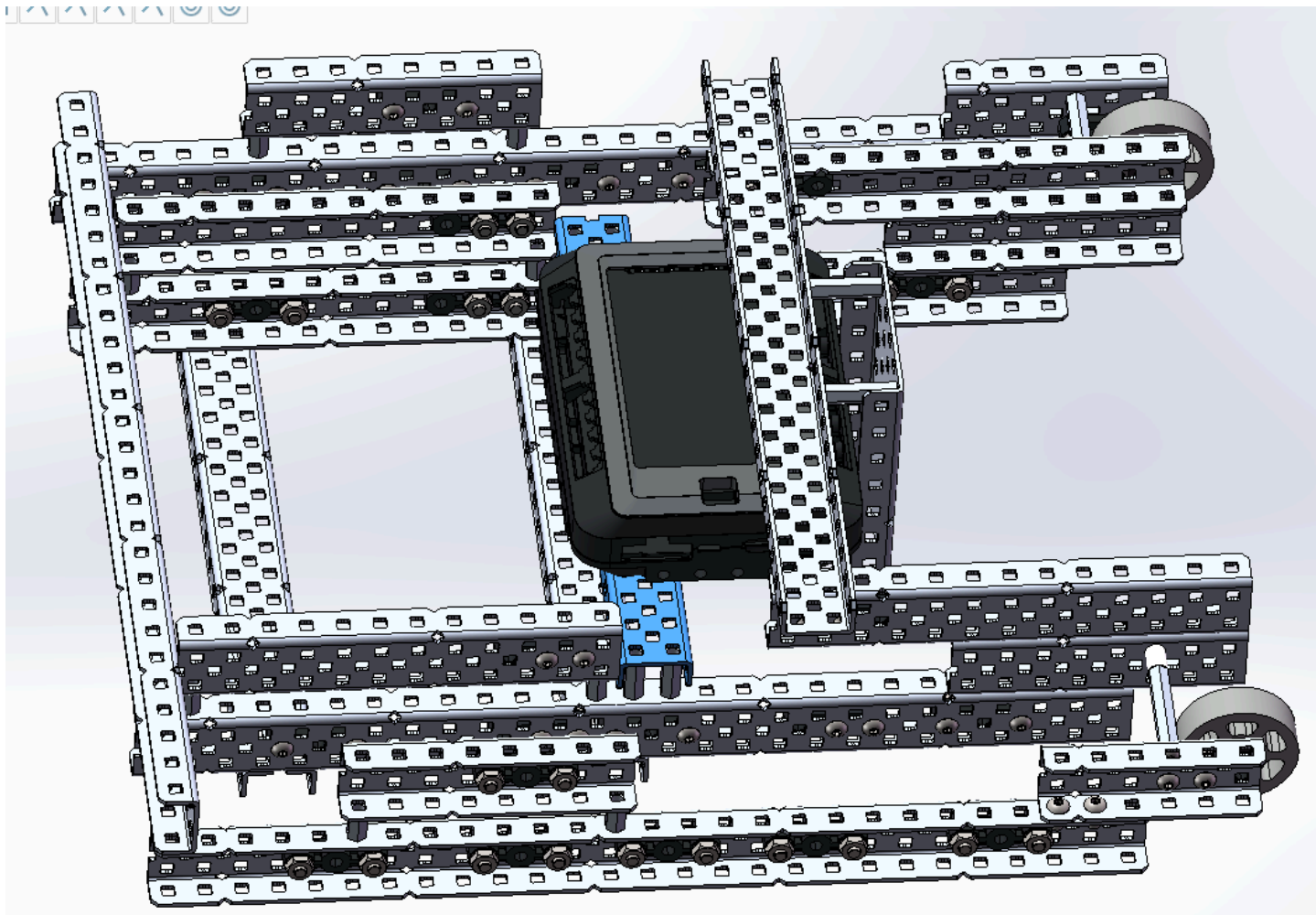
- 固定横梁5

安装0.75in螺母柱



将主控器安装到17孔窄C上面

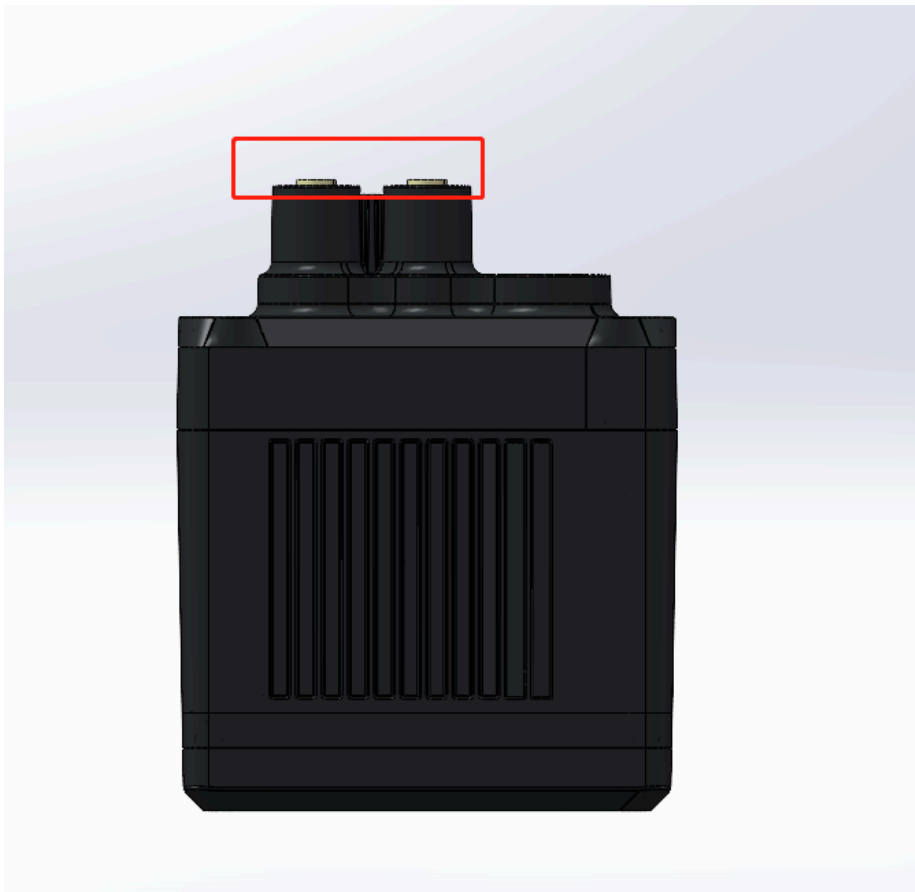




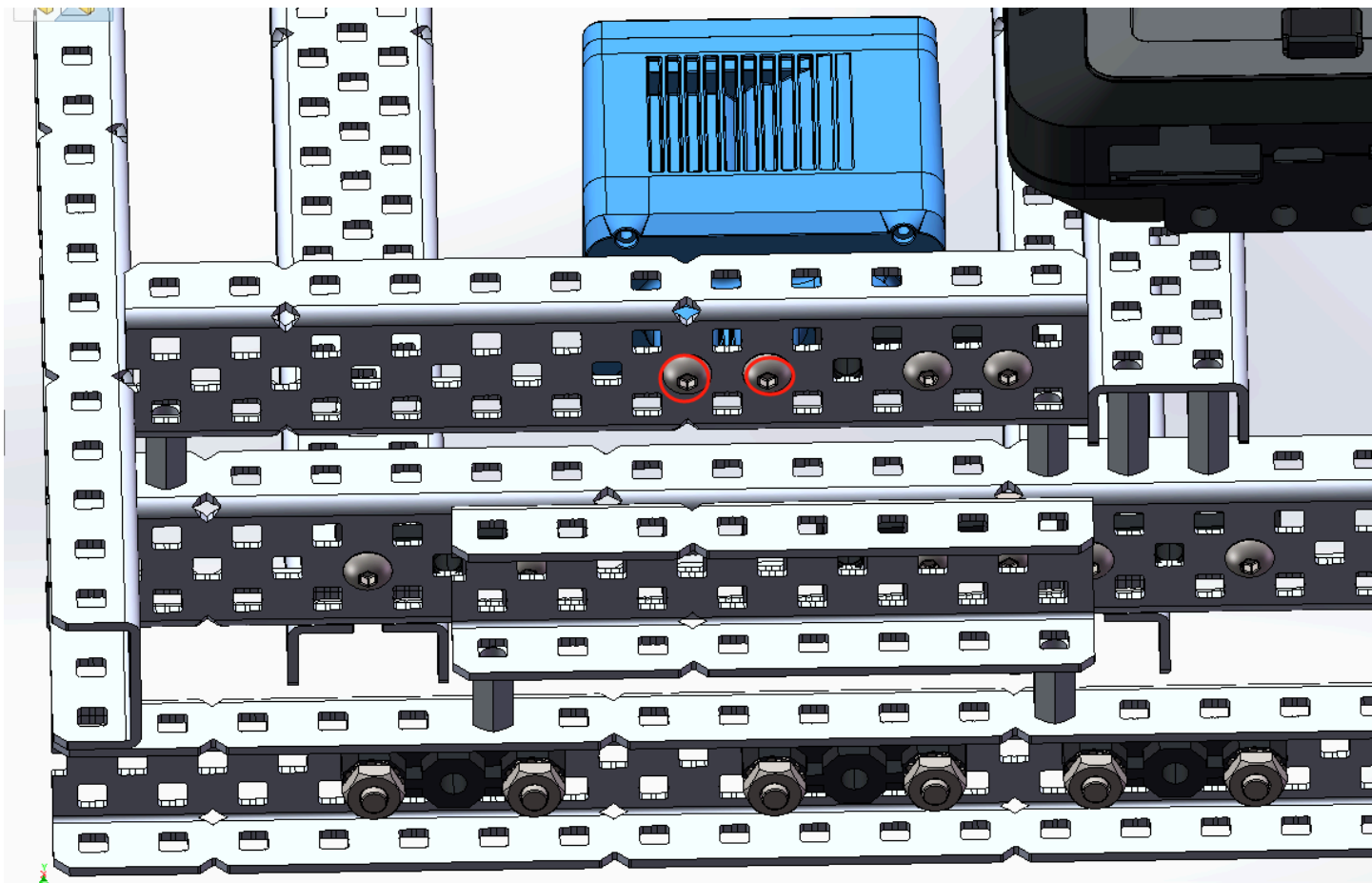
5.安装电机

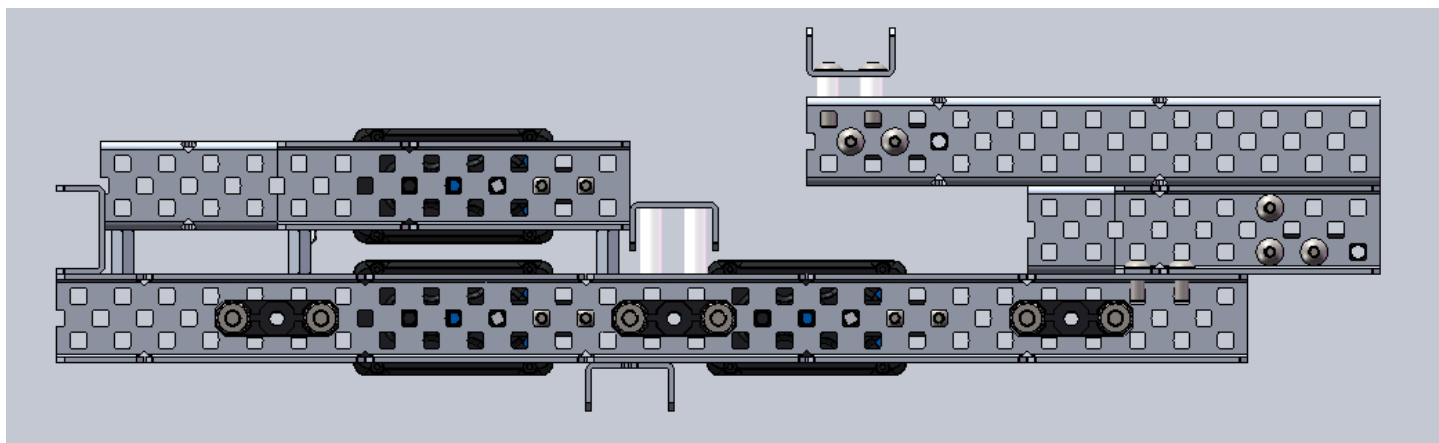
安装电机时一定要保证电机金属位置卡在铝件里面

建议使用0.5in螺丝进行固定

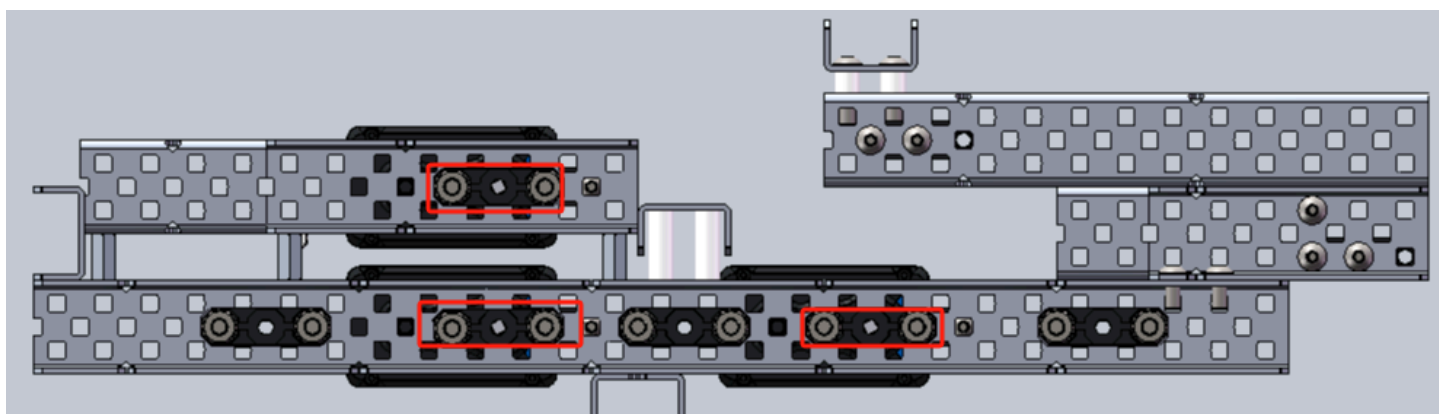


底盘一侧有3个电机，为L型排布，电机安装孔位如下：





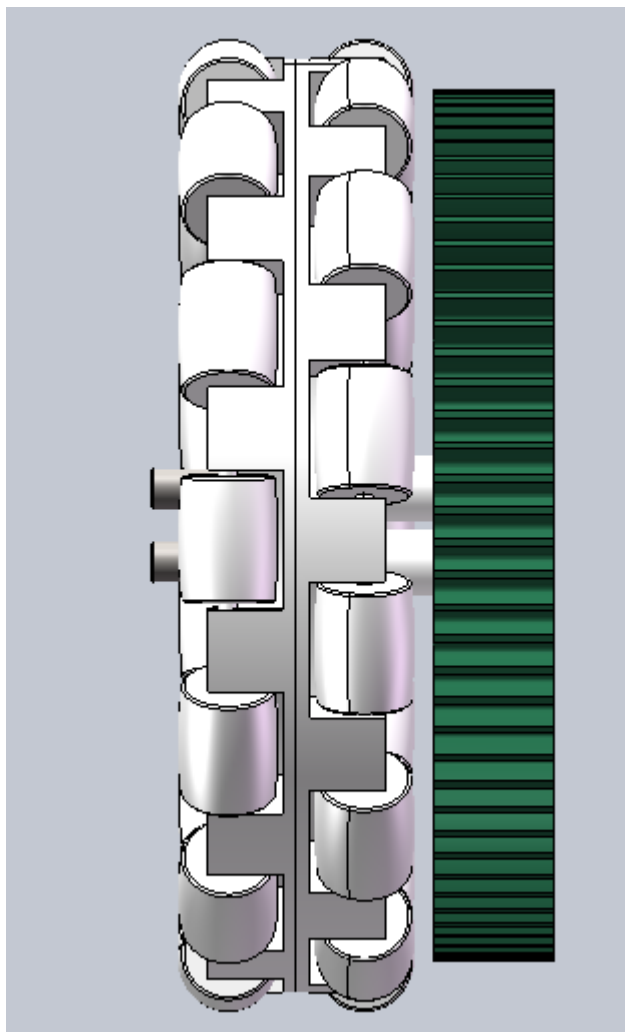
安装剩余轴承座

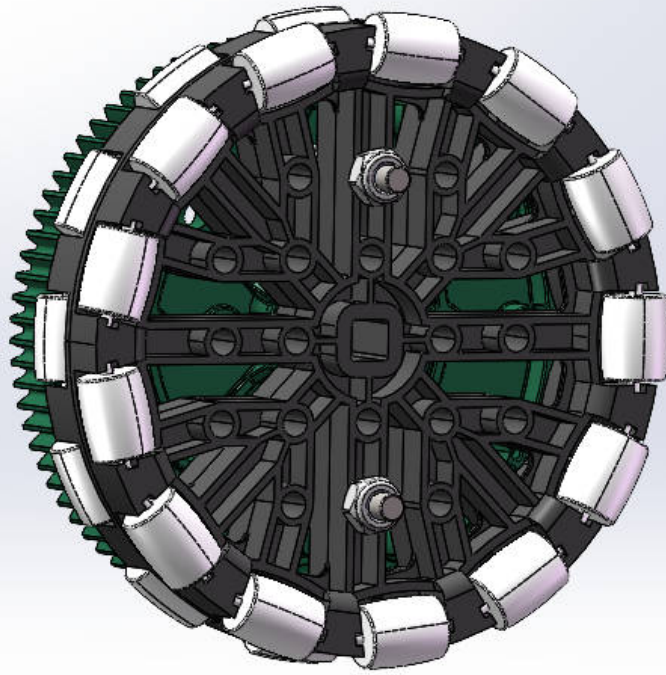


7.安装车轮装配体

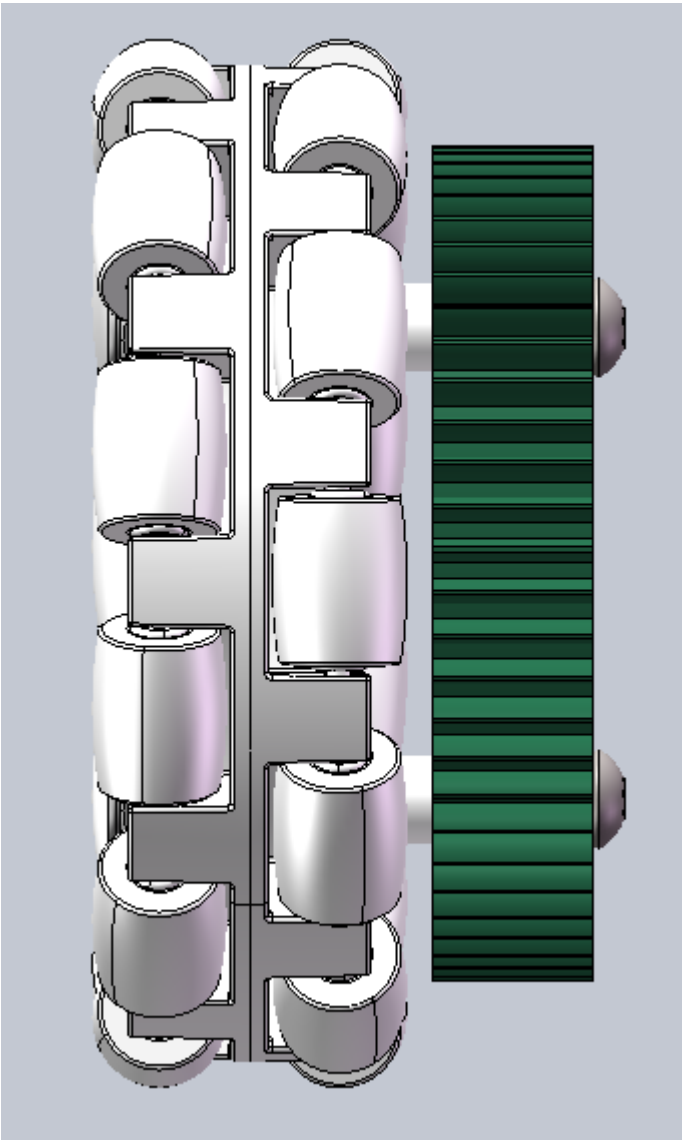
需要将4in万向轮 和 84t的齿轮合并在一起，制作4个

用1.5in螺丝将齿轮和轮子并在一起，中间需要垫入12mm垫圈





需要将3.25in万向轮 和 60t的齿轮合并在一起*2

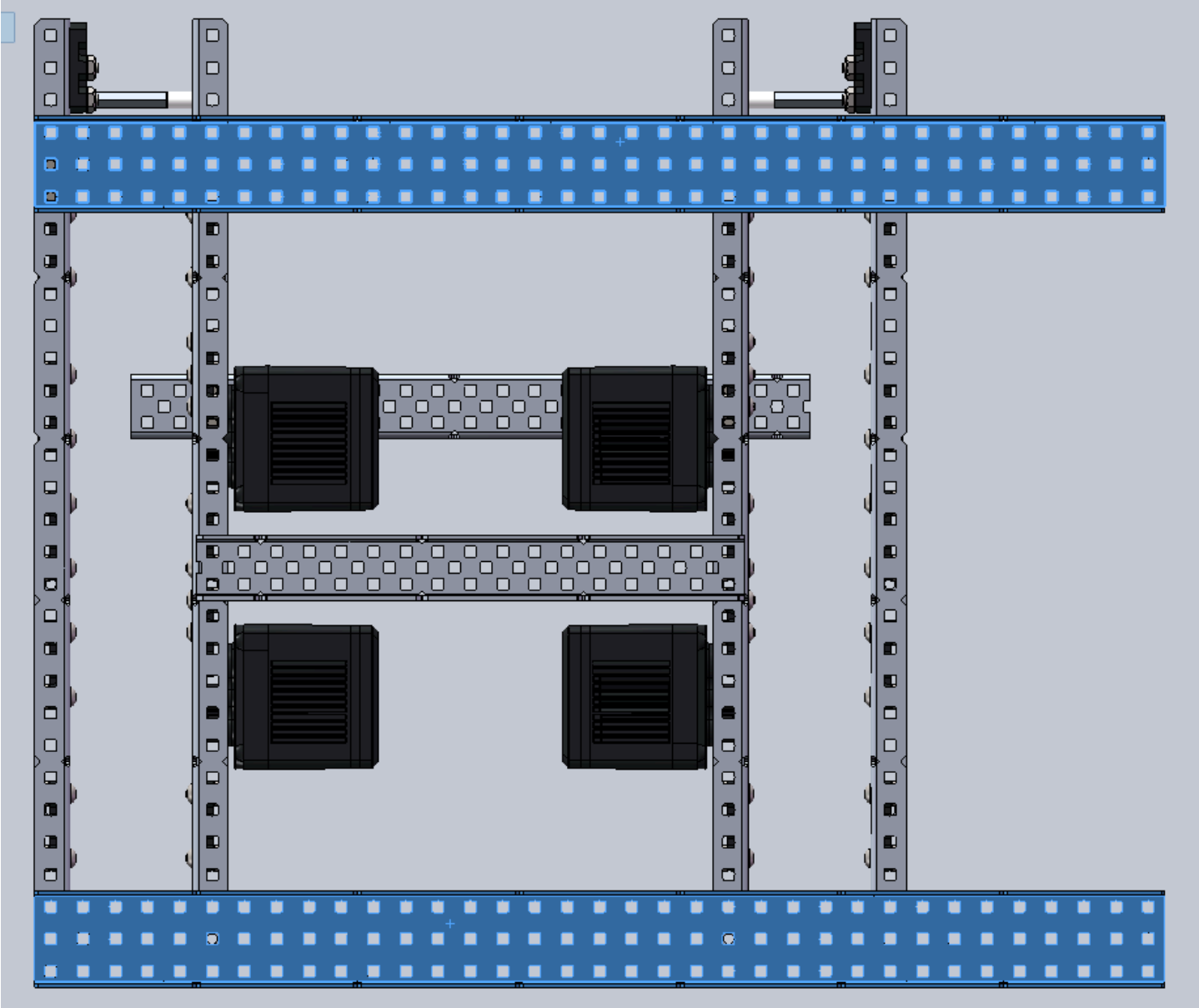


9.安装车轮

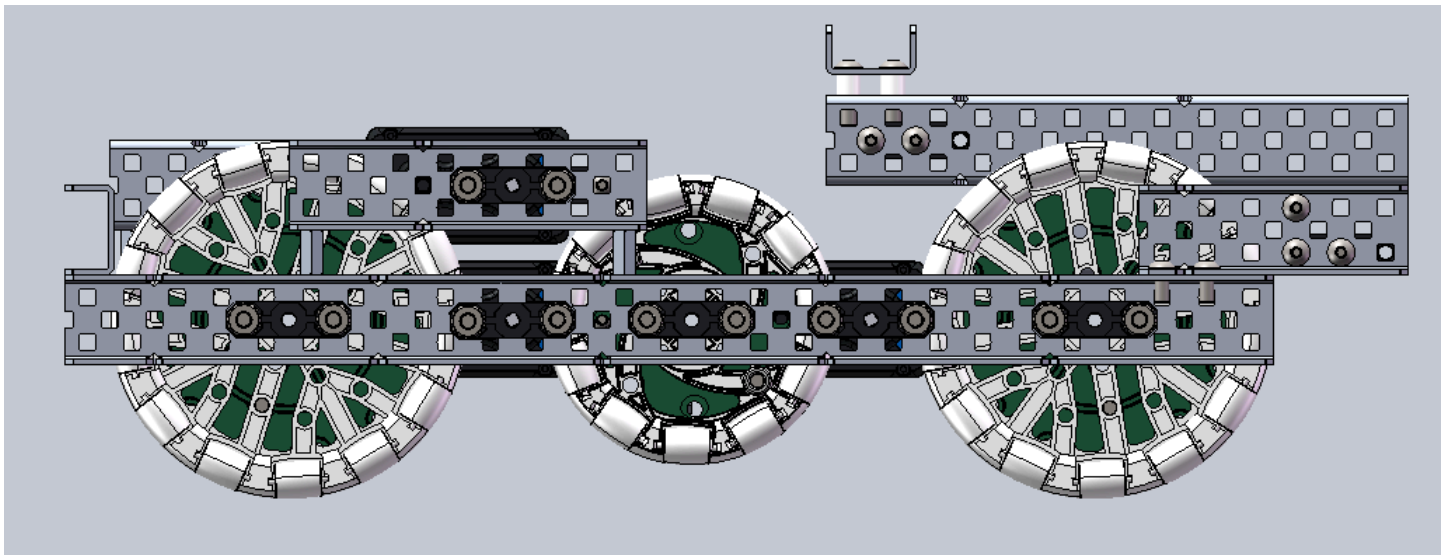
锯切轴长度

长度	数量
3.5in (88.9mm)	6

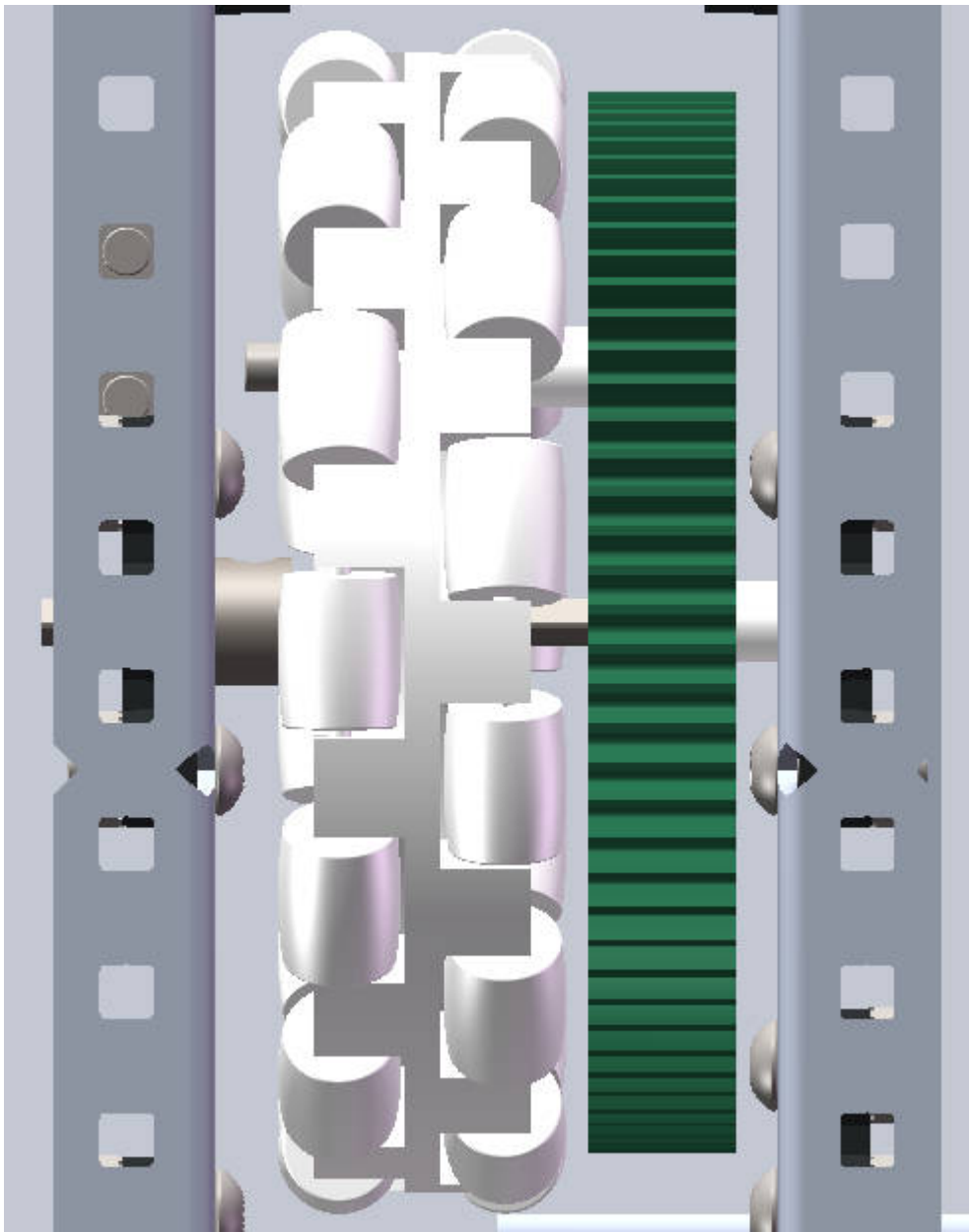
将之前安装的 固定横梁 和 固定螺母柱 拆除



4in万向轮放两侧 3.25in万向轮放中间



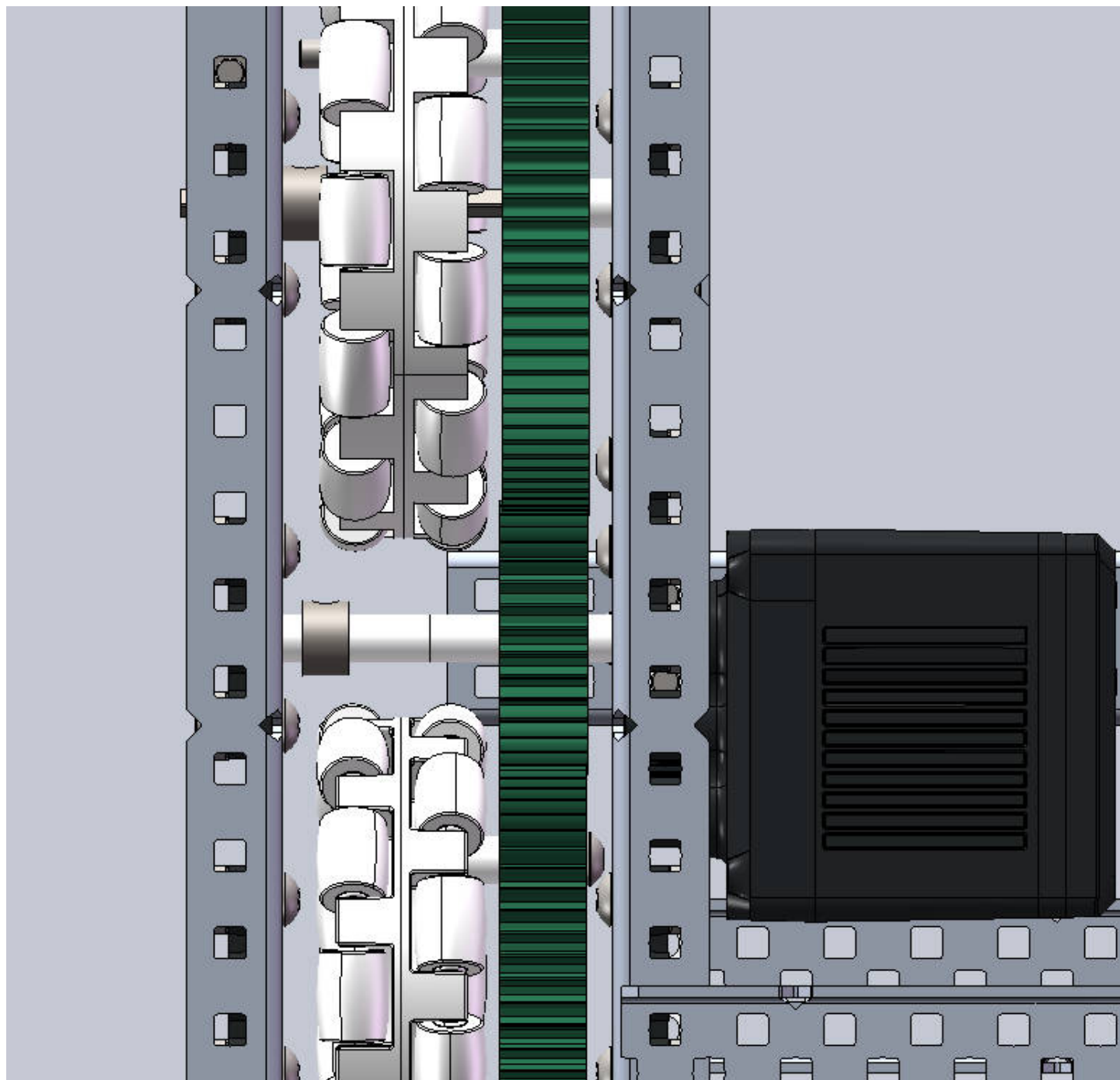
拿一根2.5in长的螺丝，穿过车轮，装入限位杯即可



安装完车轮后，安装36t齿轮

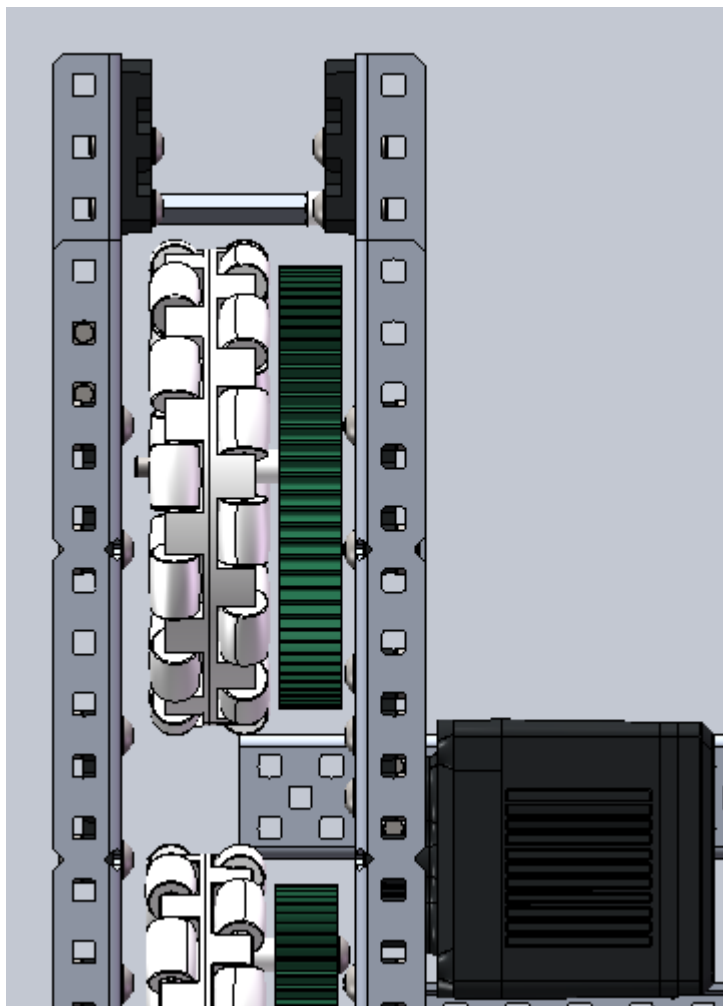
垫圈顺序（从右往左）：3mm垫圈 → 限位杯 → 12mm垫圈 → 8mm垫圈 → 36t齿轮 → 6mm垫圈

每一个电机都要装36t齿轮

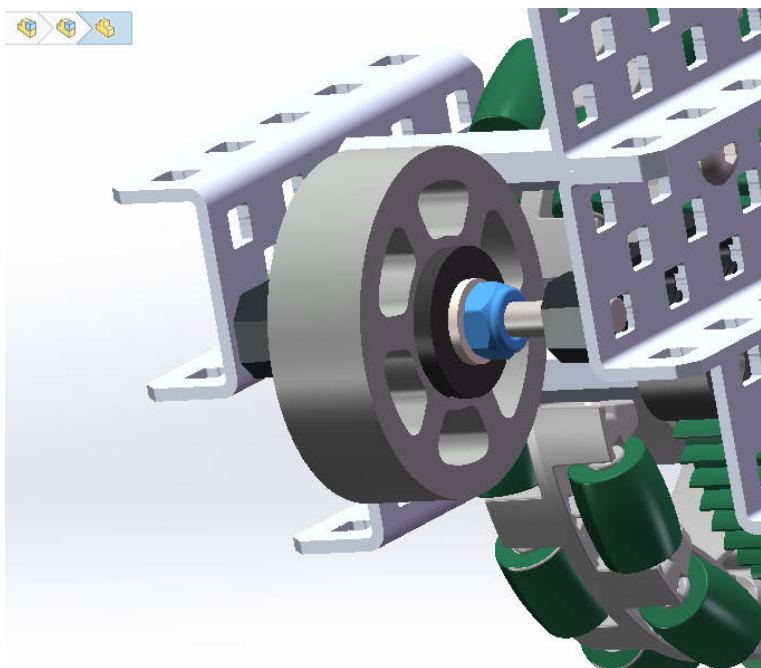
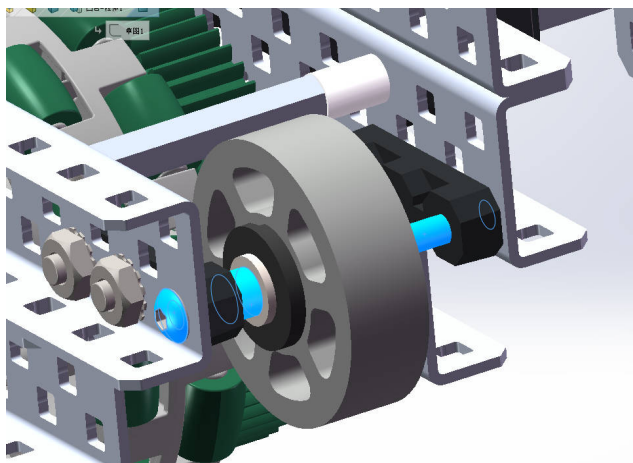


10. 安装滑轮

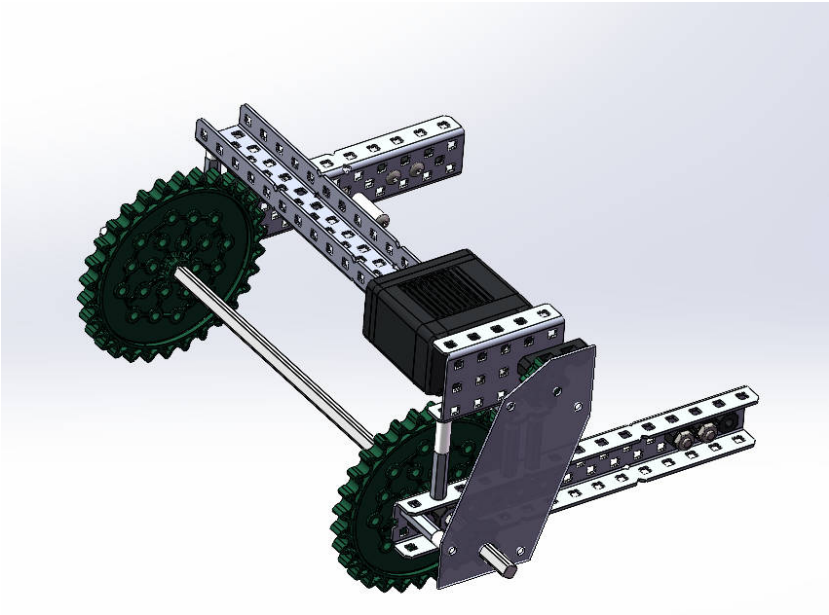
安装滑轮轴承座



- 拿一根2in长的螺丝穿过胶轮，并用防滑螺母固定（注意防滑螺母不能拧太死，需要保证胶轮可以正常旋转和滑动）



02 吸取结构



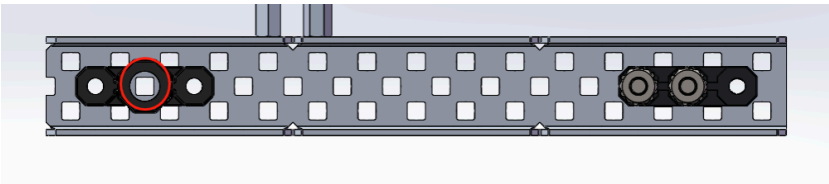
铝件长度清单

铝件名称	数量
窄C 15	2
窄C 20	1
中C 5	1

1.轴承座安装

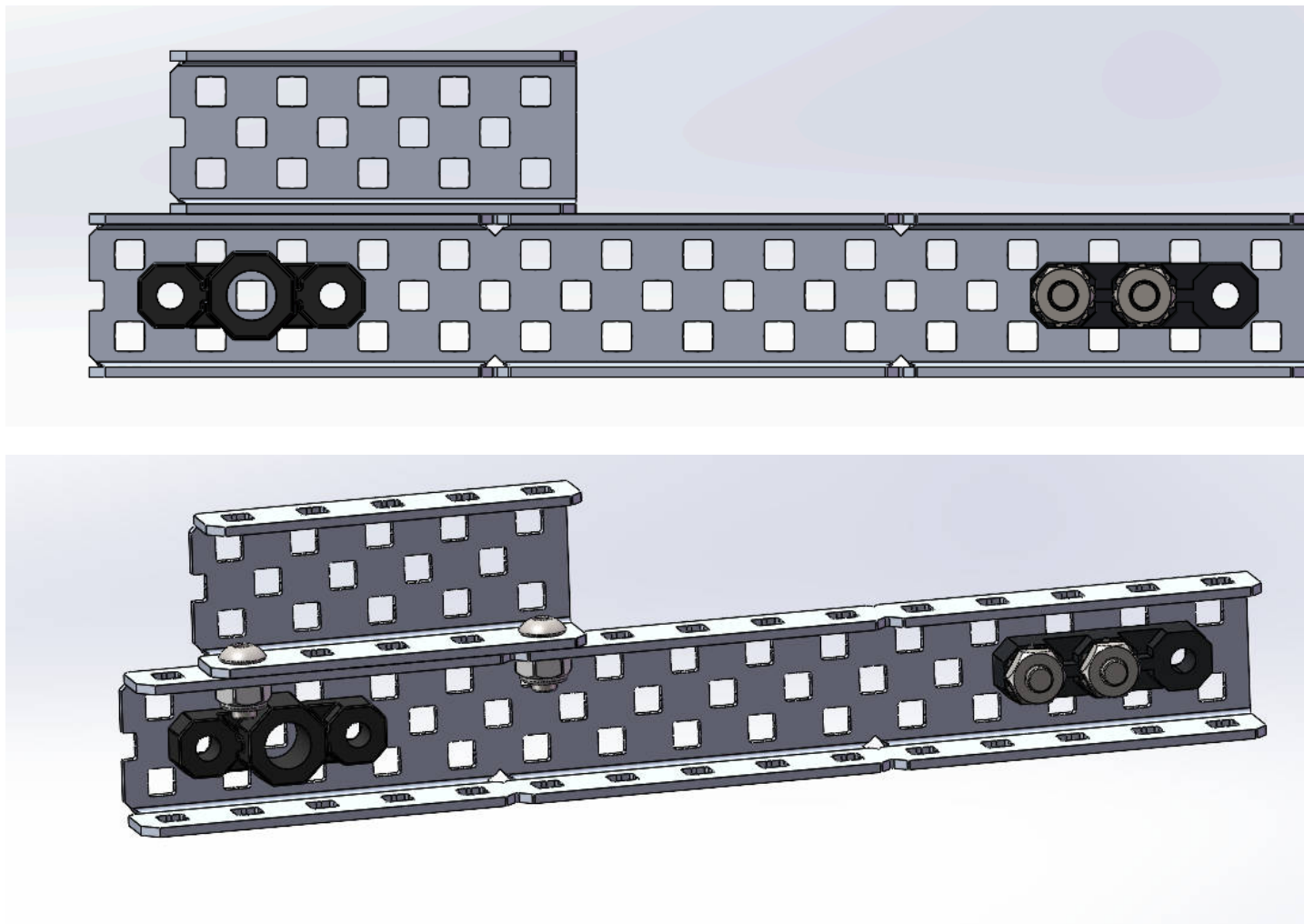
需要安装2个（一样）

先给标红的地方钻孔（扩孔的直径大小需要和粗轴的直径相同）→ 安装粗轴和普通轴承座



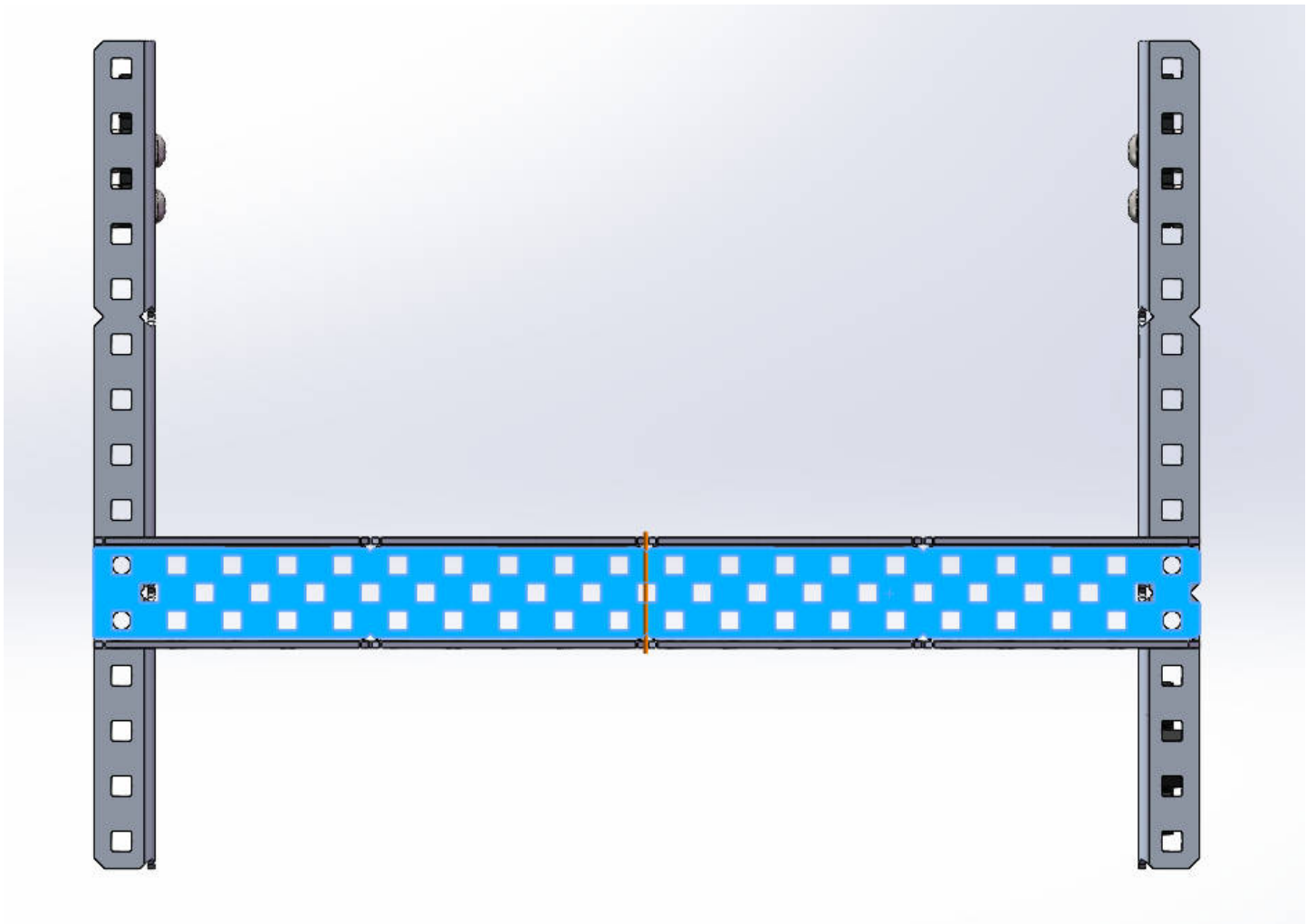
2.撑柱安装

安装1个5孔窄C

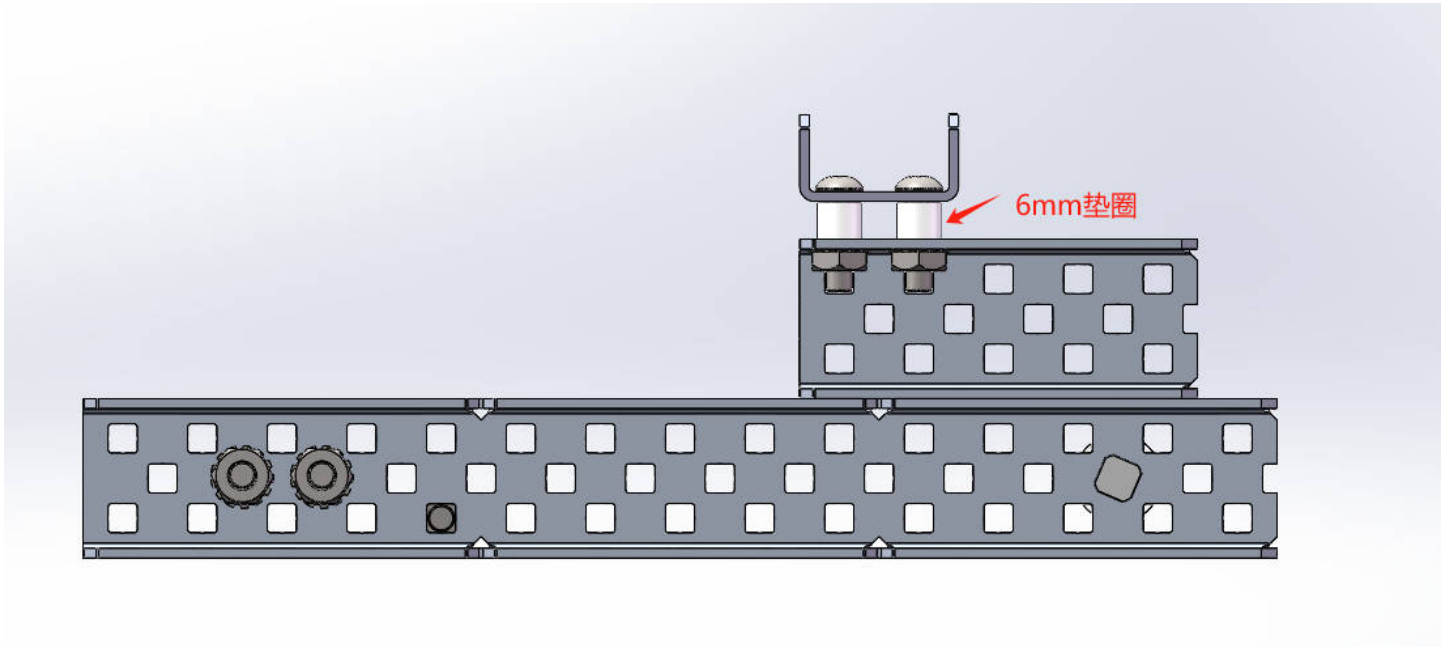


3.整体组装

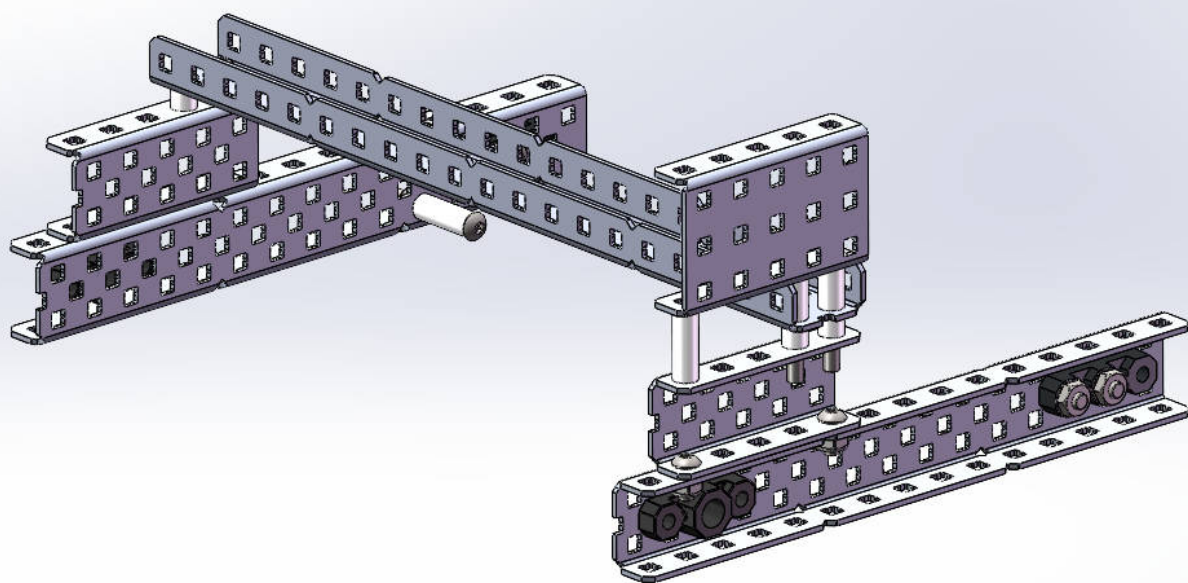
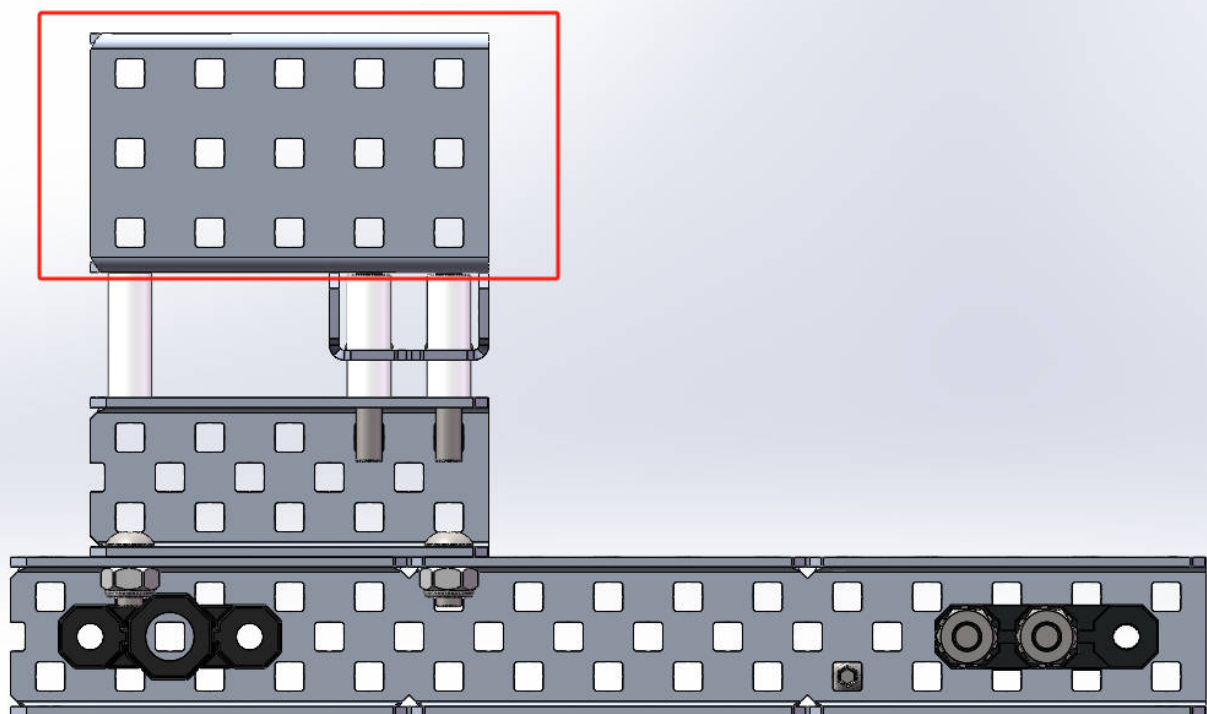
将20孔长的铝件安装在螺母柱上，中间需要垫入6mm垫圈



侧视图效果



另外一侧将5孔中C 安装在intake上面，总共3个点



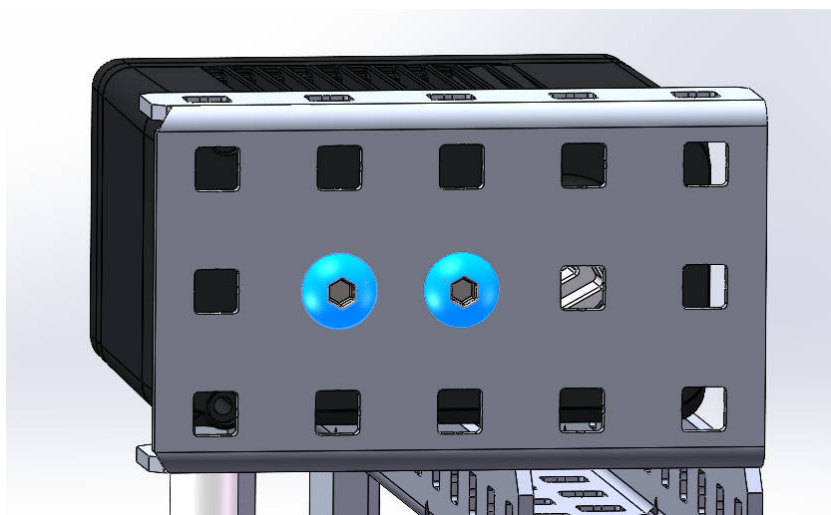
4.电机安装

先将马达齿轮箱换成 蓝色

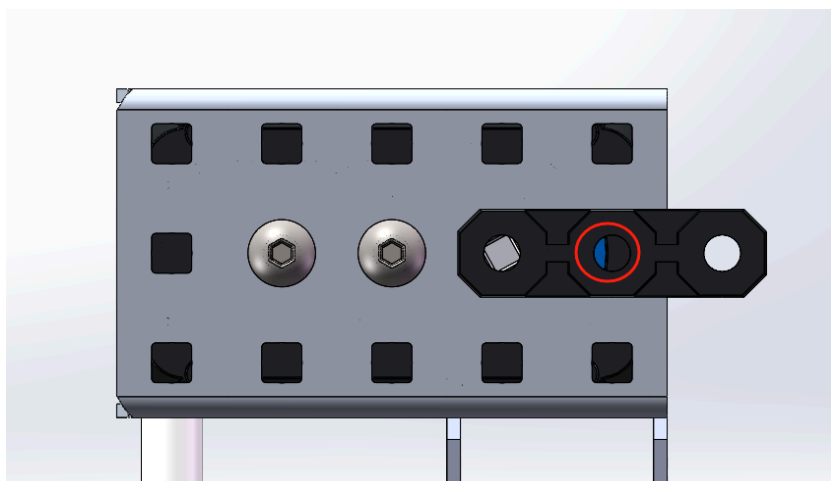
准备2根 3in 长的方孔轴，先将其插入马达中，安装垫圈和限位杯，并将限位杯锁死



按照下图位置所示安装电机

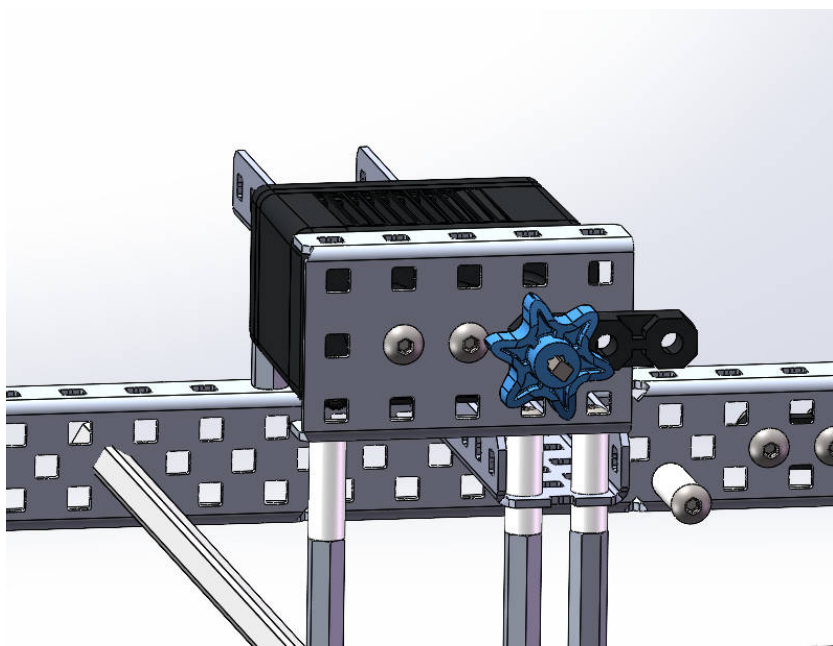


随后安装轴承座，轴承座只需要固定一个孔位即可



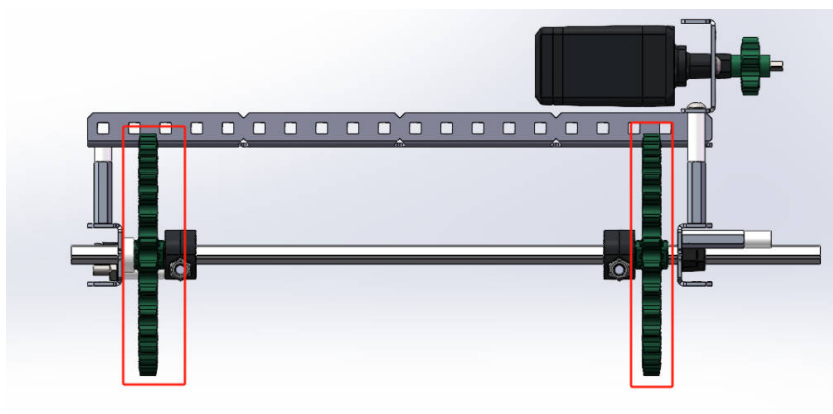
6.放置链轮

放置第一个6T链轮

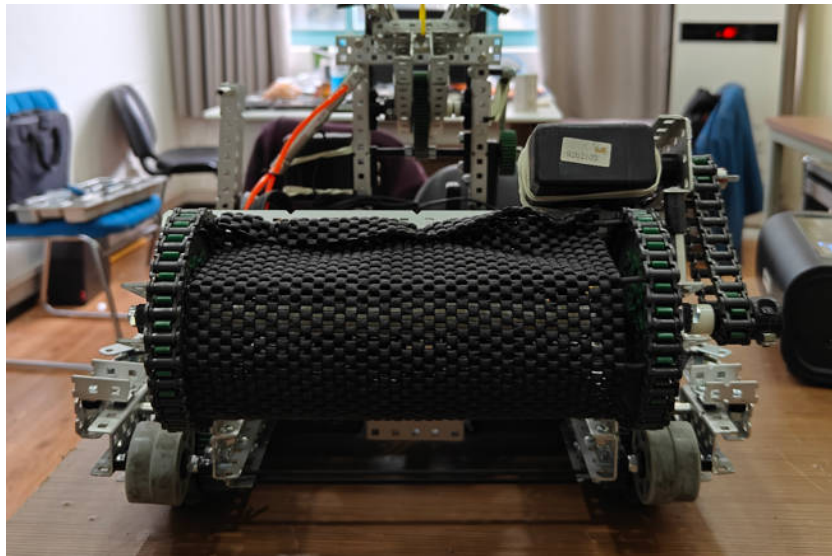
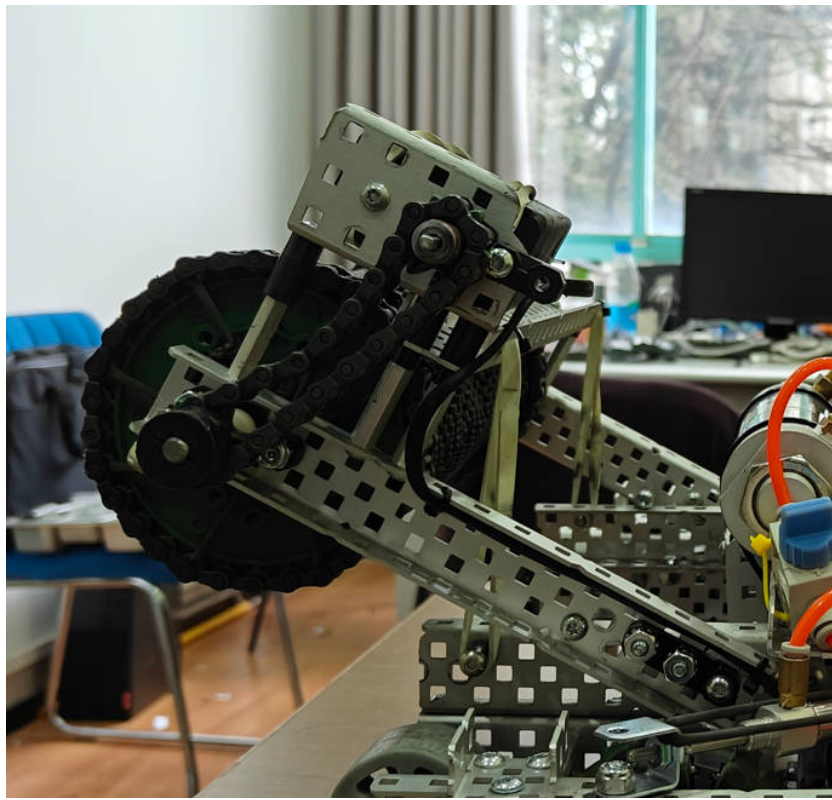


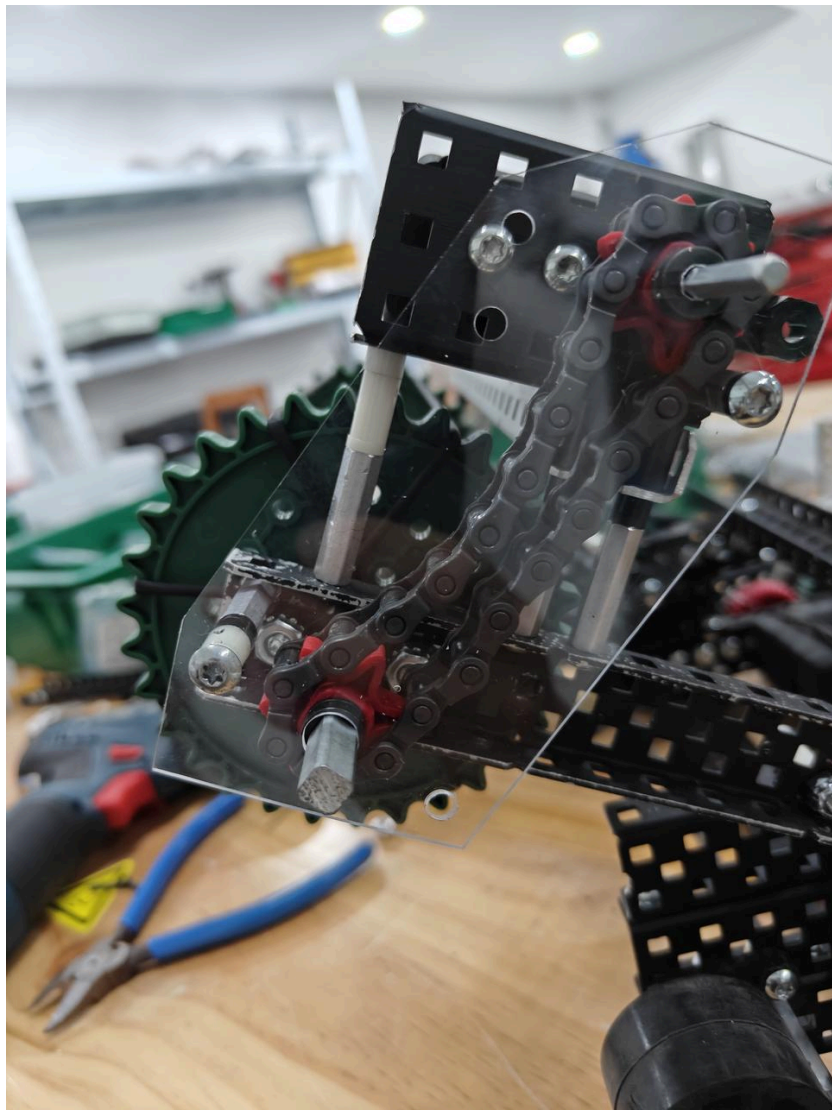
随后用粗轴塞入2个30T链轮和粗轴限位杯

中间位置需要用垫圈垫满

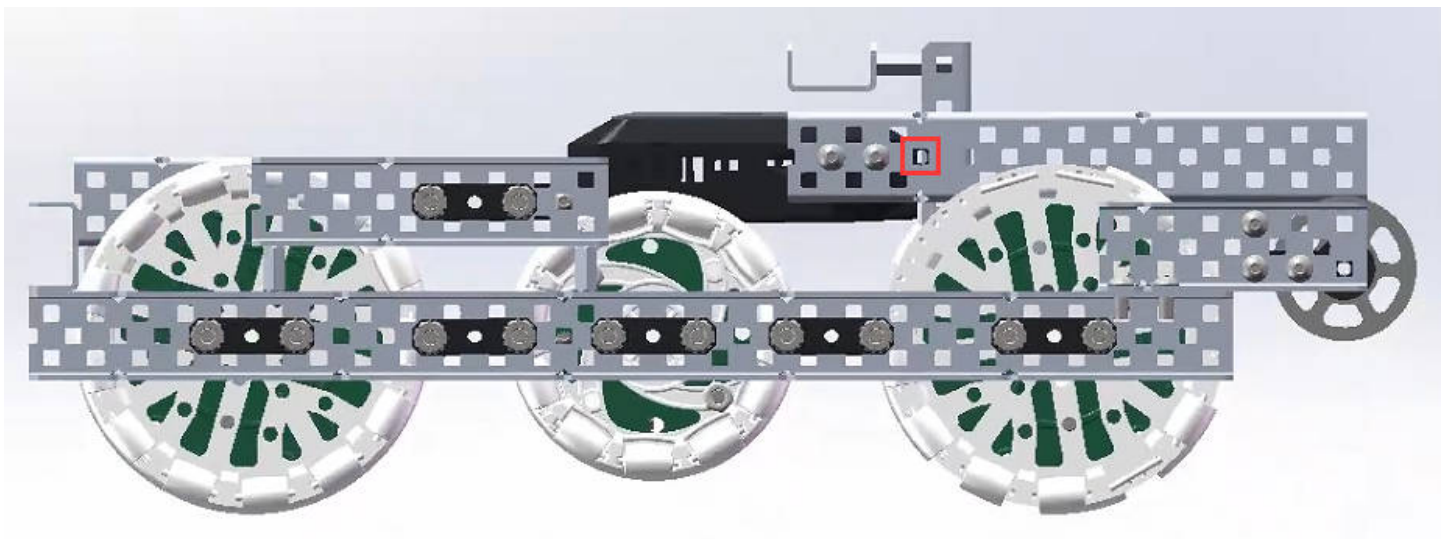


7.成品图

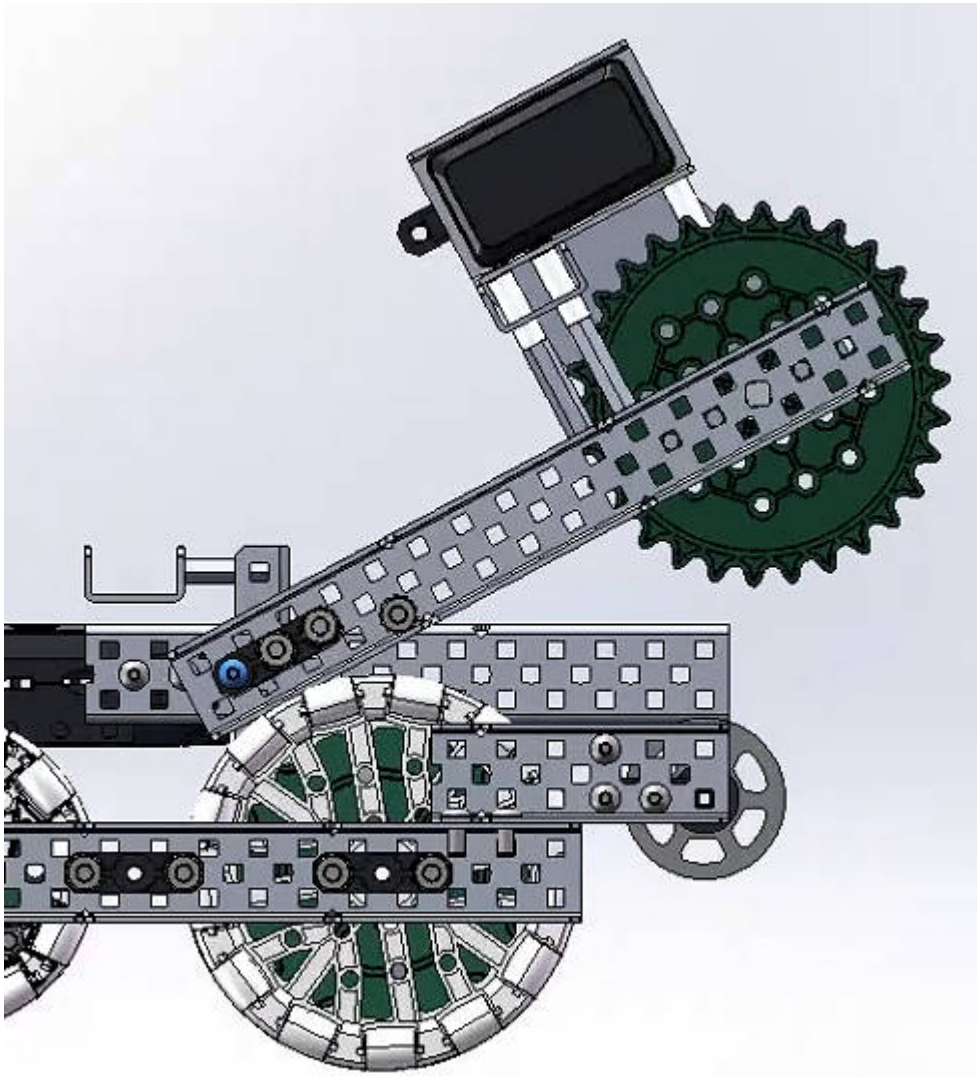


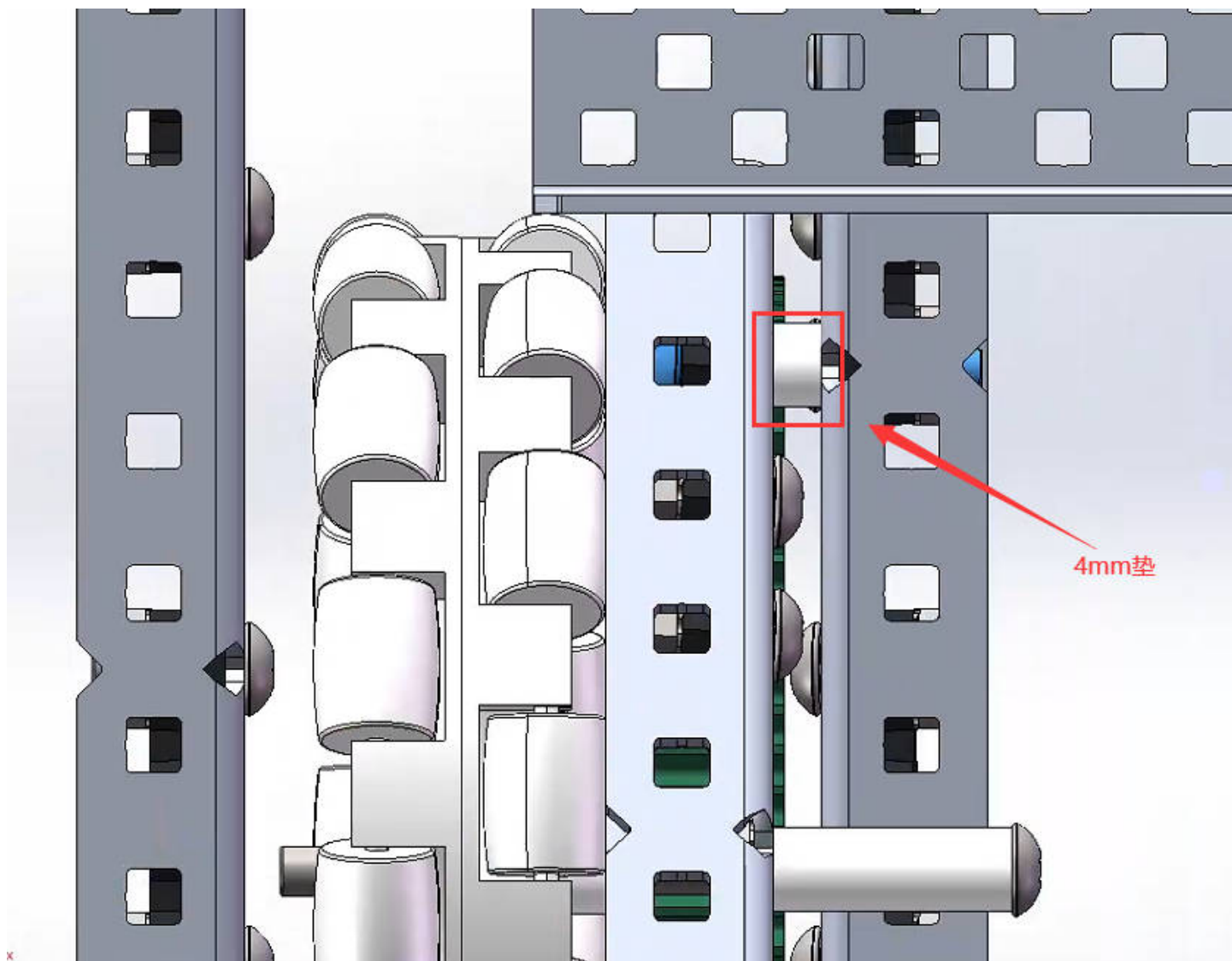


8. 安装到地盘

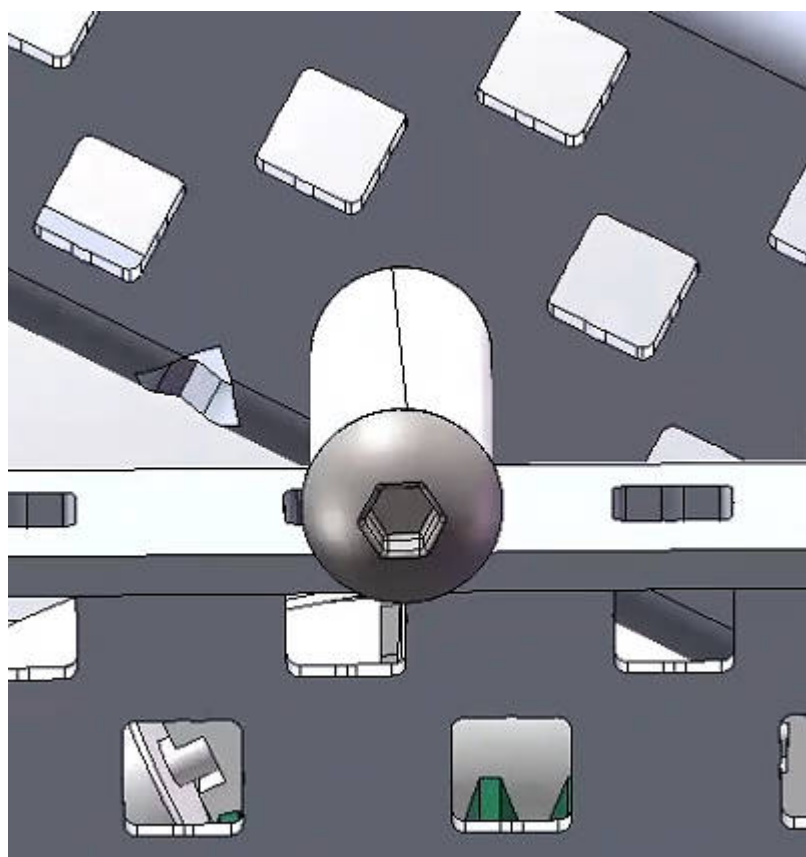


用1in长的螺丝加防滑螺母进行固定

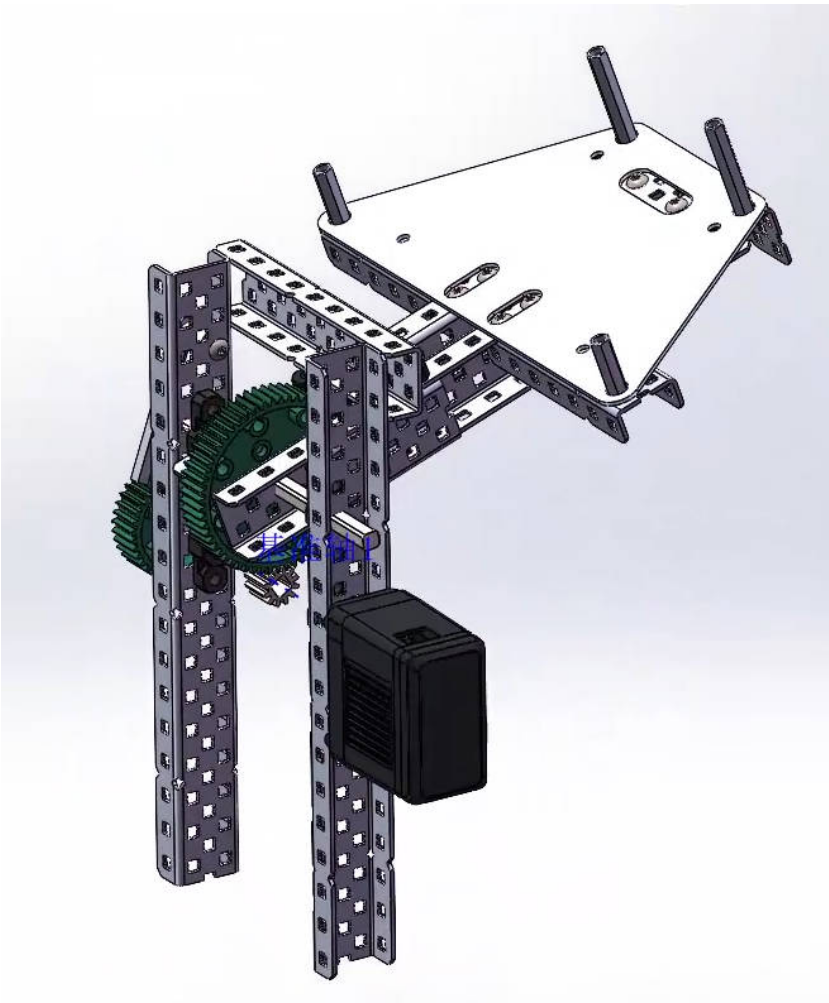




用1.5in螺、20mm垫圈加防滑螺母限位



03 发射结构2

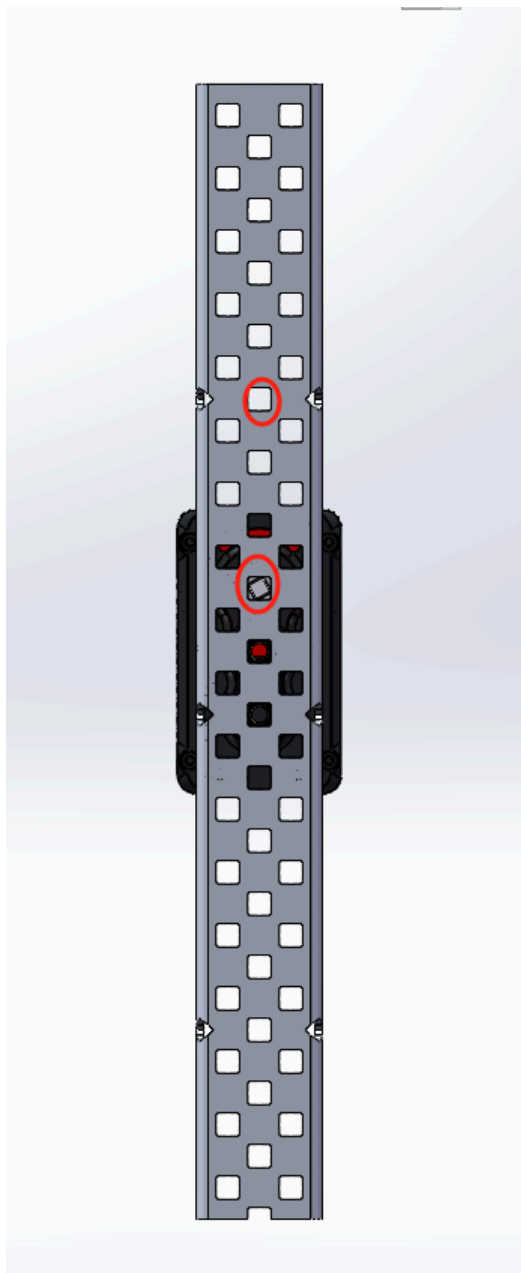


材料清单

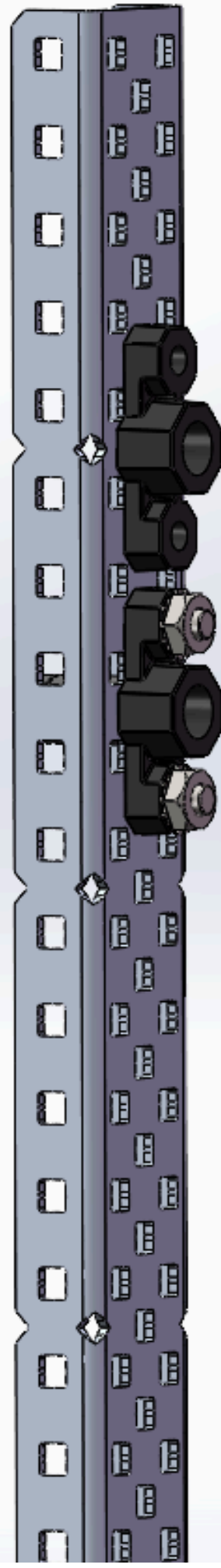
铝件名称	数量
窄C 18	2
窄C 9	1
窄C 4	2
窄C 7	2
窄C 17	2
2*2 L 7	1

1.安装轴承座

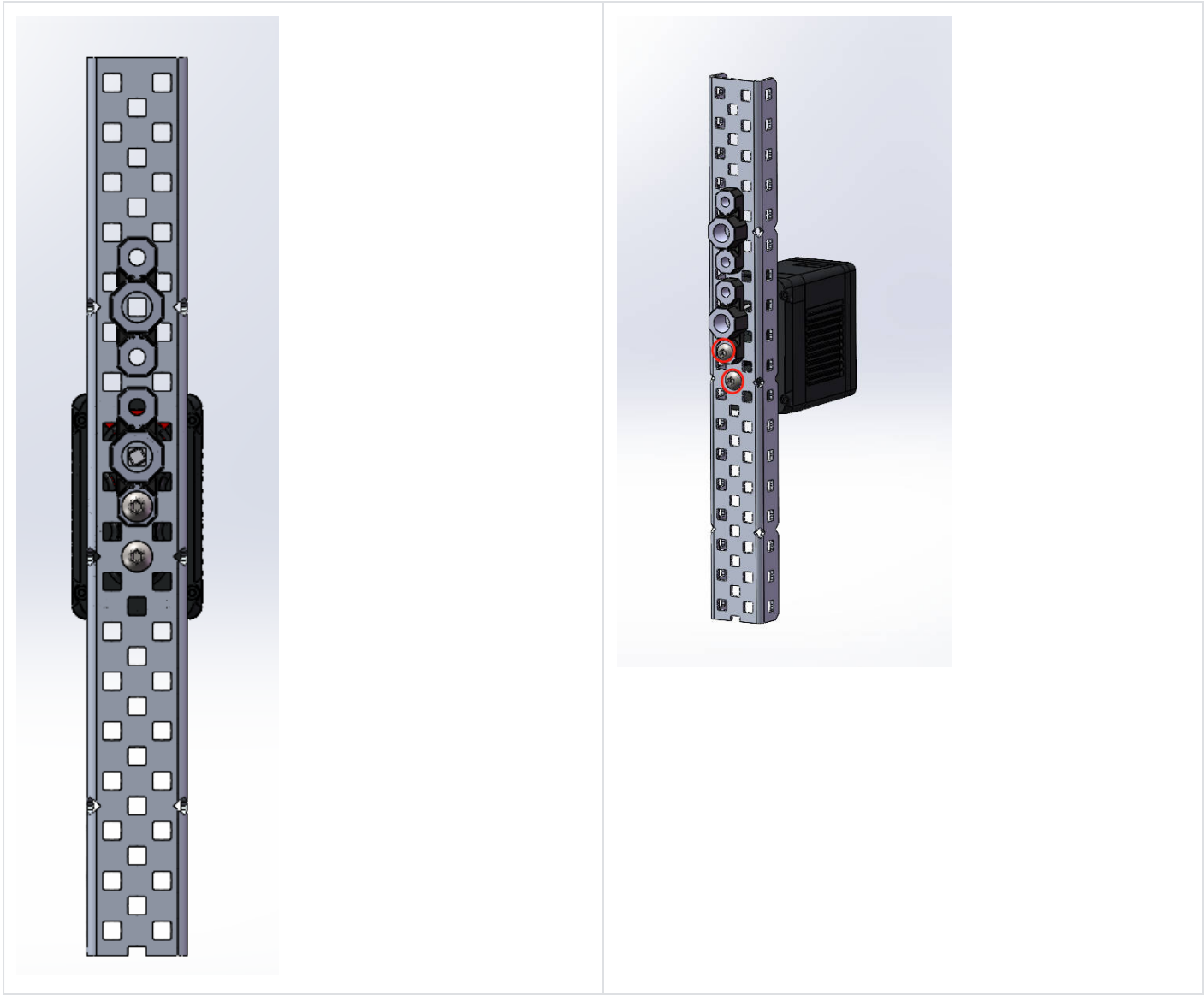
对图中标位置进行打孔



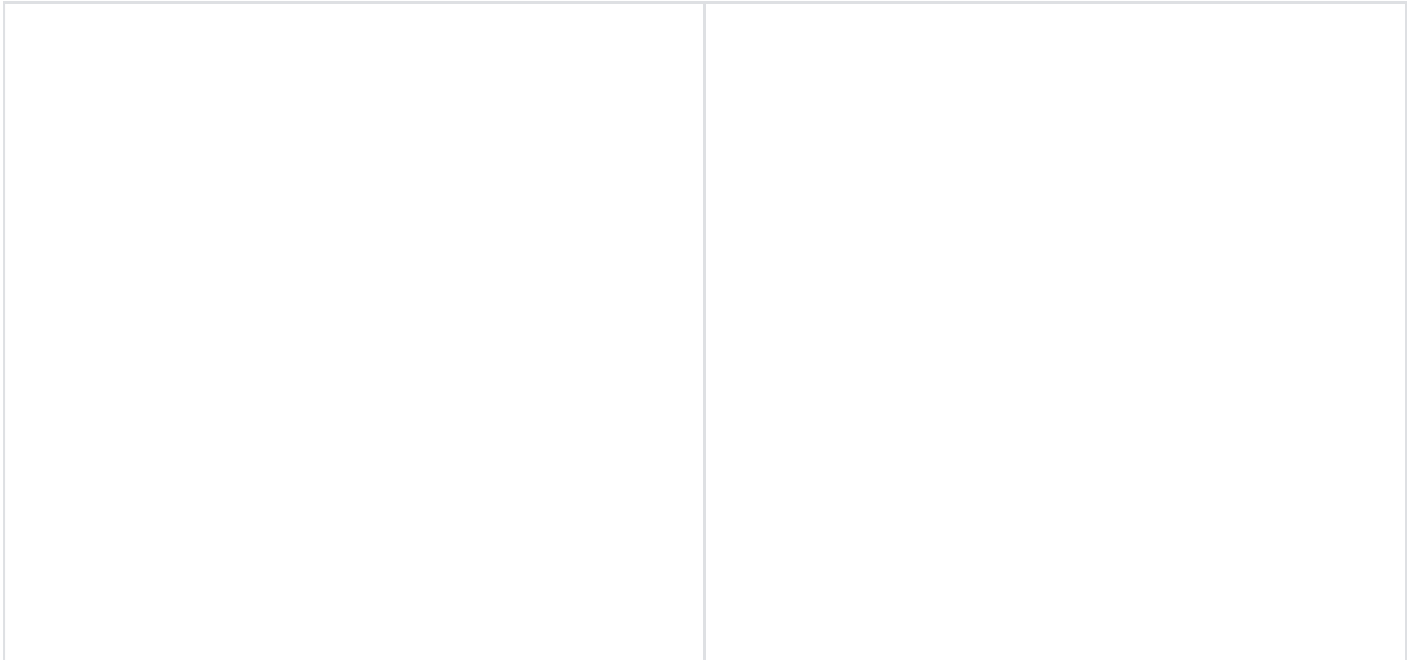
按照如下位置安装高强度轴承座，螺丝和螺母的朝向需要按照这个制作

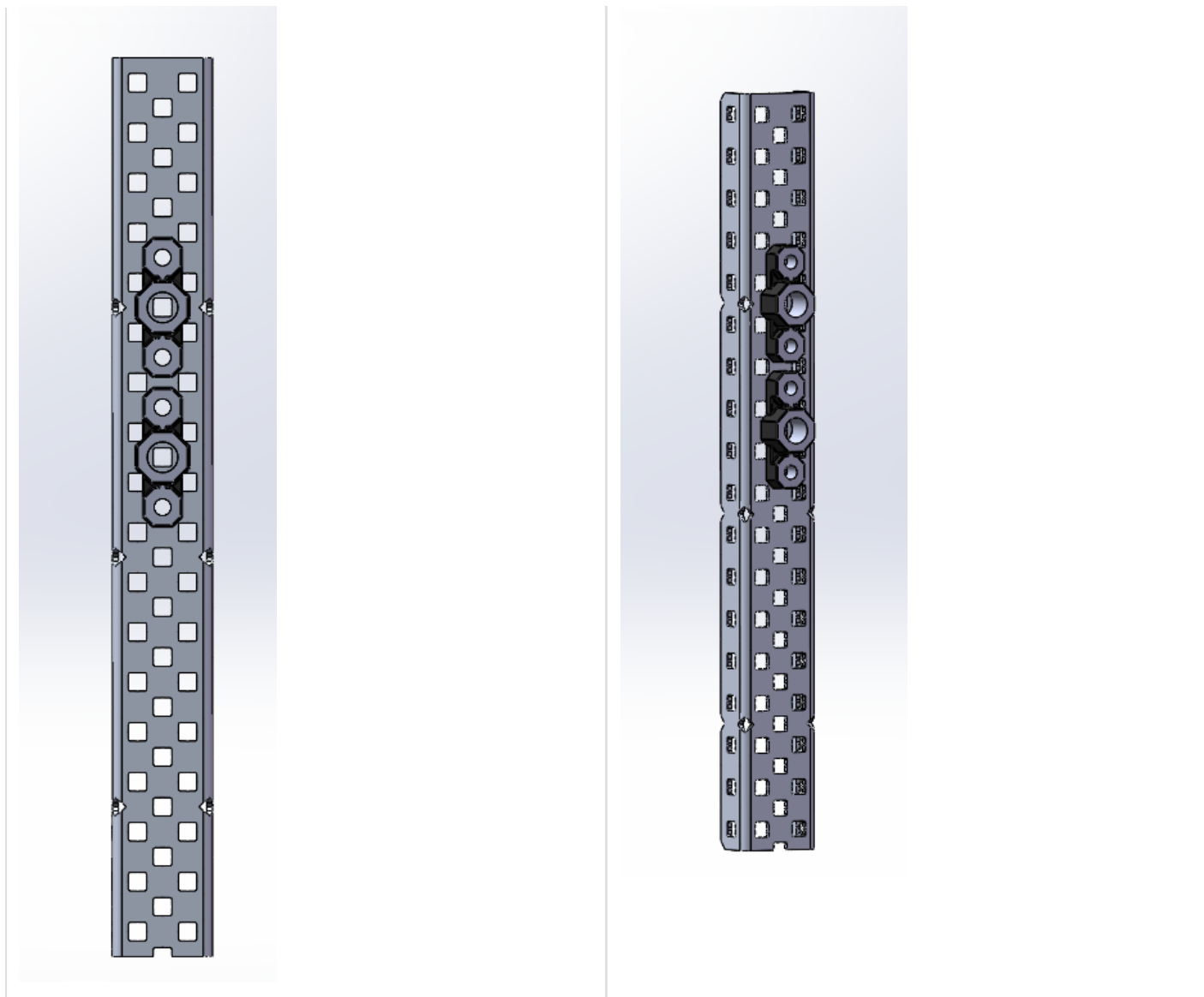


注意：第二个高强度轴承座需要和电机安装在一起



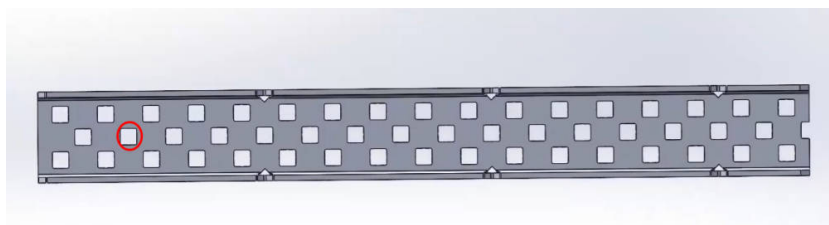
还需再安装一个装配体，只是少一个电机（需要打孔，打孔位置和上面相同）



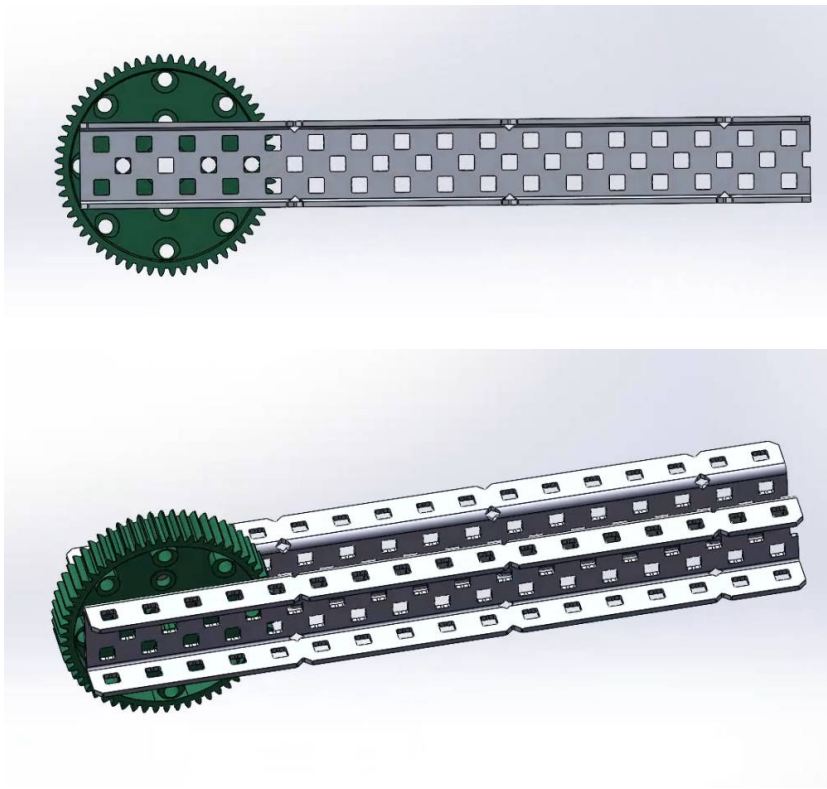


2.抛投板制作

先按照下面的位置钻粗孔（加工2个）



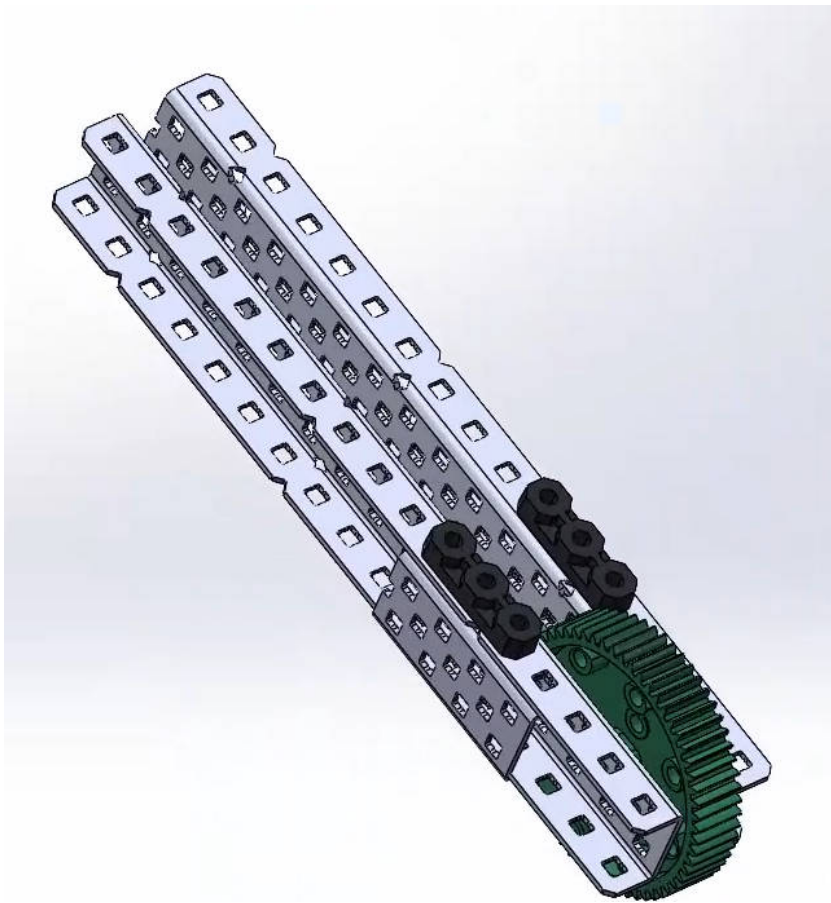
按照如下所示将2个17孔窄C与60T齿轮合并在一起（60t齿轮可能会有空隙，需要垫入垫圈，防止安装时齿轮和铝件变形）



将4孔窄C反扣安装在打孔的17孔窄C上（和轴承座安装在一起）

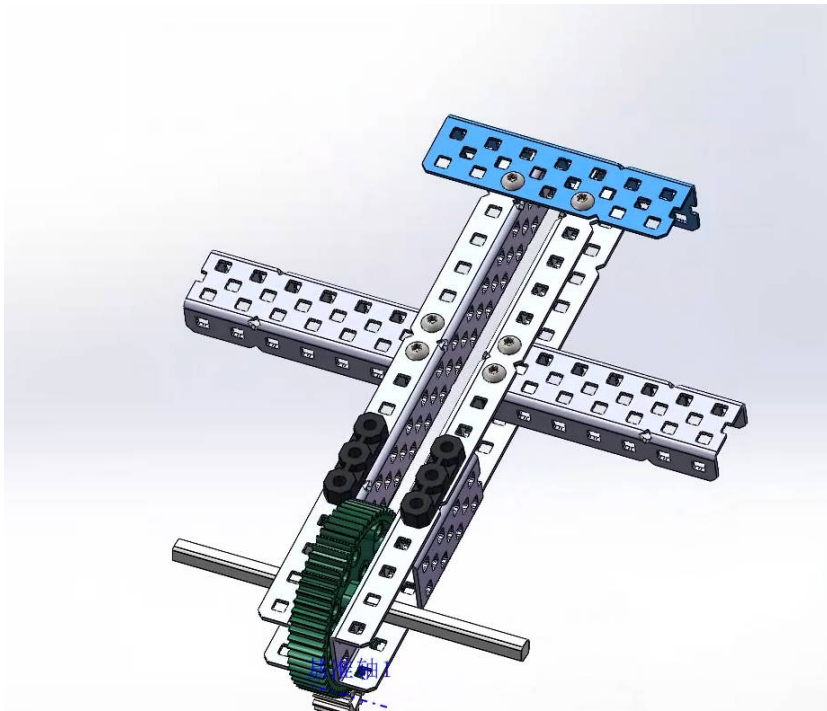
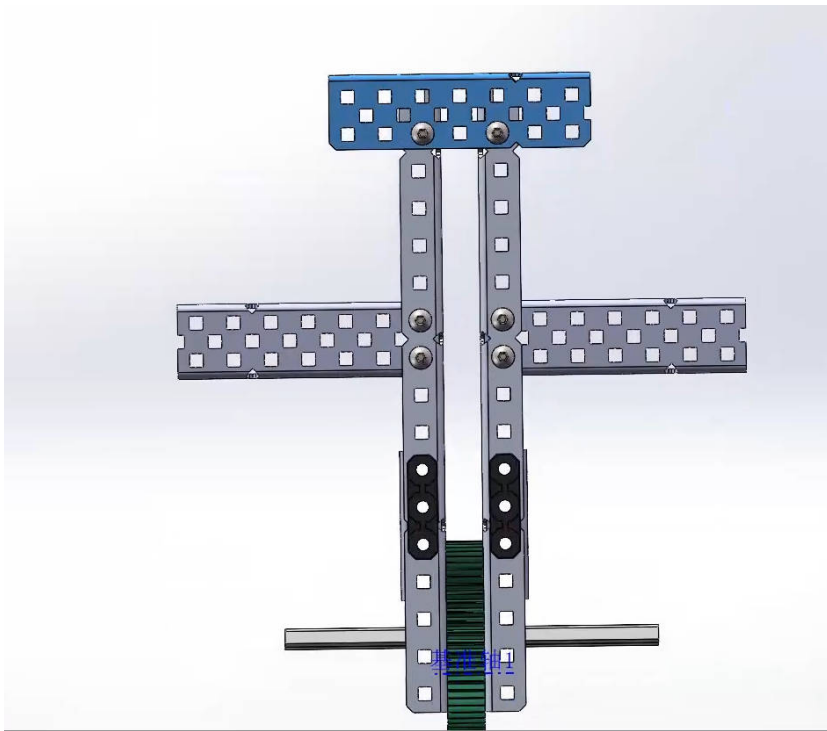


需要制作两个对称的



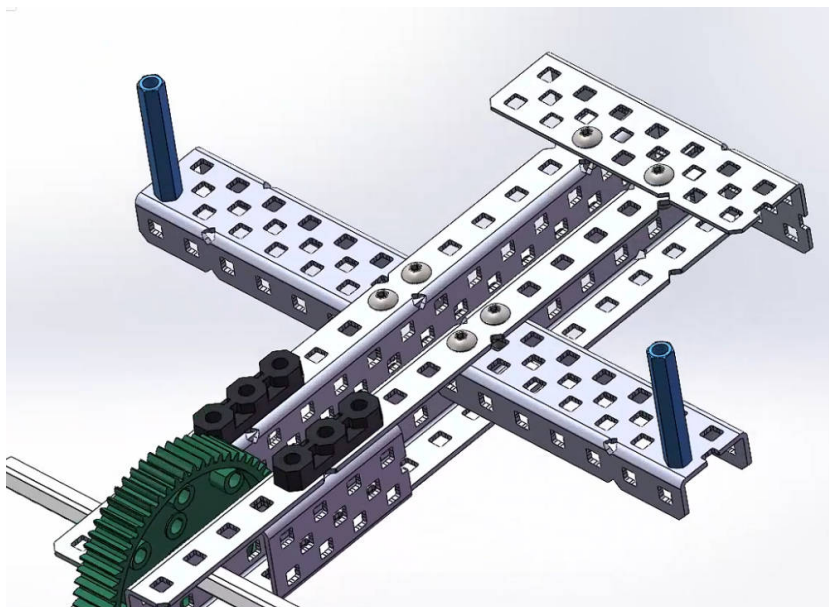
安装粽子支撑结构



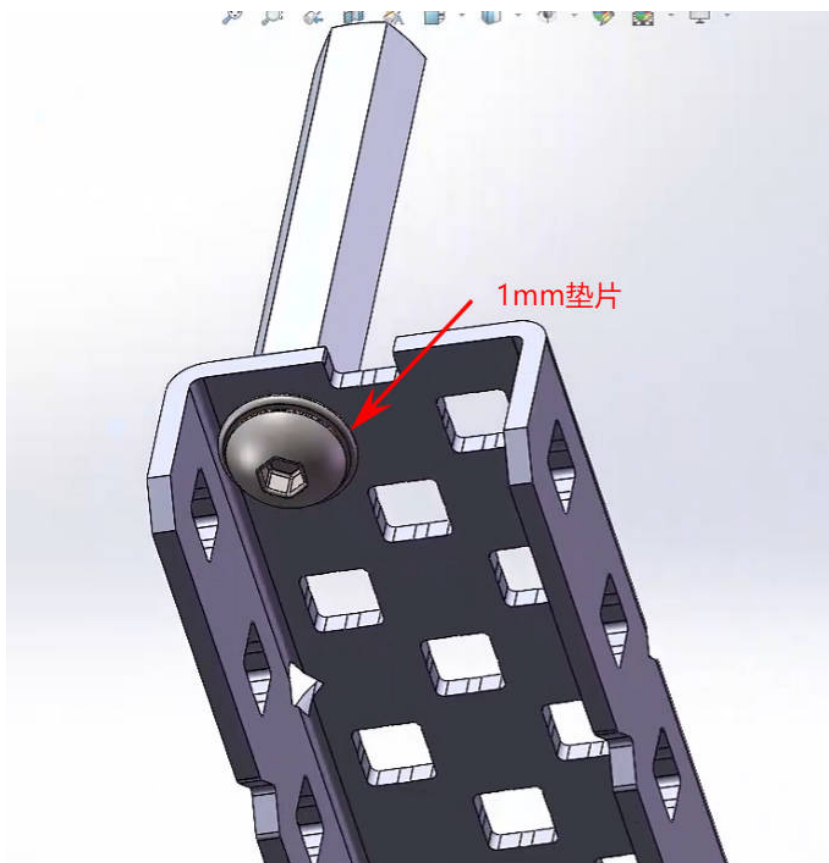


安装螺母柱

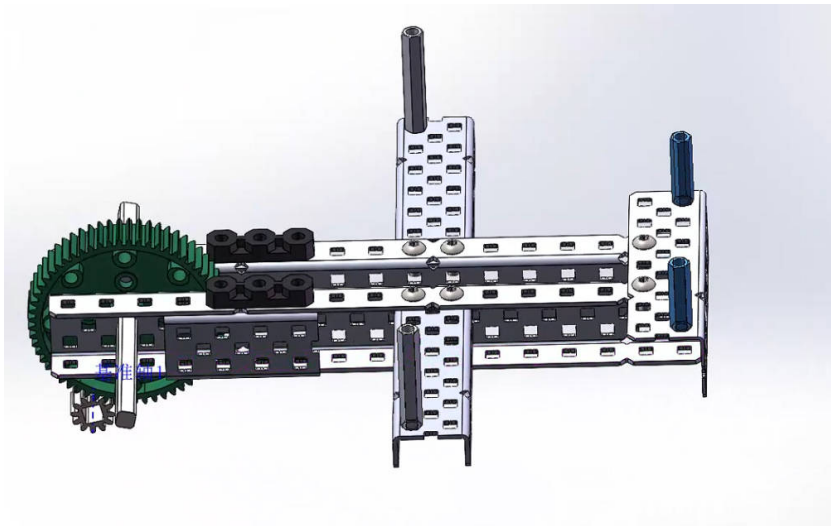
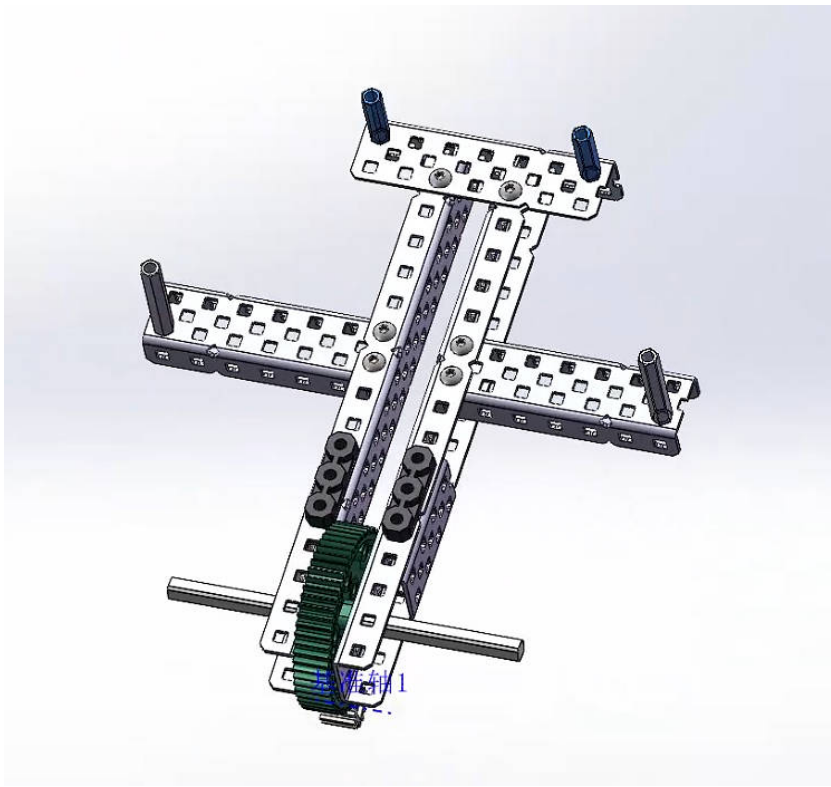
- 安装1.5in螺母柱



注：需要螺母柱时需要垫入1mm铁垫片，防止螺母柱松动

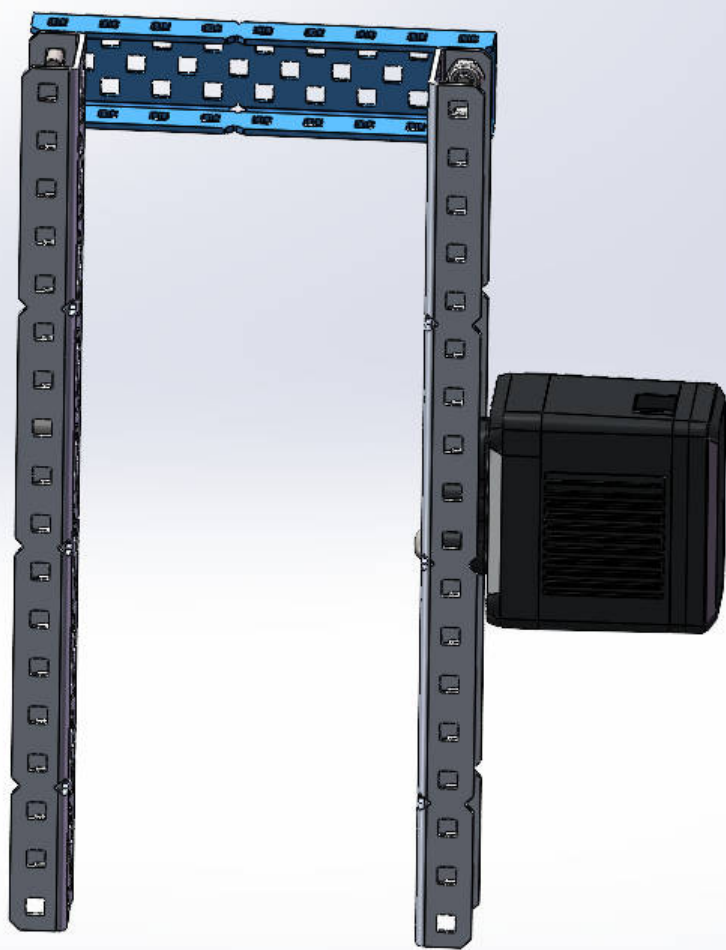
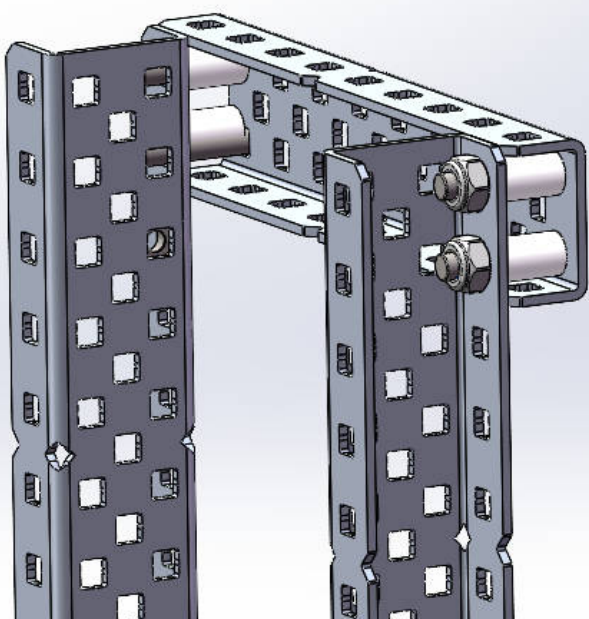


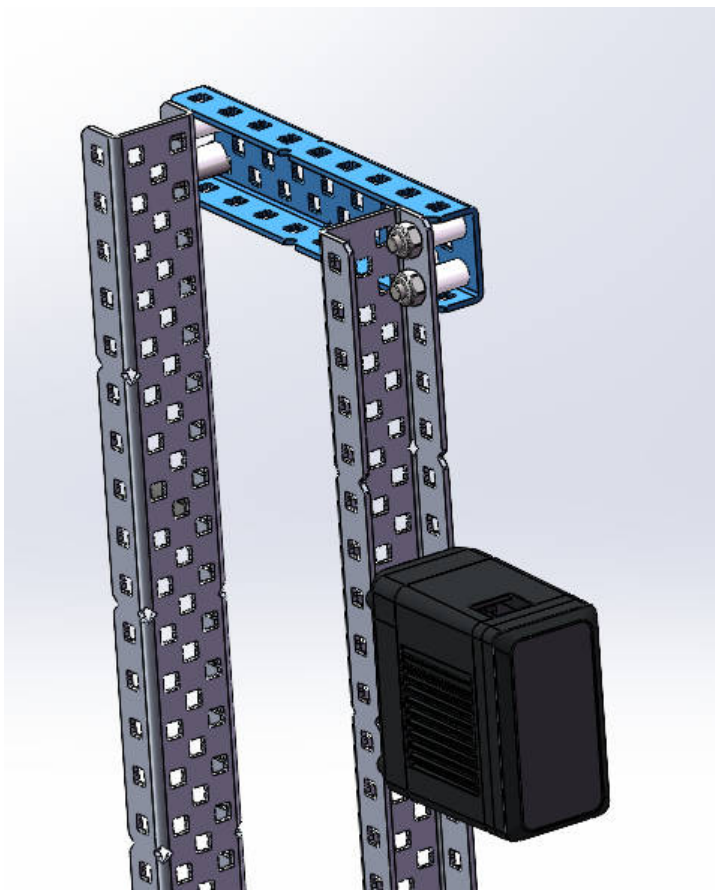
- 安装1in螺母柱



3.合并

中间需要垫入12mm垫圈





4.整体组装

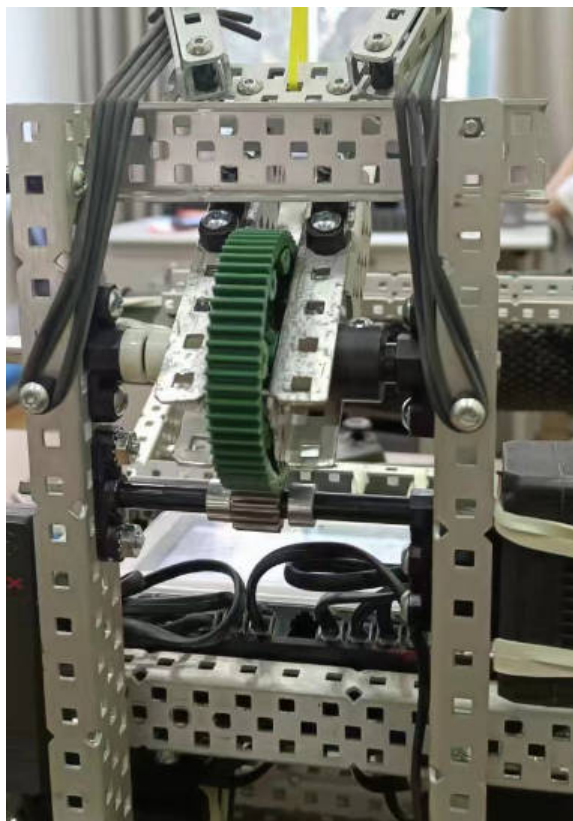
用锉刀磨掉12t齿轮中的4个齿（留8个齿）



按照下图安装

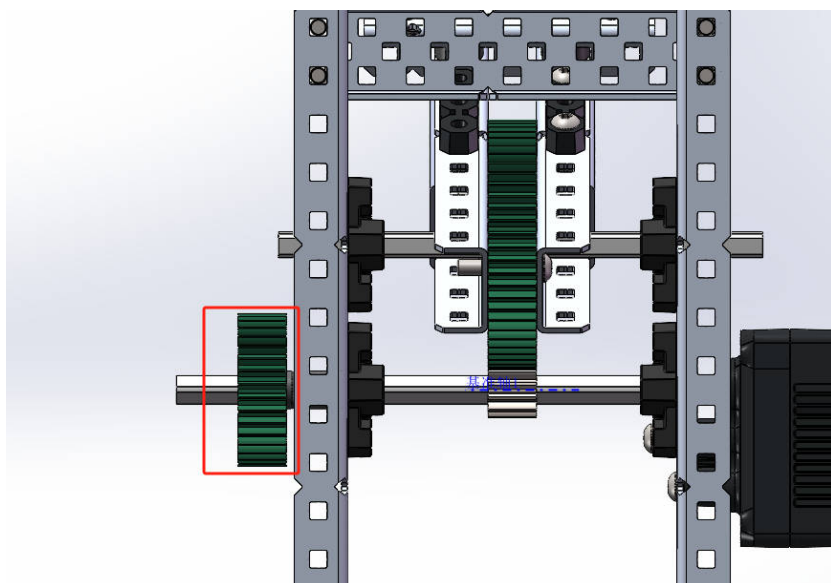
- 都使用粗轴安装（粗轴长度是5in和6.5in各一根）
- 需要保证齿轮啮合部分在正中间

- 需要安装至少1个限位杯
- 垫圈厚度自由发挥

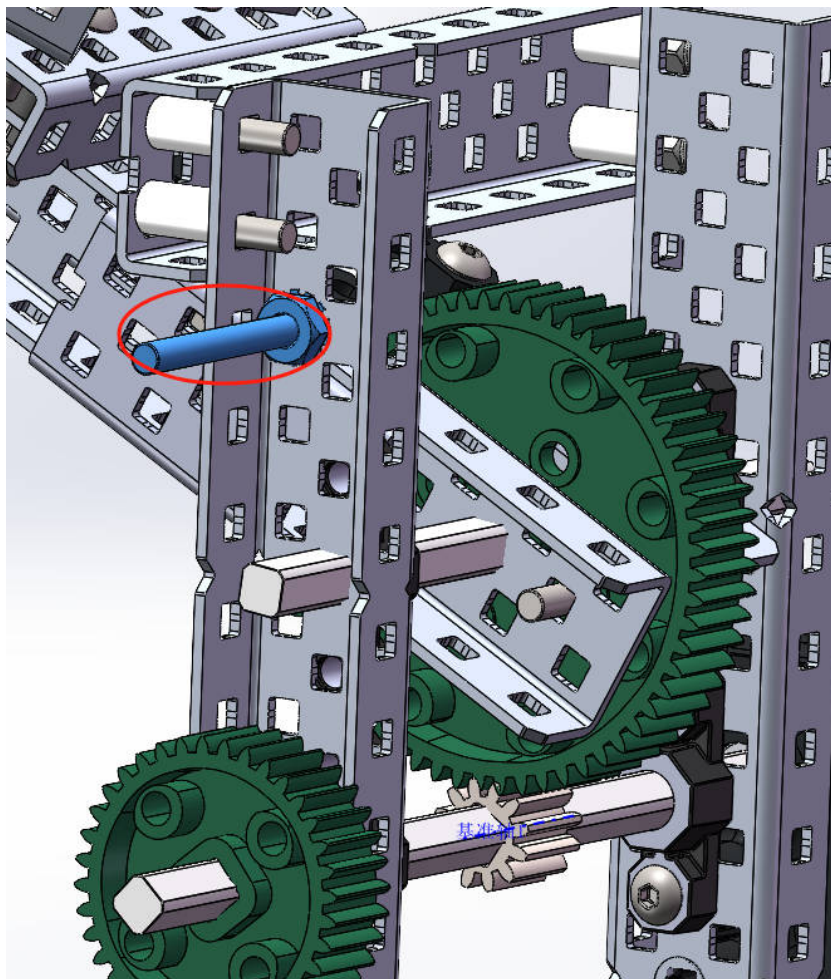


5.安装棘轮结构

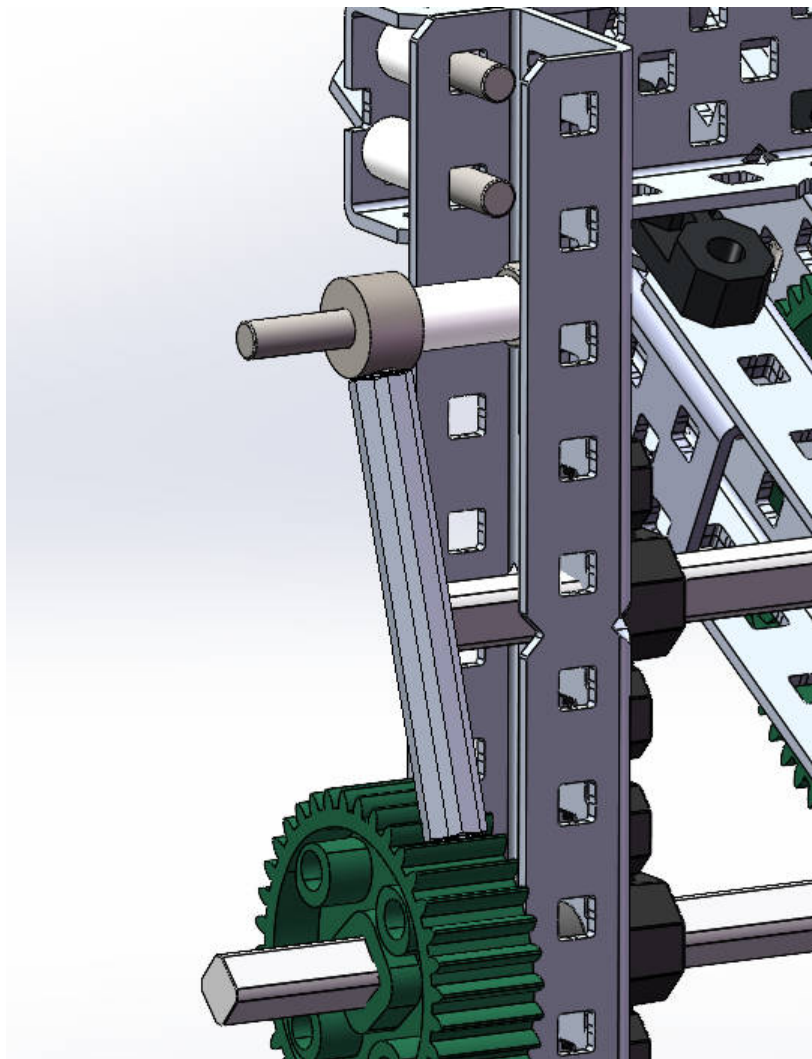
先在电机传动层安装36t高强度齿轮作为棘爪



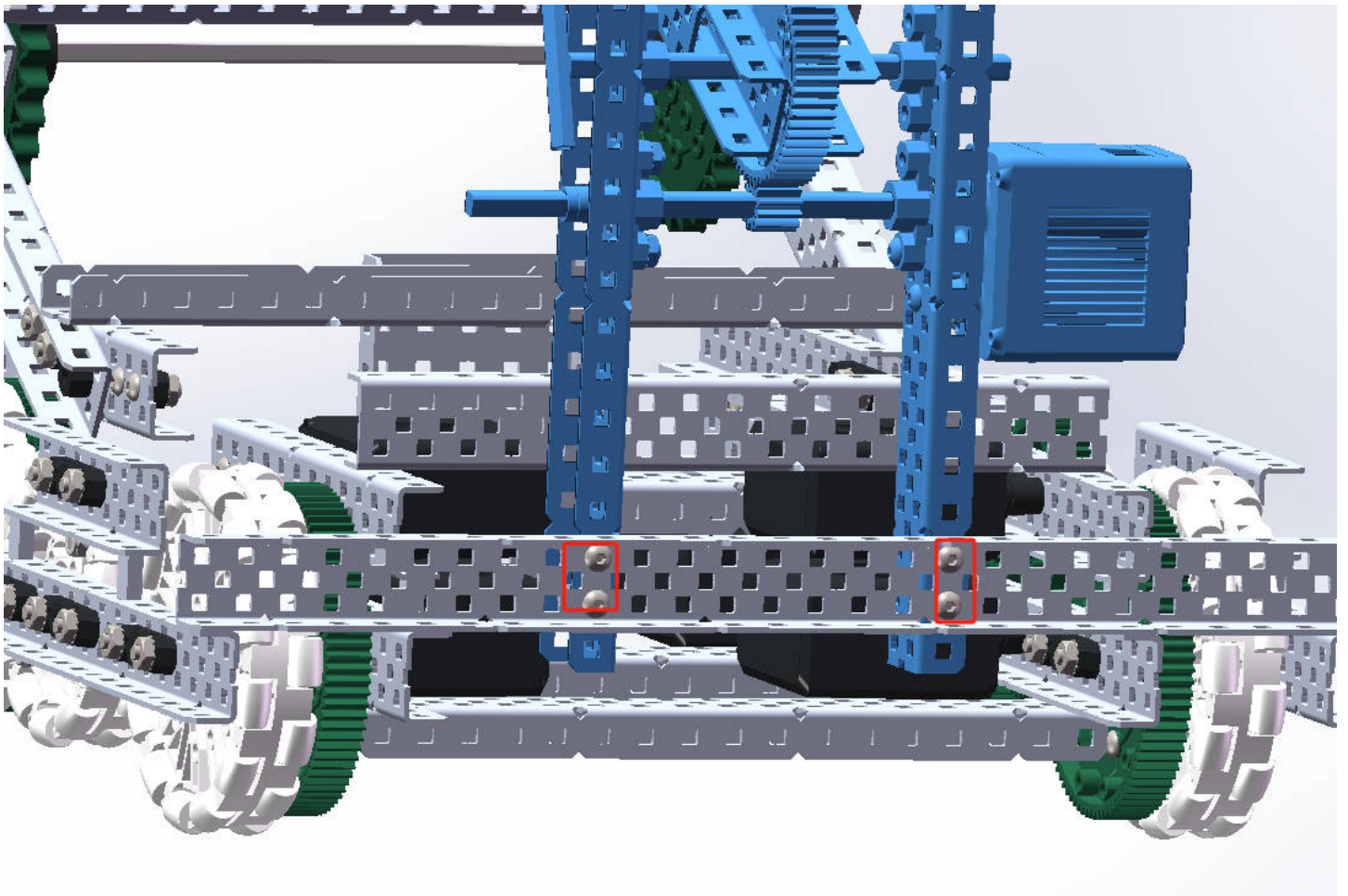
按照如下位置安装1.5in螺丝和自锁螺母



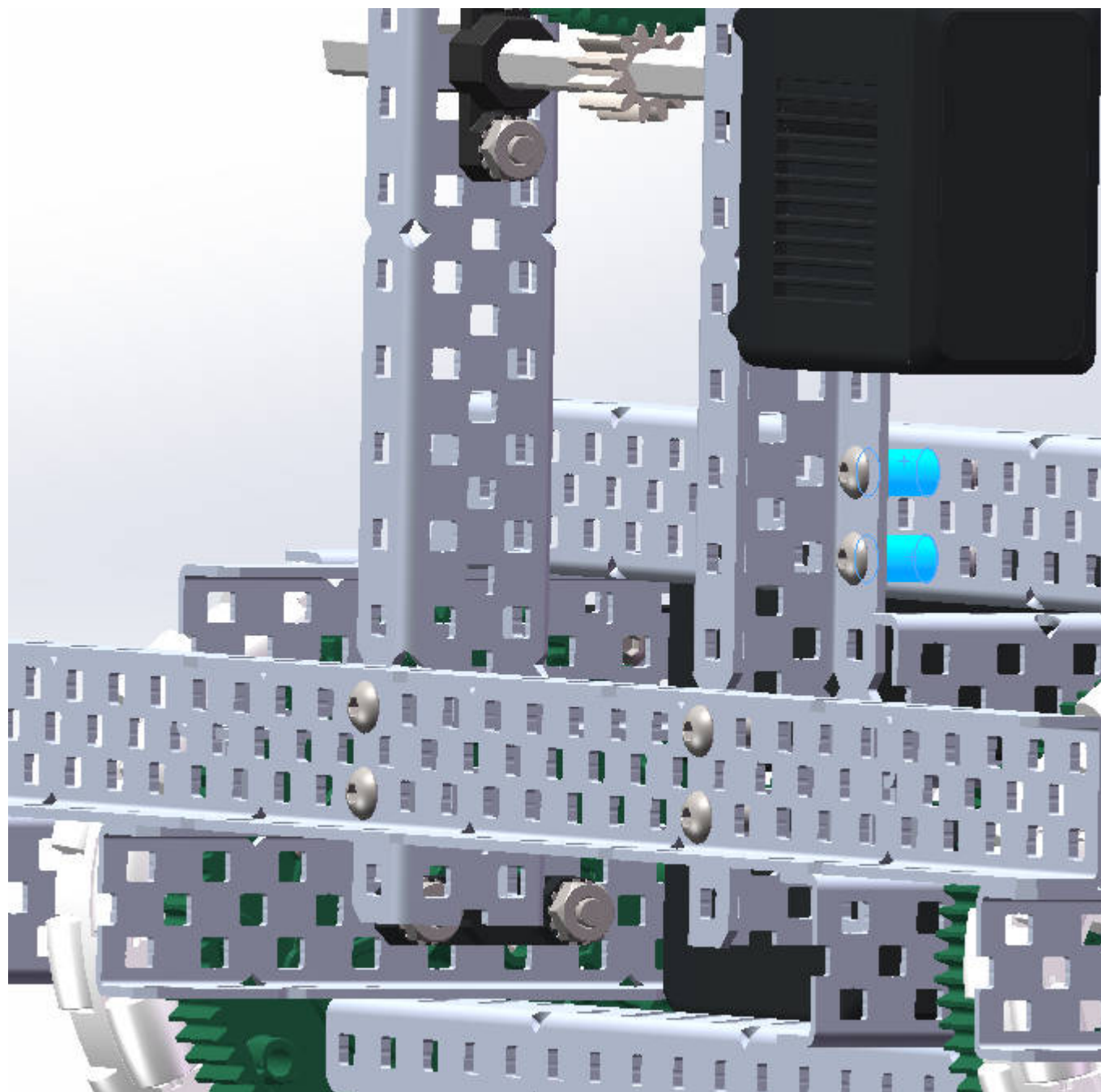
垫入14mm长的垫圈 → 限位杯 （需要拧出里面的机米螺丝，换长的机米螺丝拧入再旋入2in的螺母柱） → 防滑螺母封住



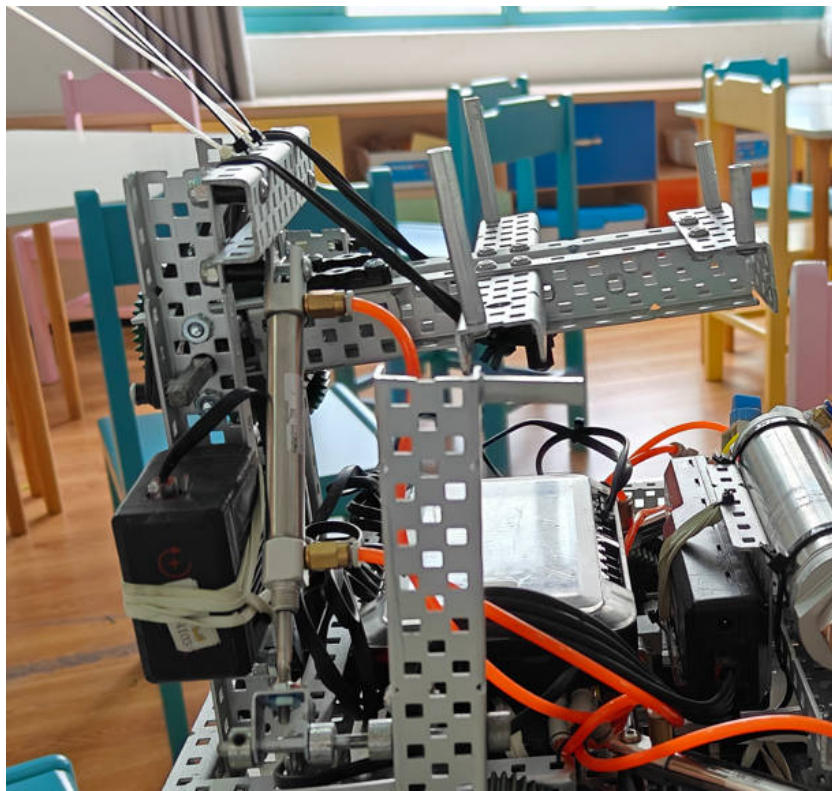
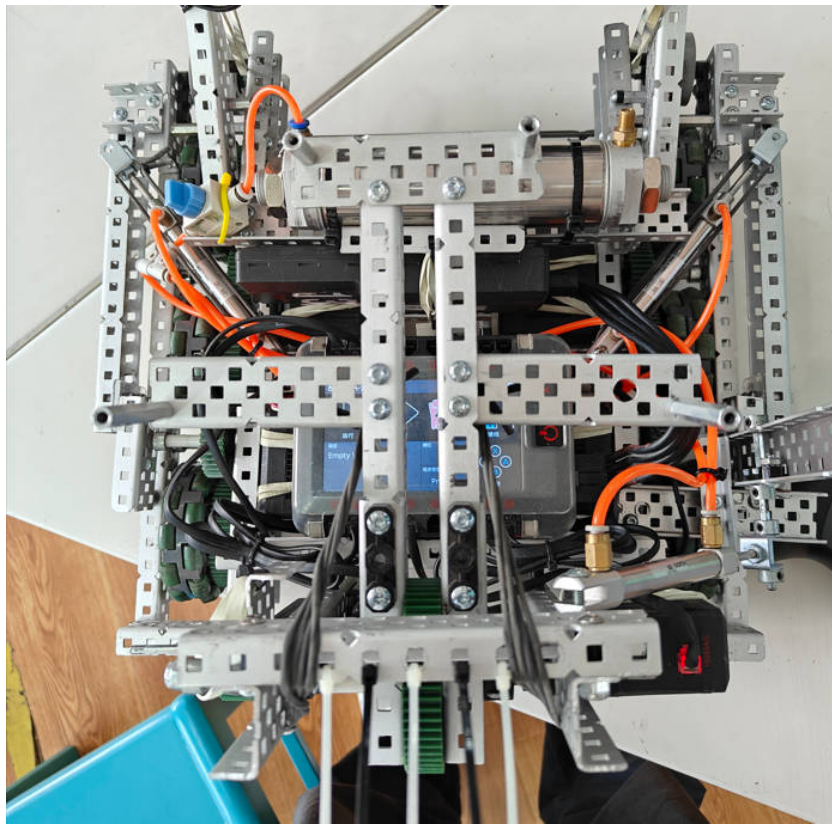
6. 安装到底盘上



需要垫入10mm垫圈



7.成品图



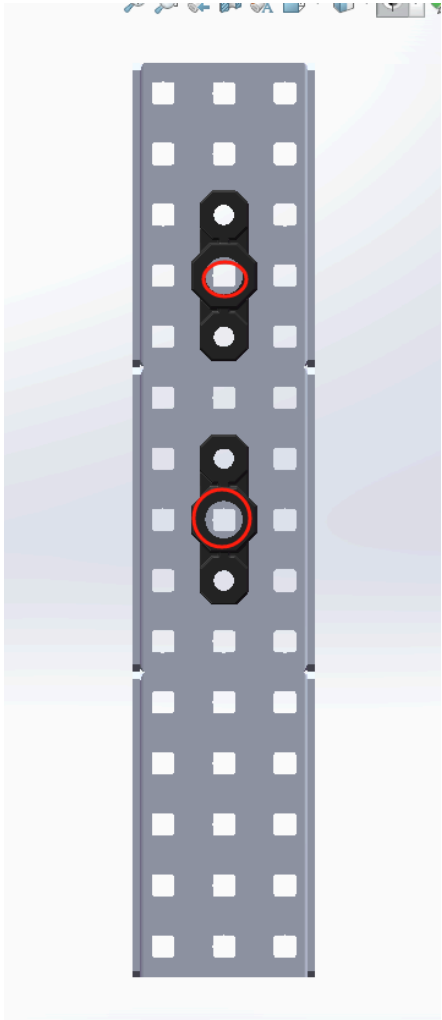
03 爬升结构

材料清单

铝件名称	数量
窄C 22	2
中C 15	2
1*1 L 6孔	4
窄C 12	1
8in粗轴	1
9in粗轴	1

1.打孔

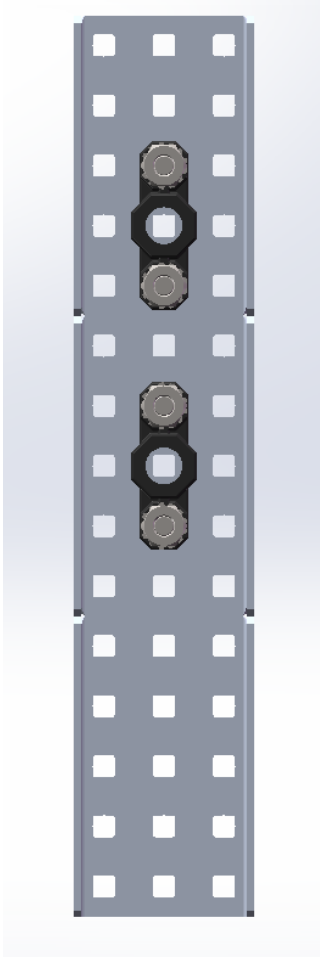
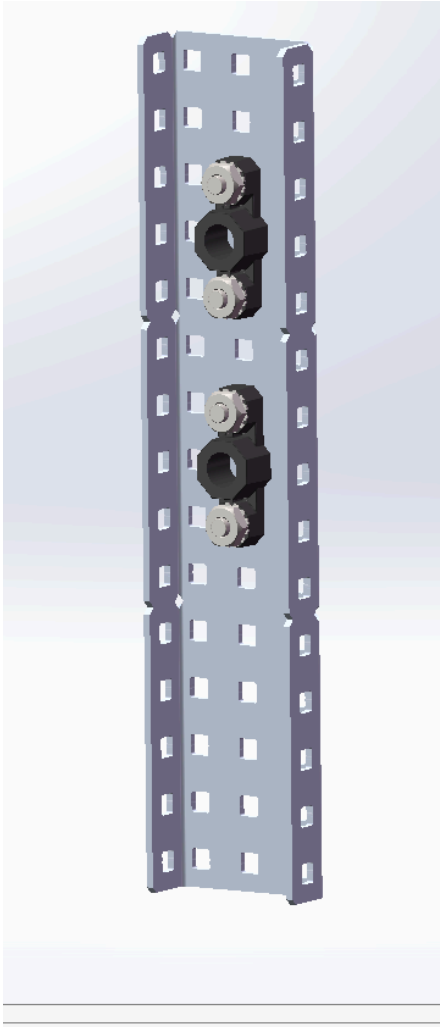
下图所圈位置进行打孔，加工2根



2.安装轴承座

按照下图所示位置安装两个高强度轴承座

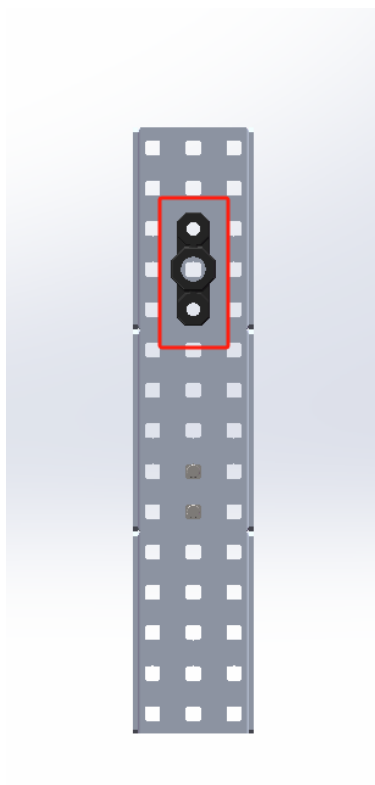
- 左侧



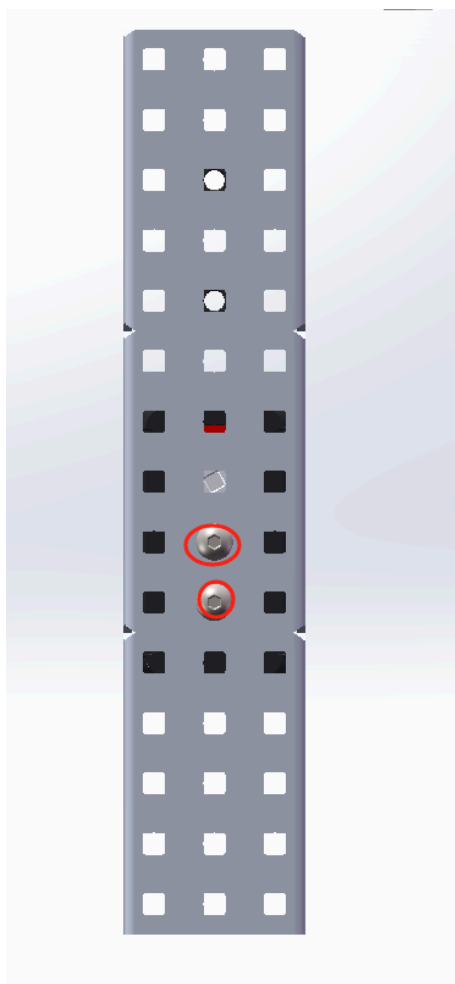
- 右侧

右侧的铝件需要安装一个电机

先将第一个轴承座安装上去



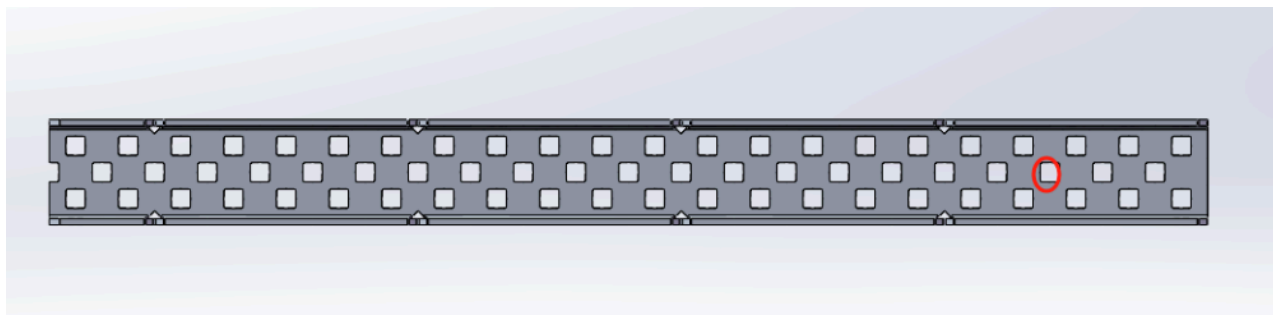
随后，安装电机



3.安装爬升杆

- 先钻孔

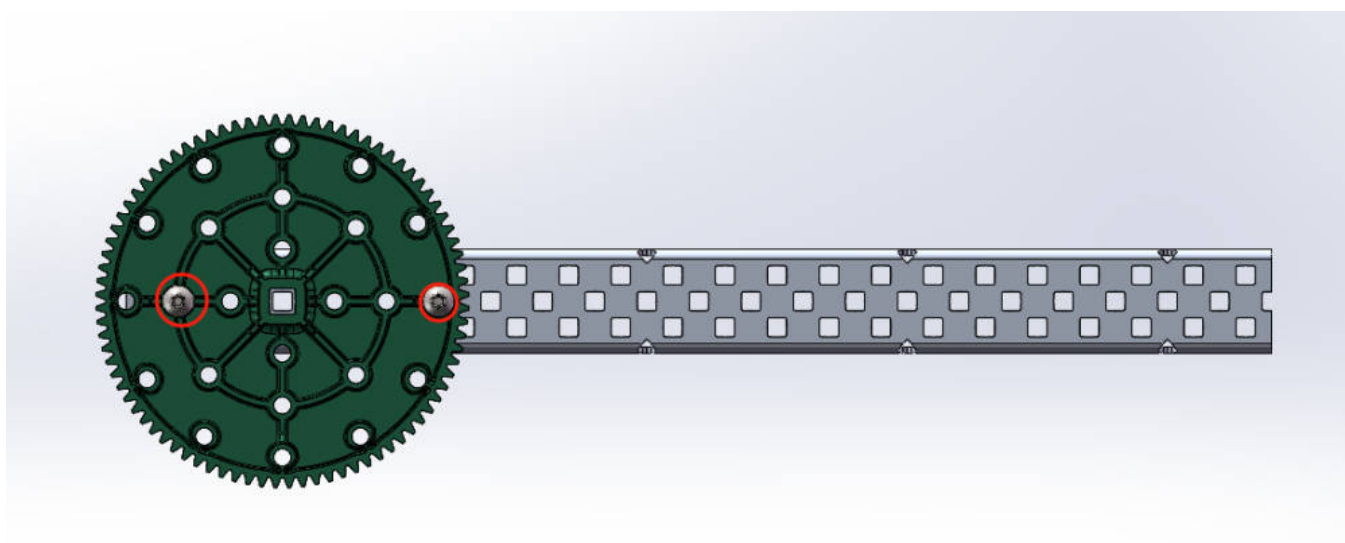
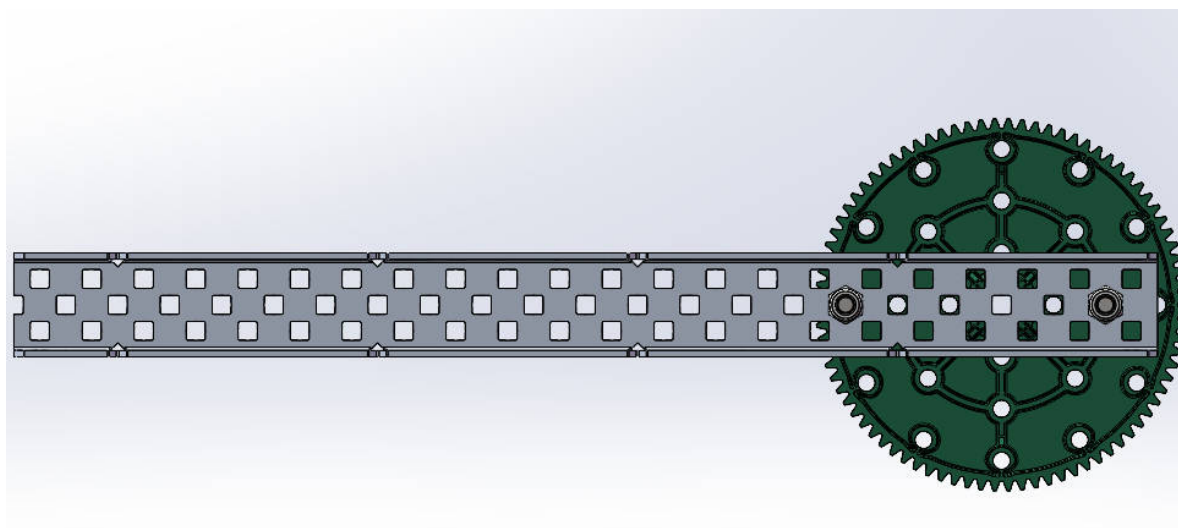
钻两个



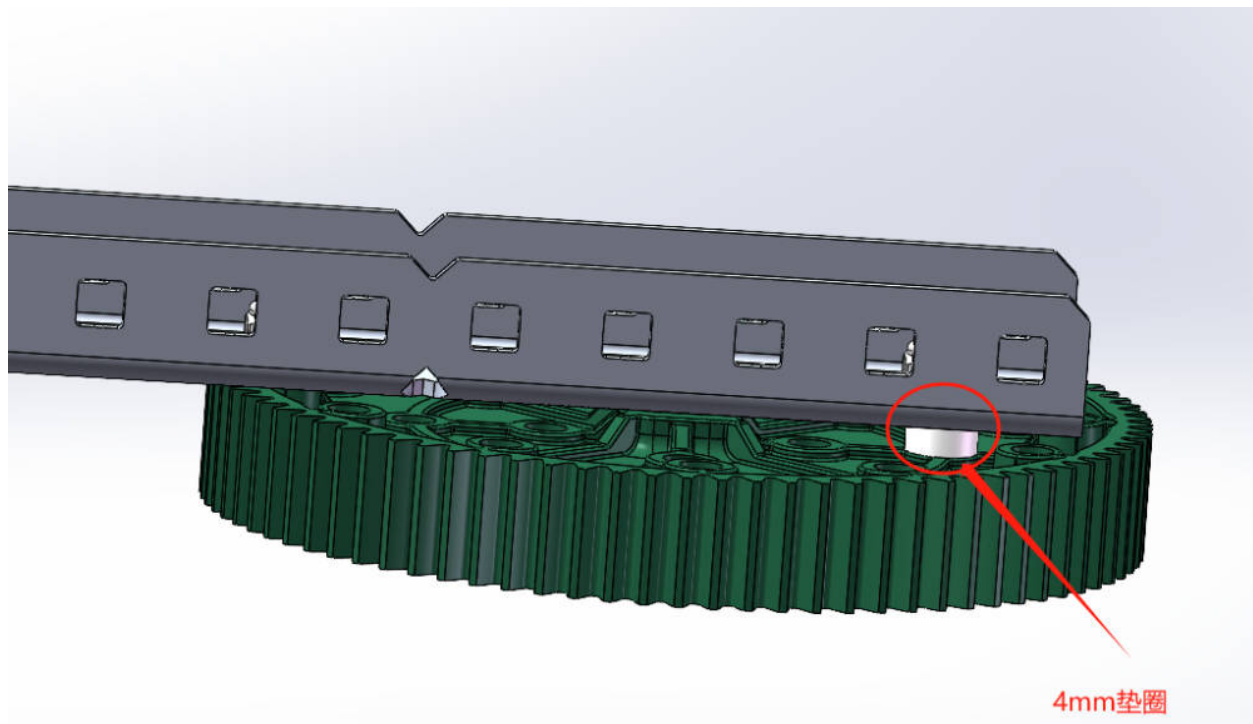
- 齿轮合并

将84t齿轮和铝件安装在一起，对称制作两个

需要安装两个固定孔

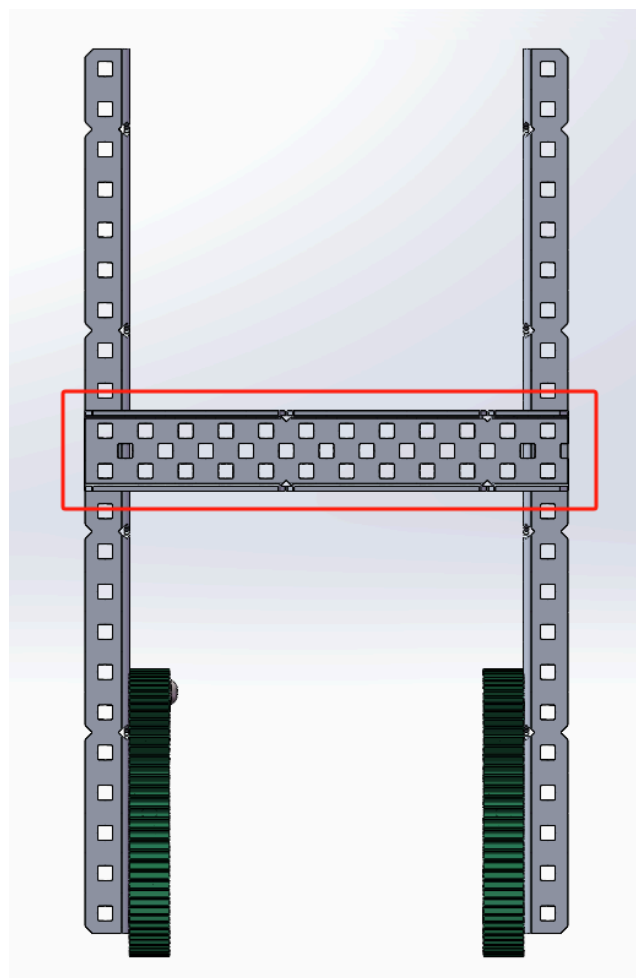
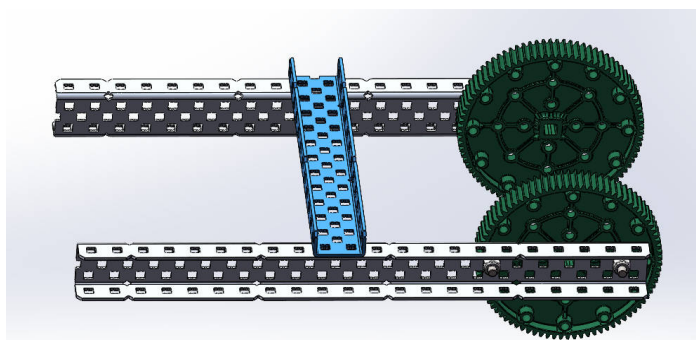


注：因为有一个孔位，齿轮存在凹槽，故安装的时候需要垫入4mm垫圈加以支撑，不然容易形变



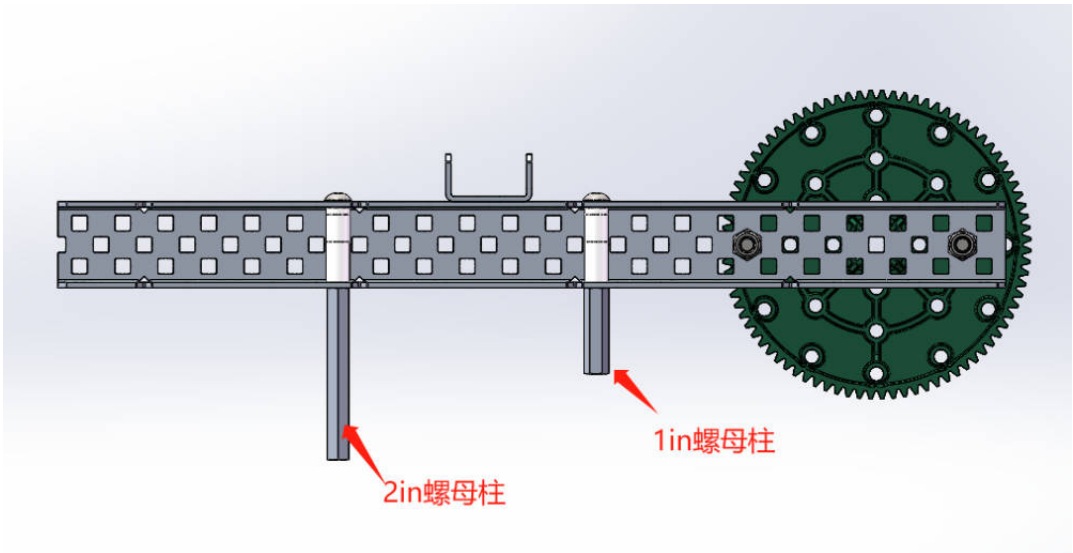
4.合并爬升杆

用12孔的窄C将两个装配体部件固定在一起

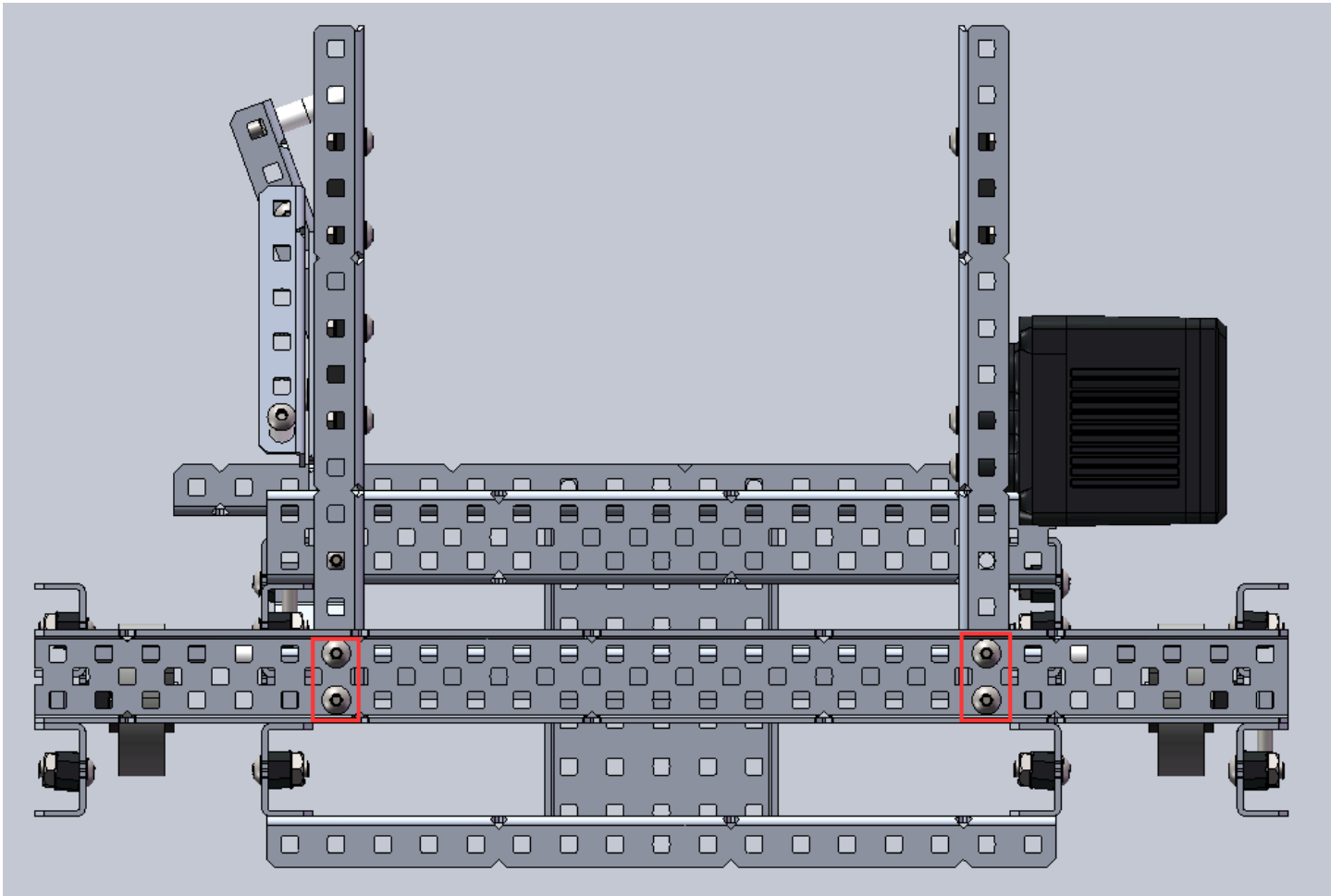


5.安装螺母柱

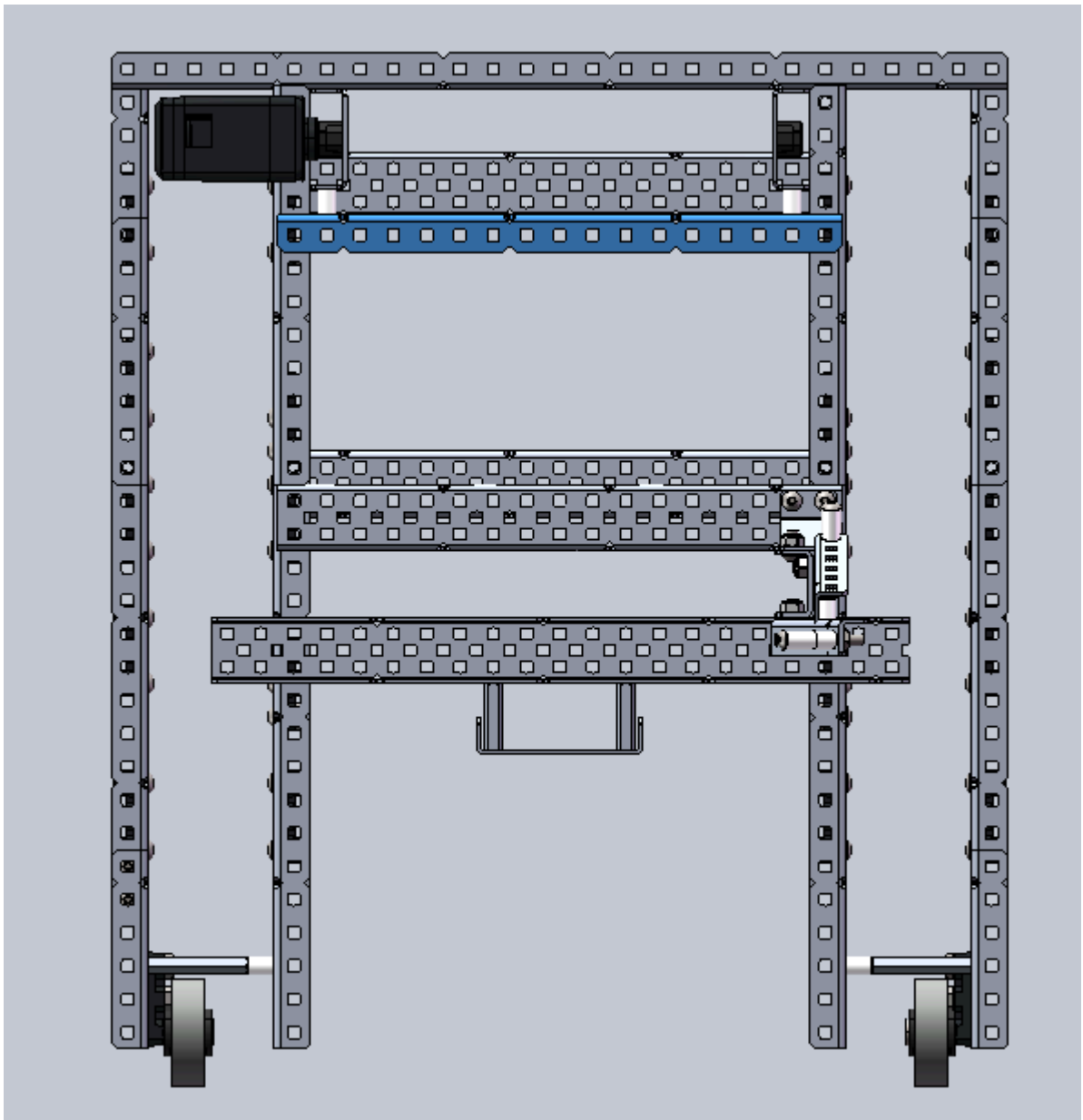
注意C型铝件里空的部分需要装入垫圈垫起来



6.安装左右固定铝件

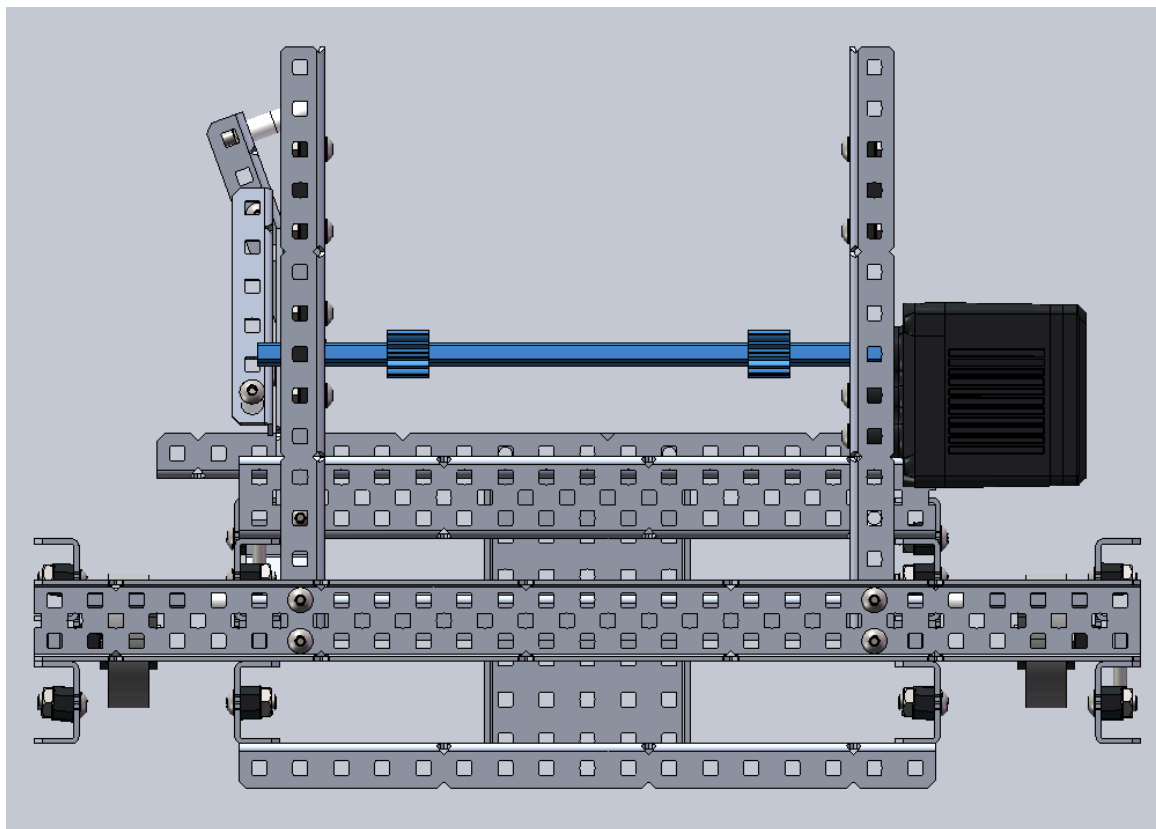
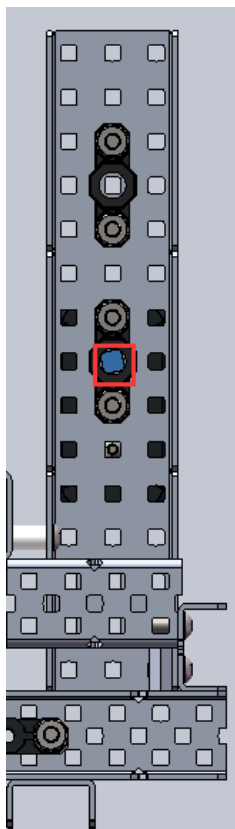


安装17格窄c横梁

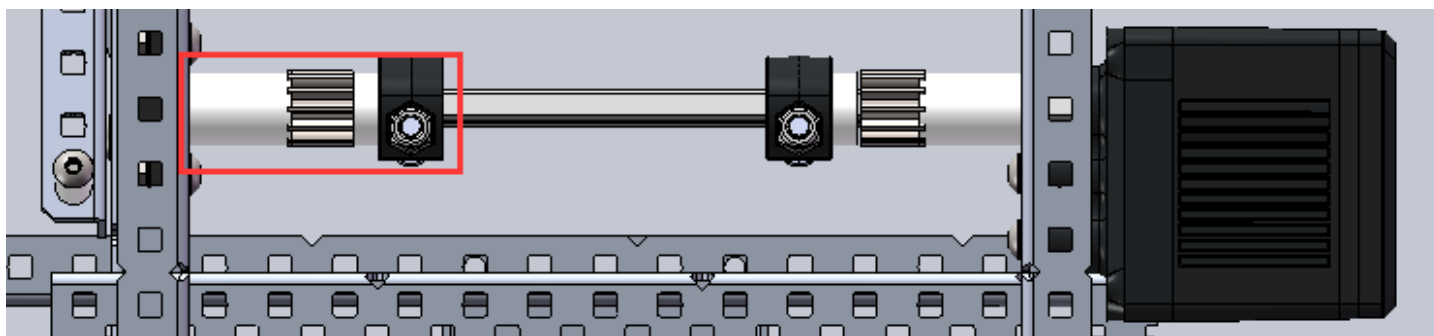


7.安装电机粗轴与12t齿轮

穿过最下面的轴承座，并装上12t齿轮以及粗轴限位

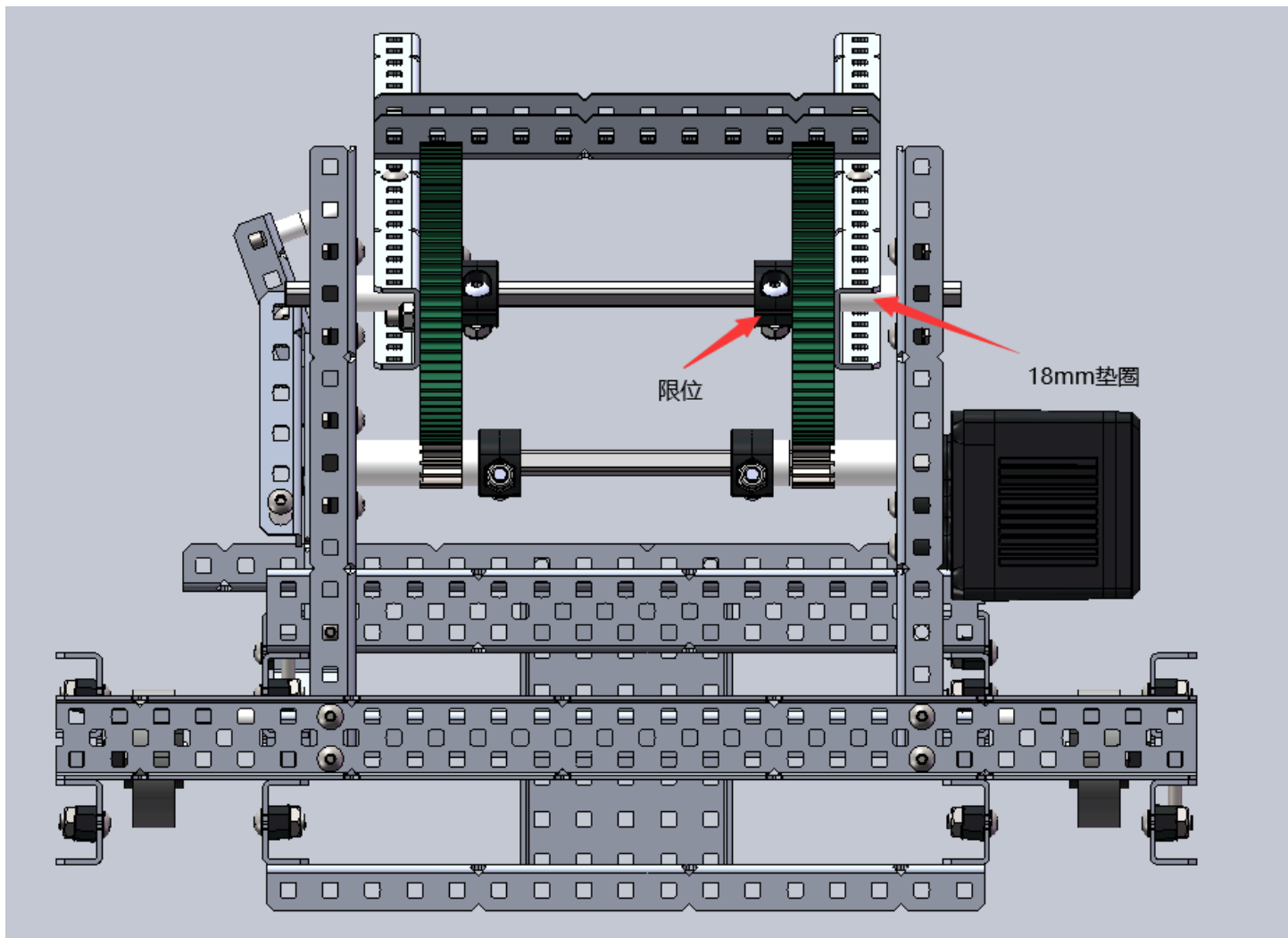
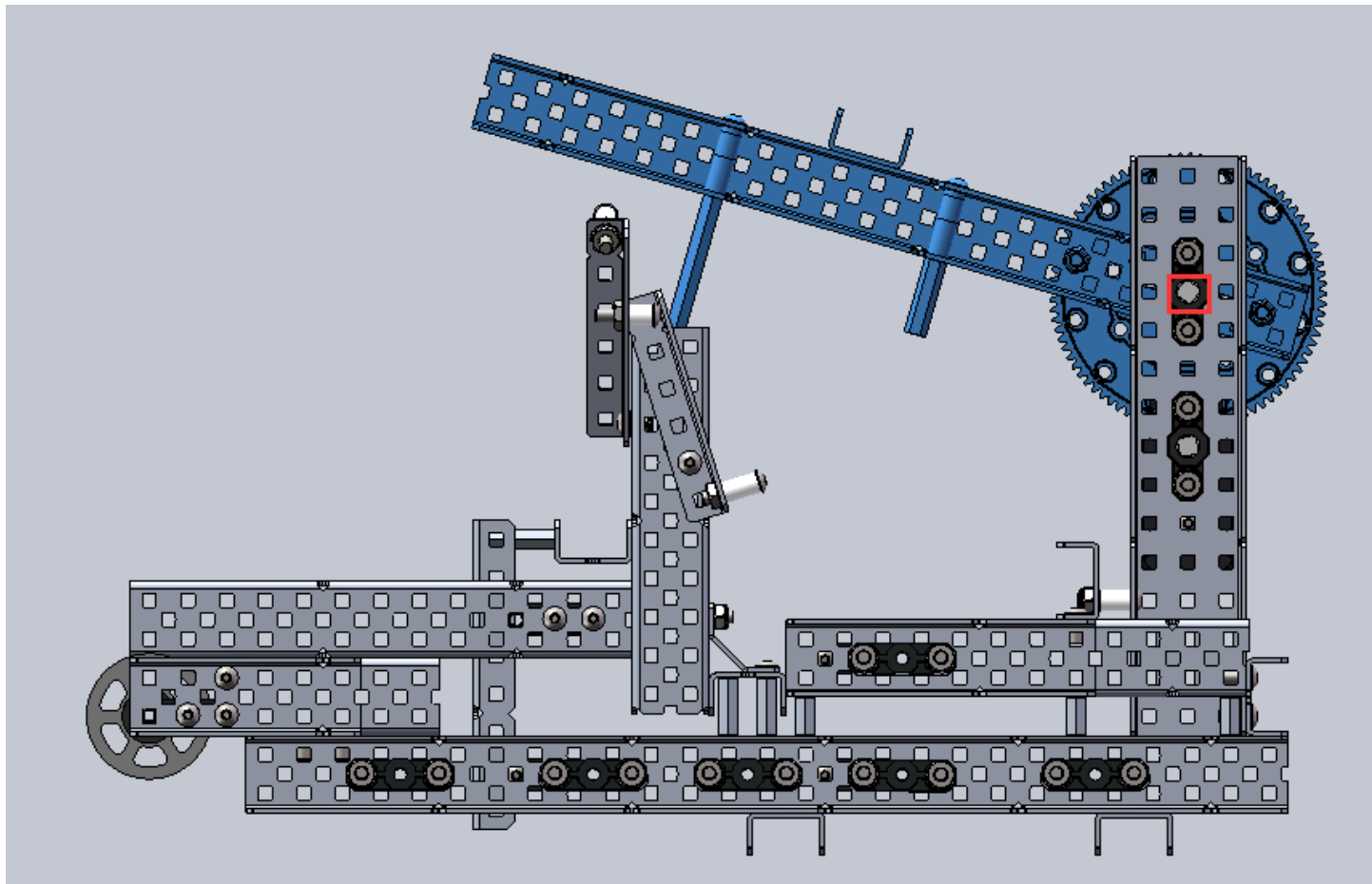


20mm垫圈+12t齿轮+4mm垫圈限位

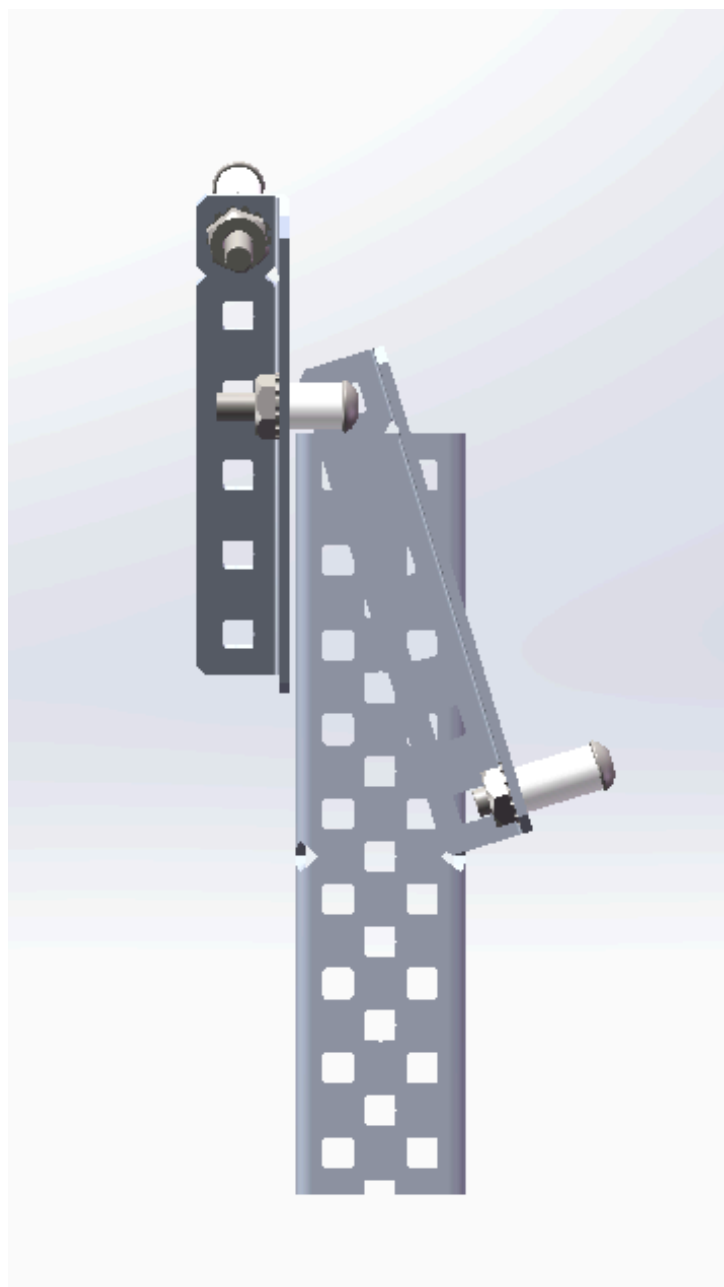
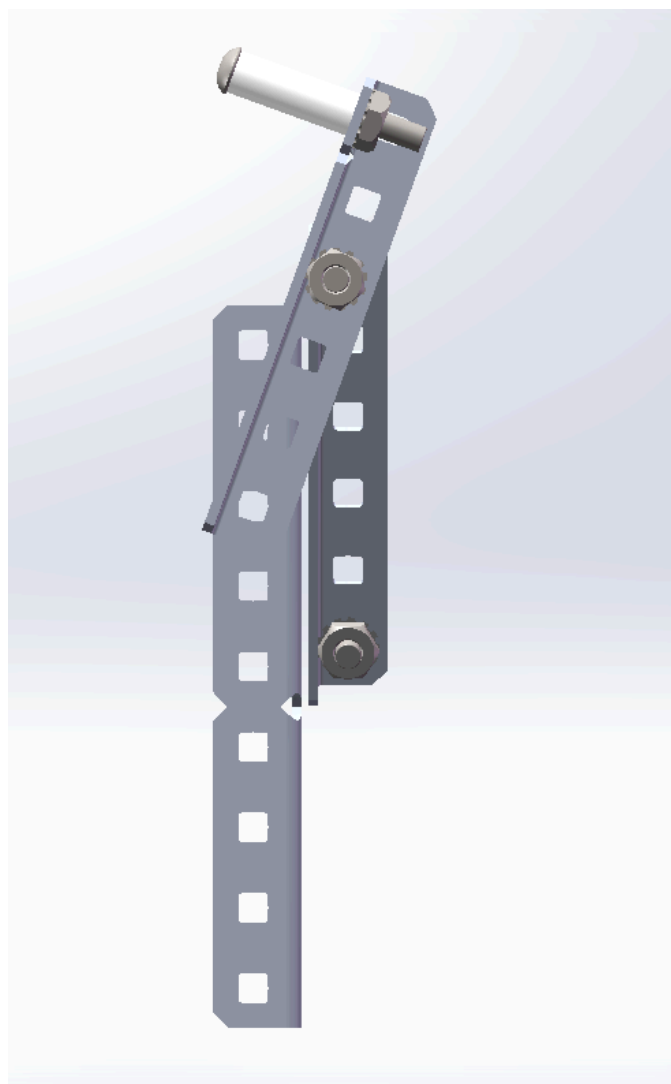


8. 安装爬升杆

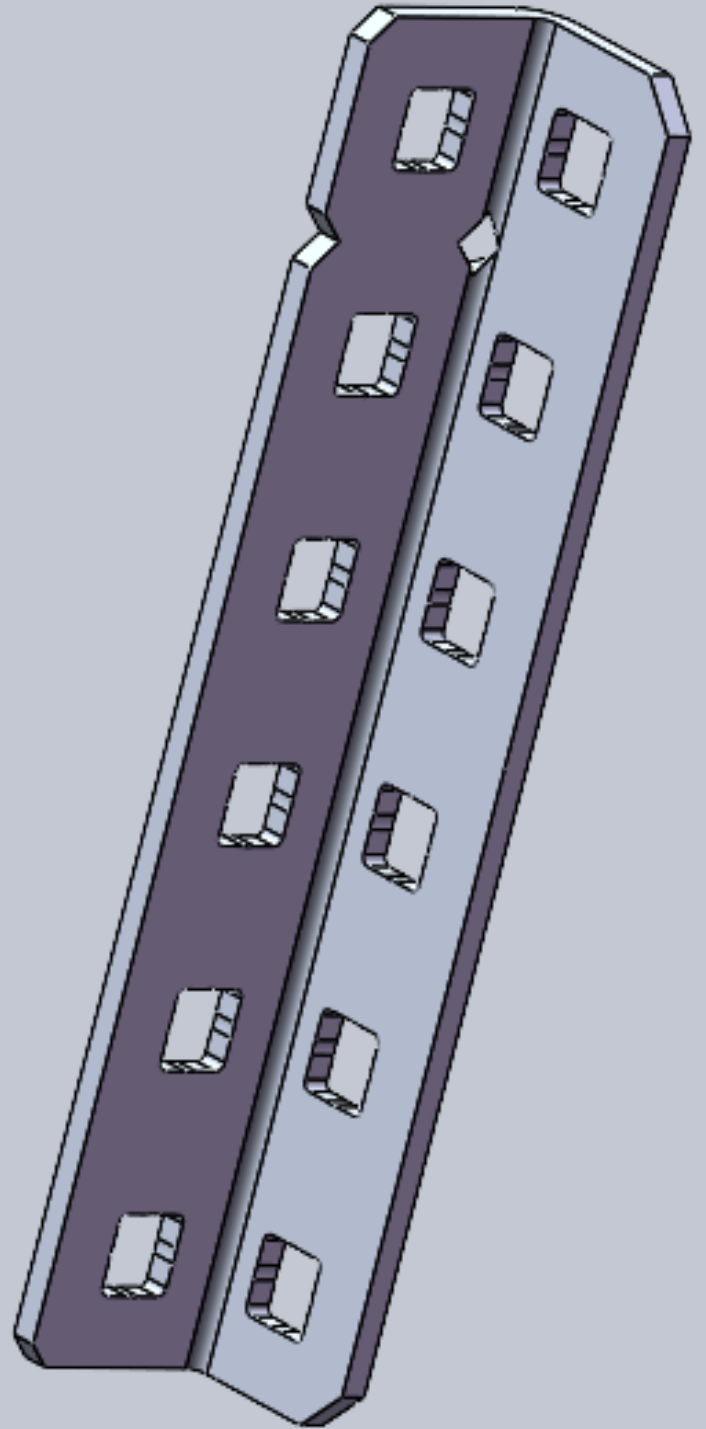
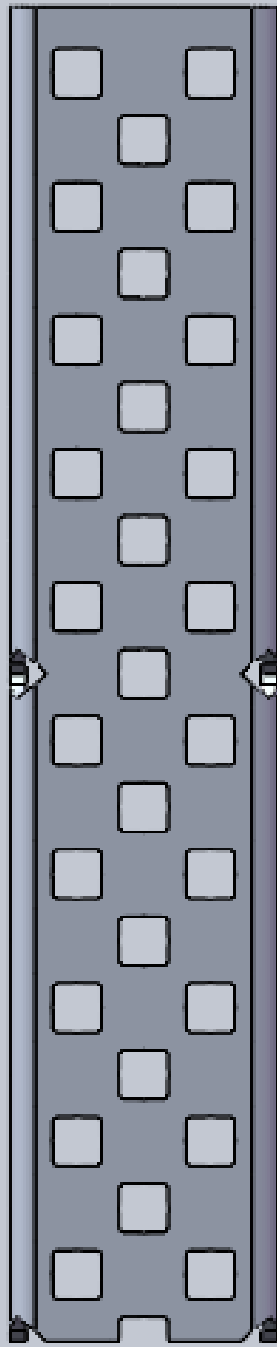
粗轴穿过最上层轴承座

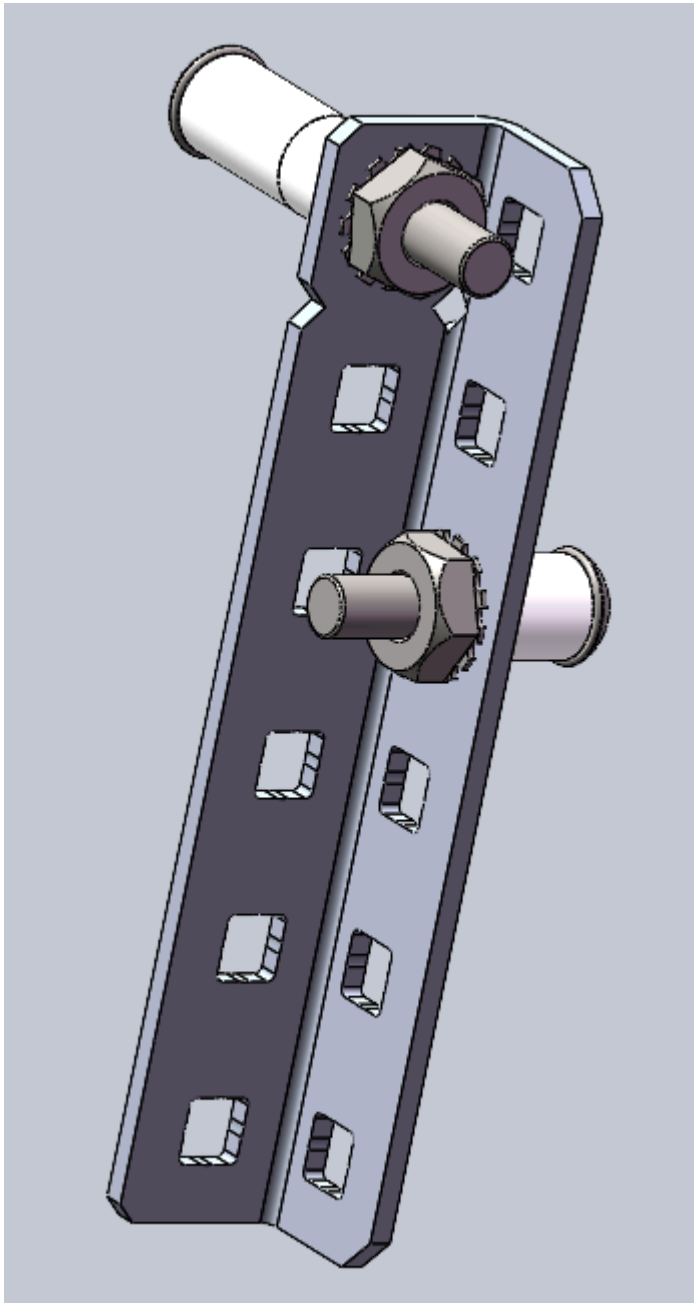


04 卡扣结构

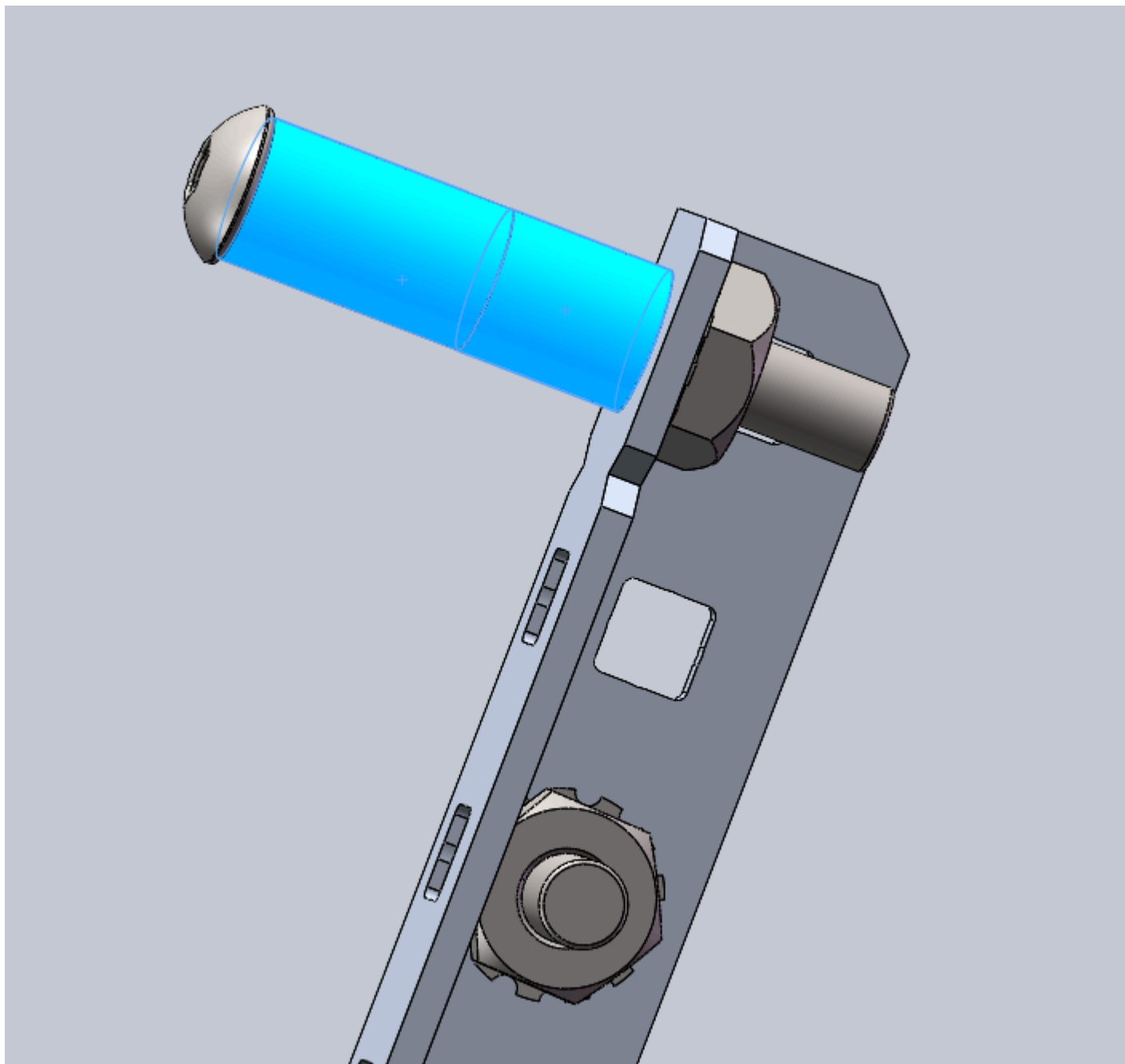


10格窄c件和2个6格1*1L件

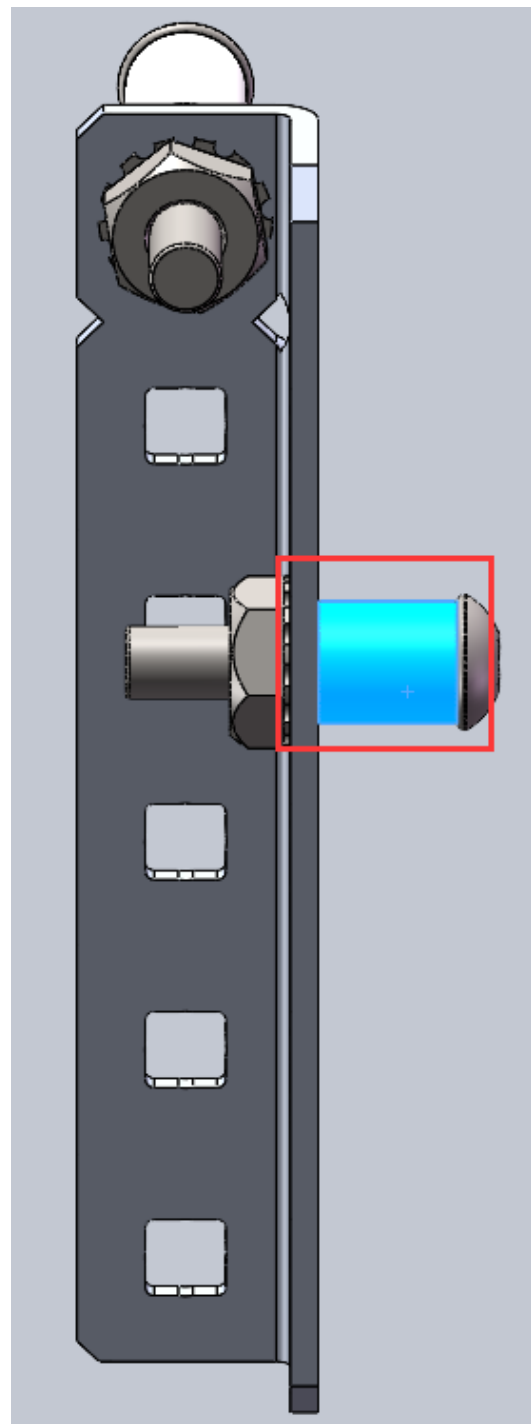
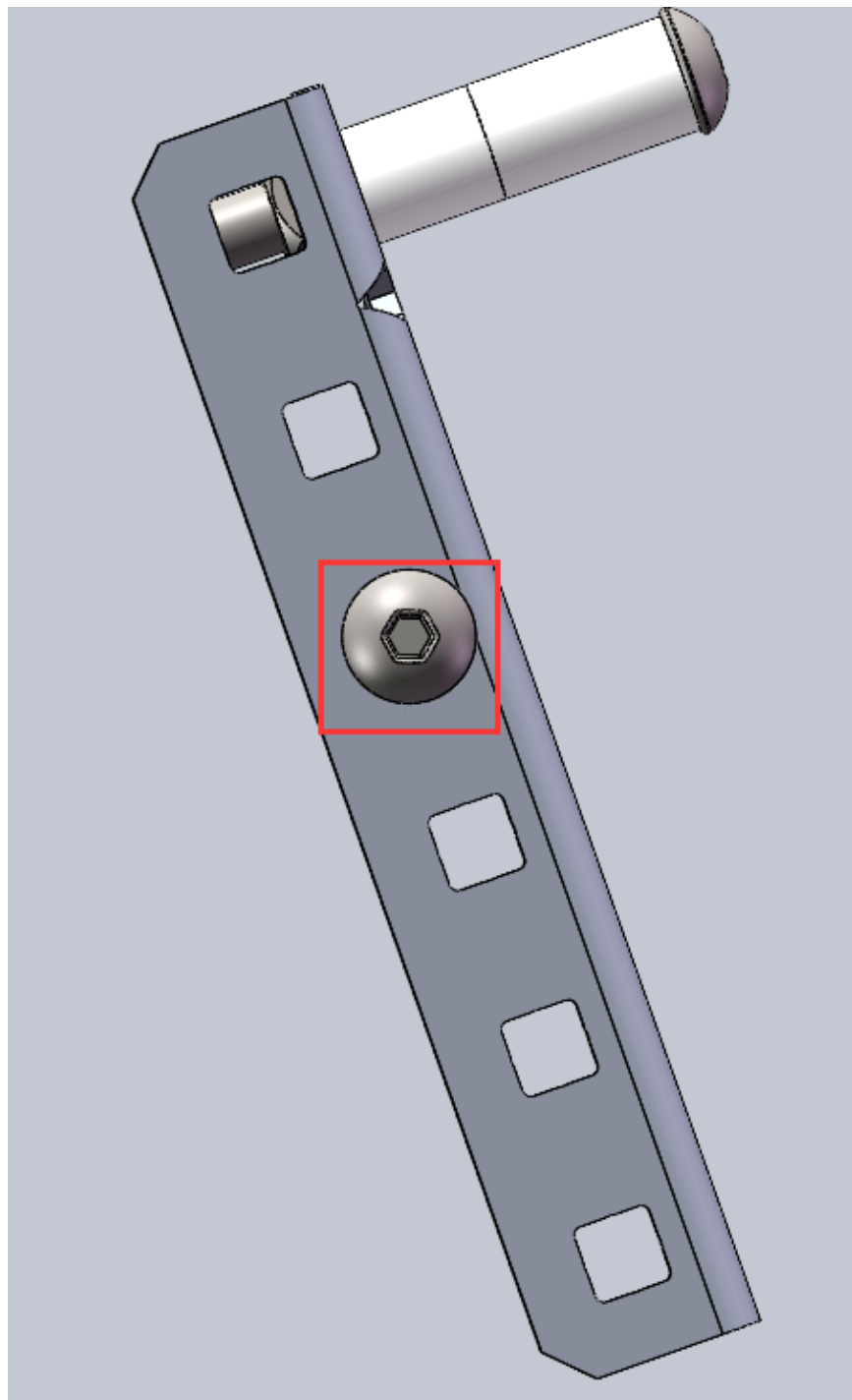




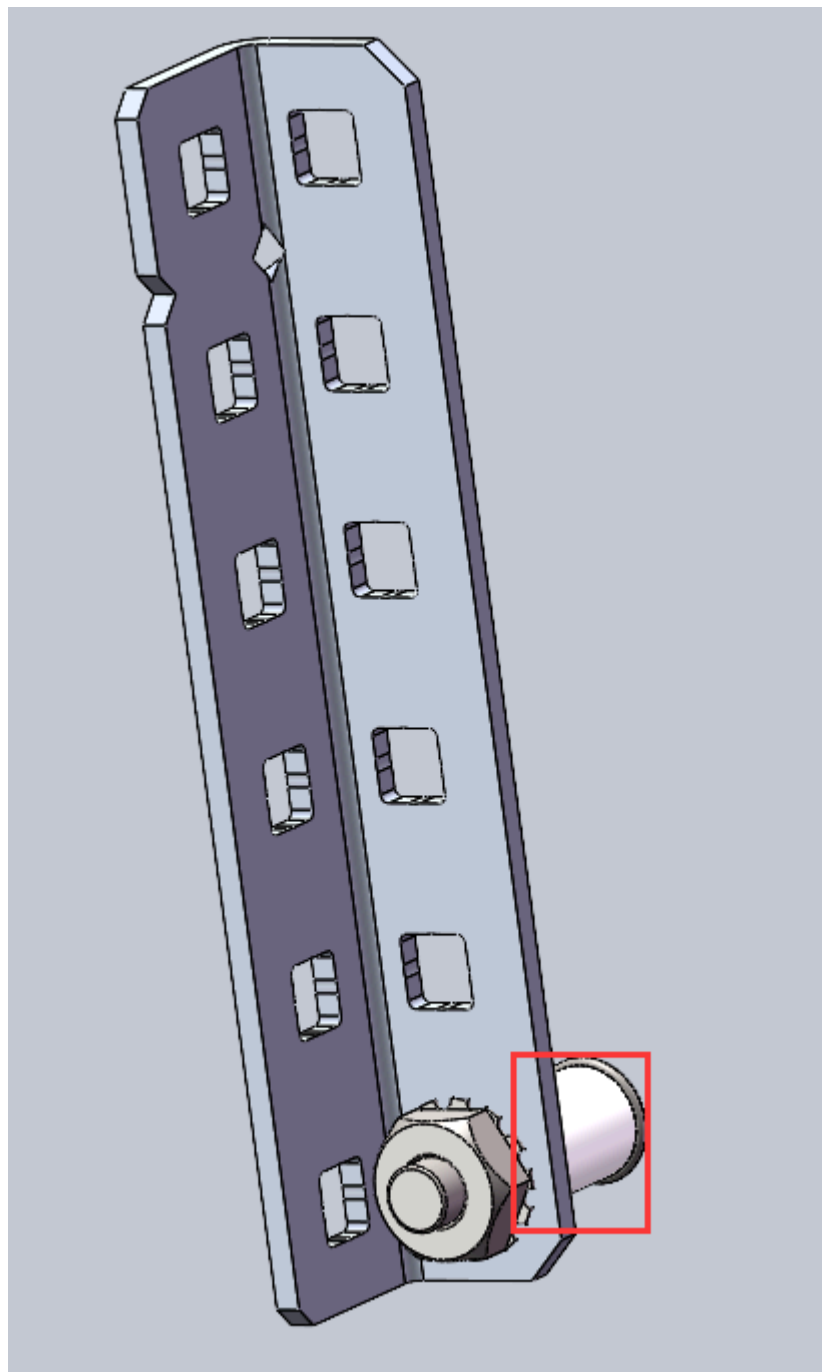
12mm垫圈+8mm垫圈



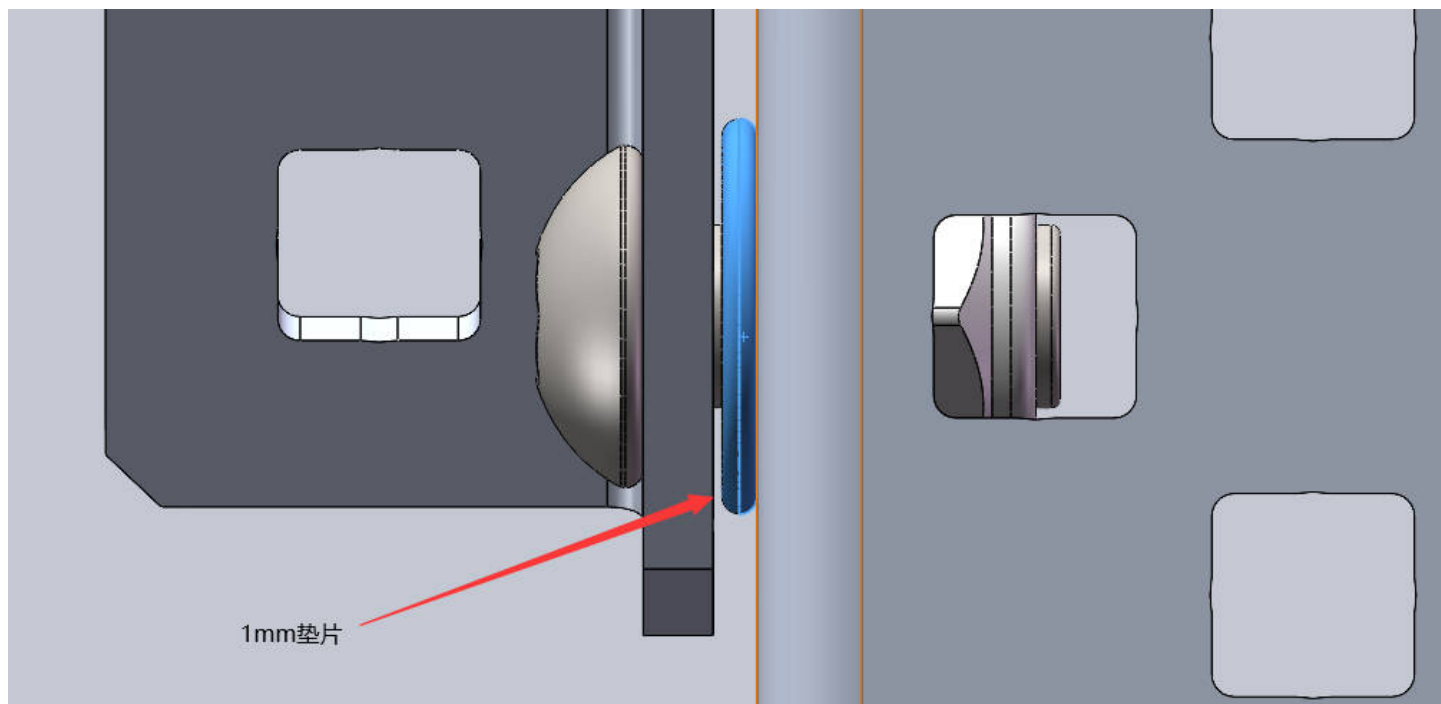
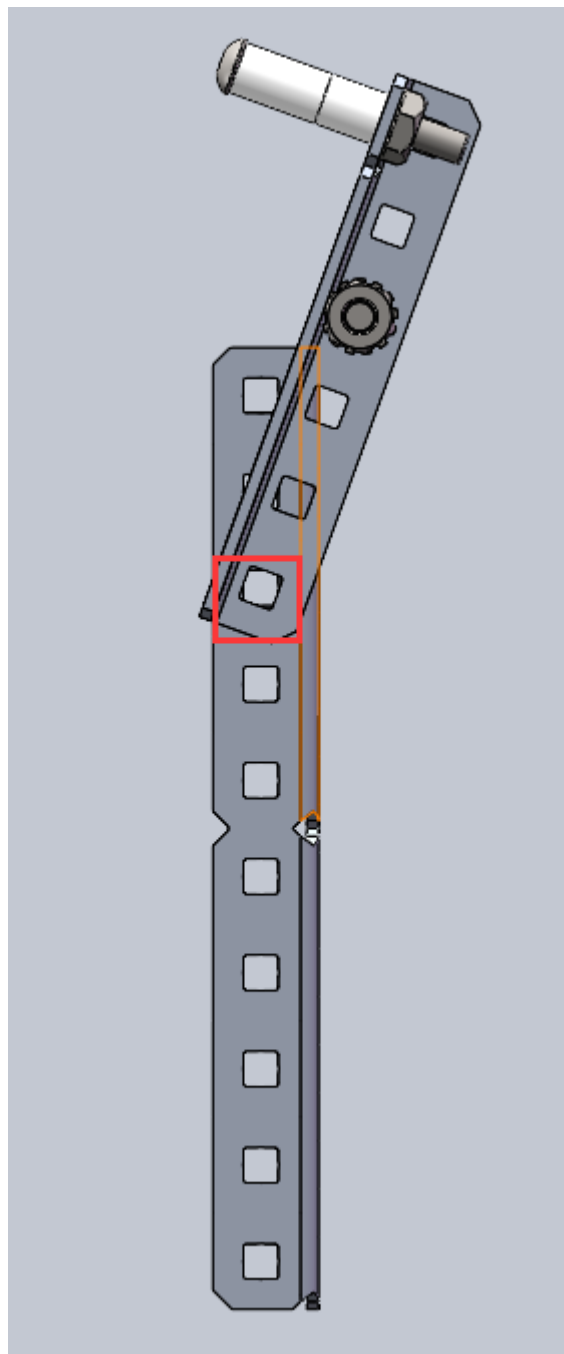
8mm垫圈

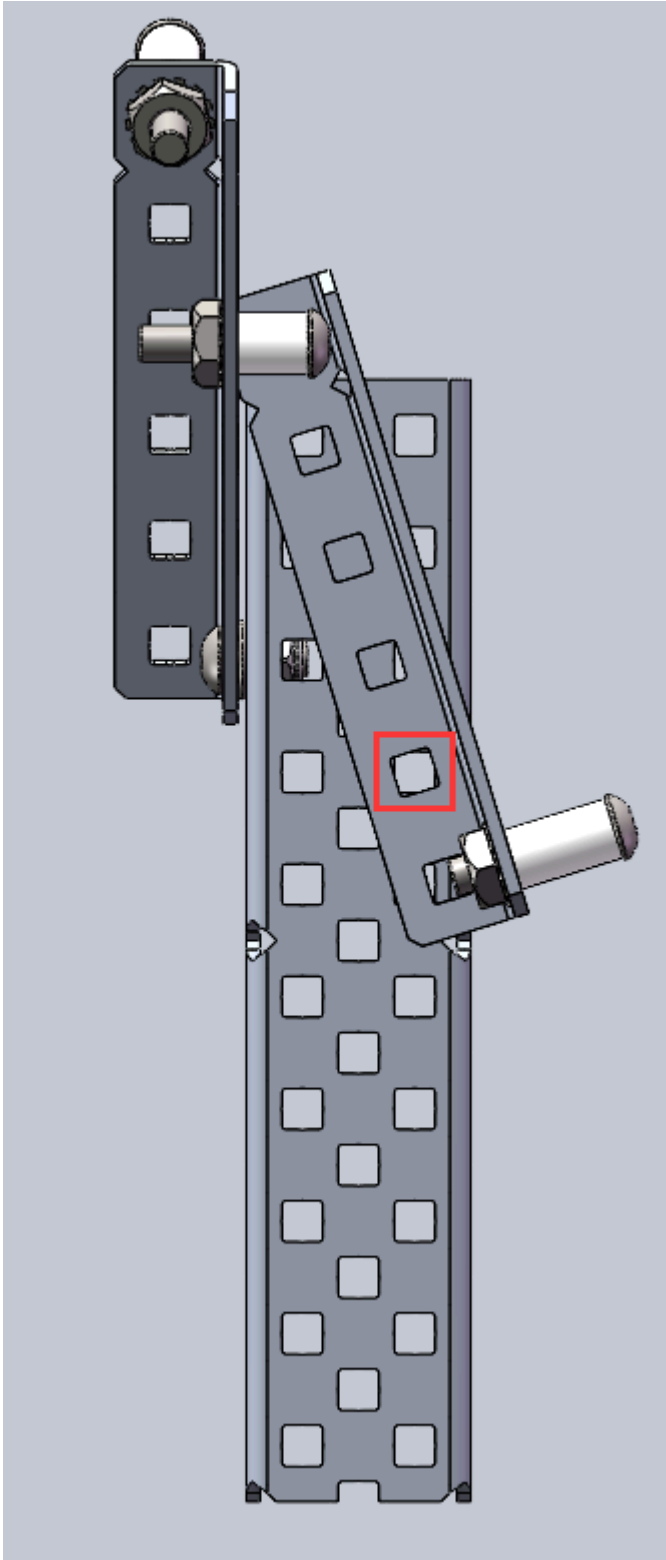


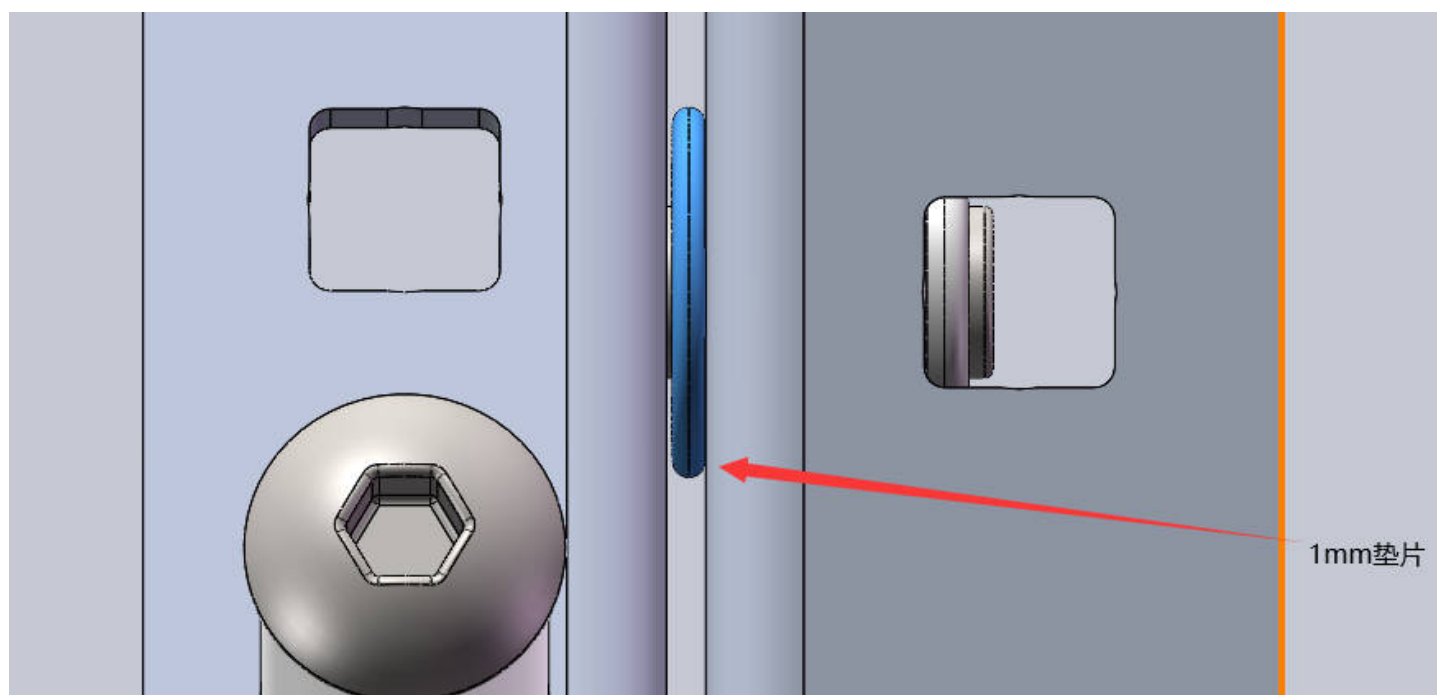
12mm垫圈



组合







安装到底盘

